The FCO Cockpit
Global Bubble Status Report
June 2022

Entrepreneurial Risks
ETH zürich

SUSTech Southern University of Science and Technology

Notification

Change of format of the FCO Cockpit Reports due to the Retirement of Prof. Didier Sornette from ETH Zurich

As Prof. Didier Sornette will retire from ETH Zurich on Jul 31, 2022 due to the mandatory age limit of the Swiss Federal law for professors, the Chair of Entrepreneurial Risks will disappear and the FCO Cockpit – Global Bubble Status Report will not be published any more on the website of the Chair of Entrepreneurial Risks, ETH Zurich. The good news is that the website content of the FCO, publications and other resources will remain accessible (but will no more be updated). The time for the URL https://er.ethz.ch will freeze indeed on July 2022!

We will upgrade the contents and format of the reports, and they will be published by the Institute of Health and Wealth, a new institute located on the lake of Lucern, Switzerland, which will be headed by Prof. Sornette.

If you are interested to follow the updates and future versions of the FCO monthly reports, please subscribe to our email list. To subscribe, follow the instructions that will soon be put at https://er.ethz.ch/financial-crisis-observatory.html

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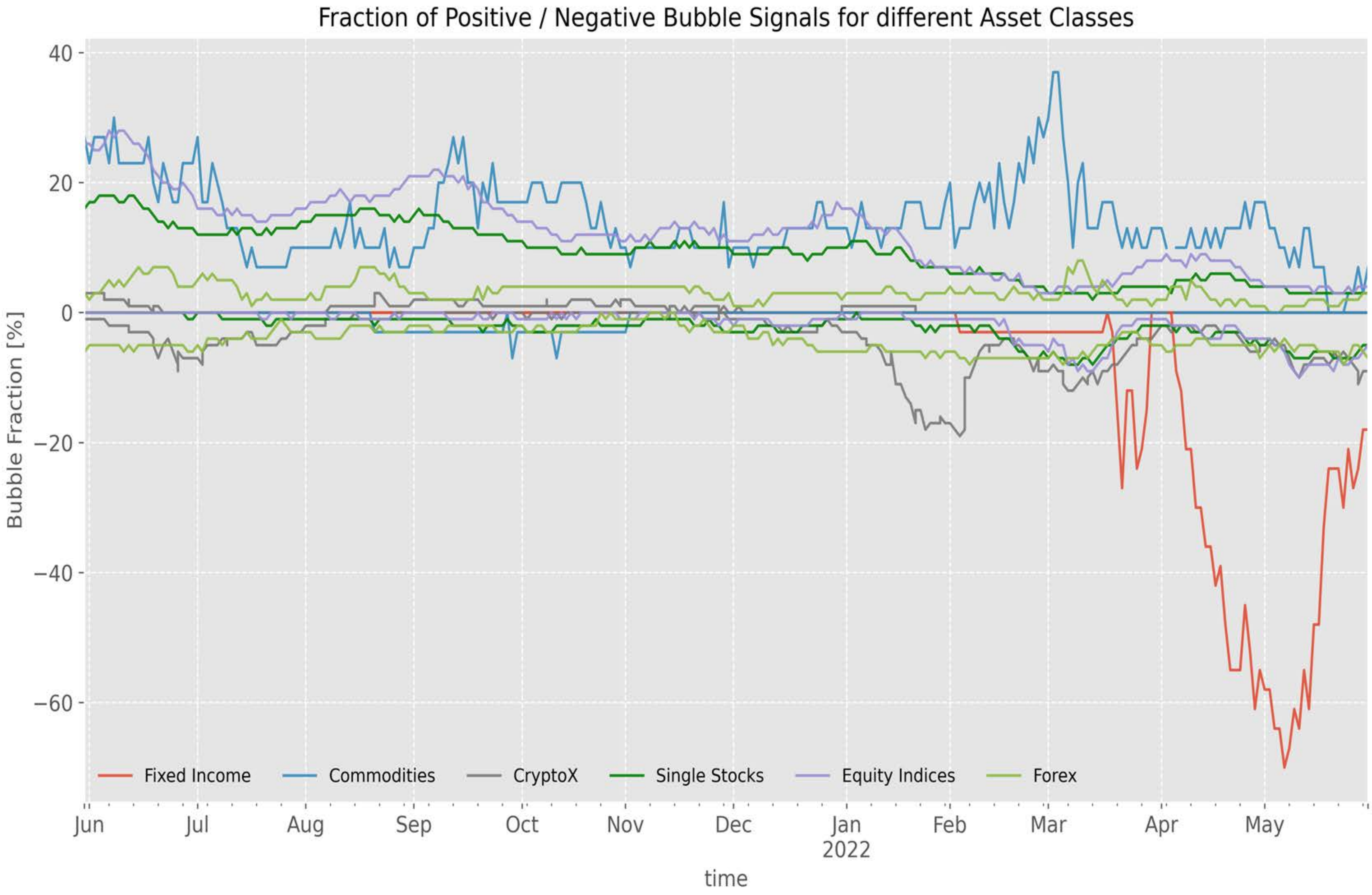
Purpose and Benefits

- The **Financial Crisis Observatory (FCO)** monthly report discusses the historical evolution of bubbles in and between different asset classes and geographies. The purpose of the FCO report is to ascertain which asset classes and sectors are **crowded** and to what degree they develop contagion risks.
- Today, the report is **used by 600+ institutions world-wide**, including universities, think tanks, sovereign wealth funds, hedge funds, family offices, private banks and pension funds.
- It delivers the big picture in terms of growing bubbles and instabilities in today's financial markets for **Chief Investment Officers**, **Senior Researchers**, **Fund Managers**, and **Independent Financial Advisors**, and all parties with **investment performance responsibility** or **managing financial risks**.
- The report is the result of an extensive analysis applied to the historical time series of about 450 systemic assets and about 850 single stocks. The systemic assets are bond, equity and commodity indices, as well as a selection of currency pairs. The single stocks are mainly US and European equities. The data is from Thomson Reuters.
- To new readers, we recommend proceeding to the appendix for more detailed information about the methodology and procedures applied in this report.

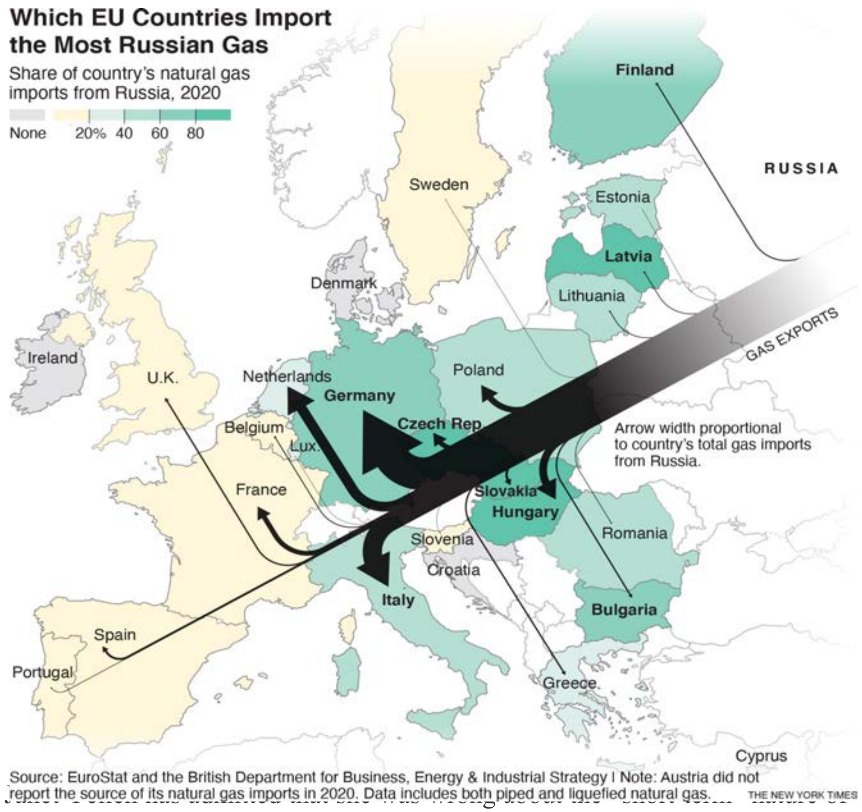

Fraction of Positive / Negative Bubble Signals for different Asset Classes
40
20
0
−20
−40
−60
Bubble Fraction [%]
Fixed Income
Commodities
CryptoX
Single Stocks
Equity Indices
Forex
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
2022
Feb
Mar
Apr
May
time

General Results as of 2022-05-31

	Region	Analyzed Assets	Fraction of Pos. Bubbles [%]	Fraction of Neg. Bubbles [%]
Fixed Income		33	0	18
Commodities		30	7	0
Single Stocks		1638	3	5
	Europe	958	3	6
	United States	668	3	3
Equity Indices		857	4	5
	Europe	52	6	2
	United States	376	4	3
	Global	383	3	6
Forex		130	3	7
CryptoX		457	0	9

- In fixed income, the negative bubble fraction has declined dramatically from 55% to 18% by the end of May 2022.
- In commodities, the positive bubble fraction has dropped considerably from 17% to 7%, while the negative bubble activity remains zero.
- For single stocks and equity indices, the positive bubble signals have both dropped slightly by 1 percent to 3% and 4%, respectively, and the respective negative bubble signals both remain unchanged at 5%.
- In the Forex sector, the positive bubble fraction as of this month remains the same at a low level of 3% as the last month, while the negative bubble signal has experienced a slight rise of 3 percent to 7%.
- In the CryptoX sector, no positive bubble activity is observed as in previous months, while the negative bubble signal continues to rise and reached 9%.

FCO Market Outlook



Four Horsemen of the Apocalypse

The four horsemen of the Apocalypse from the Old Testament embody the punishments from God. The first horseman rides on a white horse, and he might bring the plague; The second horseman is on a red horse, and he is the creator of war; the third horseman rides a black horse, symbolizing famine; and the last horseman on a pale horse, bringing Death. From an historical perspective, there is a logic behind the sequences: around the time when a civilization reaches its peak, the population density increases with strong social interactions, augmenting the likelihood of epidemcis. The ruling elites may stoke conflicts with neighbouring countries to distract public anger from domestic pains. Then, the war would occur, and military powers collide, aiming to take control of the vital resources. However, the food (and energy) supply might be hammered due to the fierce battles and severe social disorder. Consequently, the lack of essential living elements will escalate conflicts, resulting in more deaths. Biological researchers name similar patterns in ecosystems as “intraspecific competition.”

Chaotic World

Europe is now in the middle of a chaotic geopolitical transition, and its leaders seem to have rushed their decision about its energy transition policies. It is a sad fact that the EU does not have the luxury of being tough against Russia due to its strong reliance on its natural resources. Still, the EU has decided to embargo Russian oil (cutting 90% of Russian oil imports by the end of the year), leading to further geopolitical antagonism, as we mentioned in our May monthly report. Although the EU and US do not directly fight in the Russo-Ukrainian War, they are trying very hard to have an economic war and natural resource conflict with Russia, which is just starting to show in growing costs for them and even more for the rest of the world. Vladimir Lenin once said, “every society is three meals away from chaos .” We reckon that the EU and US will not have famine at first, given their economic structure and wealthy status. Still, Africa, the Middle East, Asia, and other regions will endure the consequences of the EU’s decisions.

Fix income: In line with our previous reports, the U.S. 10-year bond yield might keep increasing, in a context where Janet Yellen has admitted that she was wrong about the “short term” nature of inflation. FOR YEARS, our FCO monthly report has criticized the MMT, and central banks' money printing business. We have been alarmed about the inflation risk and regime change of the monetary policies since Q3 2021.

U.S. Equity: The technology sector (such as Amazon, Tesla, Meta, PayPal, etc.) and retail sector (Walmart) have shown signs of labour surplus, and they have either begun a headcount freeze or started to layoff their employees of some of their departments. When the leading companies begin to shrink their business, investors should be very cautious about the future perspectives of the U.S. economy. In addition, we notice that smartphone production will reduce due to the weak demand worldwide in 2022, according to the latest word from production insiders. Thus, we suspect that a Recession will come in the EU and even in the US , sooner than later.

Commodities: The energy sector is always capital-intensive, requiring constant huge Capex (capital expenditure). Today's output was determined by the investments made at least two years ago. A new production line, e.g., a crude oil refinery, will need a multi-billion-dollar investment and 5-7 years to mature, and oil companies make their budget planning based on long-term perspectives, not on just the current price. However, the growth of ESG policies has limited the financing of the traditional fossil fuel industries for the past few years, which restricts the industry's long-term perspectives. In addition, the COVID also shrunk further the Capex since 2020, further damaging the potential production. Fossil fuel companies will not suddenly invest massively in response to today's oil, gas or coal spot price to boost their future productions, given the strict Carbon emission requirements (a targeted reduction of 45% by 2030). So, when the U.S. and E.U. decide on their embargo on Russian oil, gas, and coal, they create a man-made supply-side energy crisis. Besides, the Russian oil business relies heavily on western extraction technology, which will no longer be available due to the sanctions. Moreover, if the E.U. does not demand Russia's oil and gas, Russia has to shut down its wells in Siberia (Russia does not have more capacity to store the gas and oil after the next few months). So, once the wells are shut down, they cannot be put back in productive mode soon without western technology and secured demand. Besides, Saudi Arabia and other middle east countries are likely to have anticipated such outcomes, so we believe they will have less intention to help the U.S. to control the oil price, considering their economic interests. The E.U. has made such rush decision without considering the potential catastrophe risks. Thus, we repeat our previous words: no one will win this war but the arms dealers (and other well positioned economic actors).

Currencies: Emerging market entities have a very different fate. Energy-rich countries such as Saudi Arabia and Brazil have made massive petrol-wealth gains. In contrast, energy-hungry countries like Pakistan and Sri Lanka suffer from high energy prices and have run out of U.S. dollar foreign reserves. The current high leverage worldwide indicates the fragility of the emerging markets. So we will be no surprised to see debt crises and currency crises in the near future..

"Saving the Planet" and the "Energy Transition"

Strategic energy analysis for the future

with Ali AYOUB, Wolfgang KRÖGER, Euan MEARNS, Andrej STANKOVSKI, Spencer WHEATLEY,…

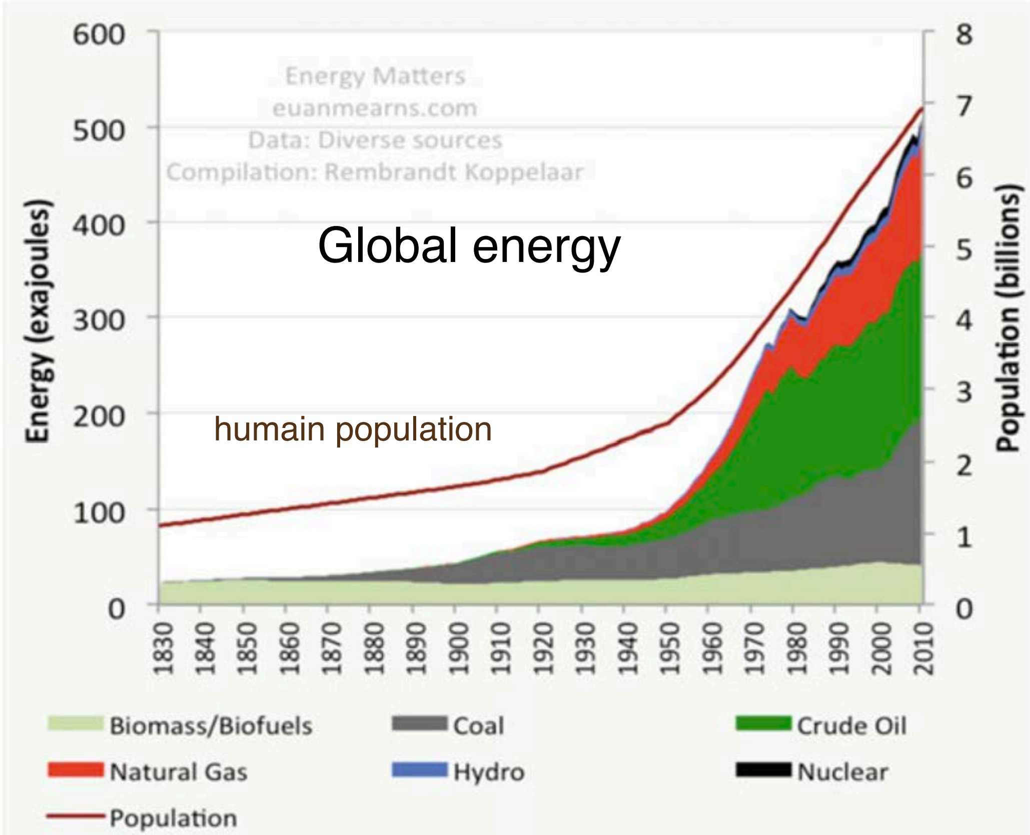
Energy transition is an economy of **wealth replacement** (and not more of wealth growth): **analog to WWII**: destroy-reconstruct-destroy-reconstruct…

=> **impoverishment of the 90% bottom** and increasing inequality

=> transition from a society of abundance to a **society of scarcity**

=> **secular inflation**

=> Risks of social instabilities and **demographic backsliding transitions** toward extremes (political extremes)



need of physics, thermodynamics, engineering, demographics, economics and not just "aspirational infantile fantasy or dreams"

Where are we going?

<u>2020-2030</u>

Endemic debt in perpetual growth?

Decadence and progressive loss of dominance of the West

De-globalization and polarization of the world into **two "empires"**

Conjunction with the exorbitant costs of the "**energy transition**" (we are not ready: it is an **energy replacement** with **no creation of new wealth** like in WWII)

Inflation is coming back sharply (multidimensional and sectoral) and **will last**; risk of hyperinflation

Inflation + monetarization => **Accelerated impoverishment of 90%** of the population

Interest rates will stay low, or if they rise, **central banks will intervene** even more directly in bond and equity markets

Risks of **social instability and democratic transitions to extremes** (e.g. political extremes)

Possible technological transition to **digital currencies** which allow **all exchanges to be controlled**, a transition associated and "justified" by the advent of "**universal basic income**" to calm the popular masses deprived of rights

Acceleration of **individual losses of freedoms "for our good"**: Western societies increasingly imitate the Chinese model while defending themselves against it

<u>2030 to 2070</u>

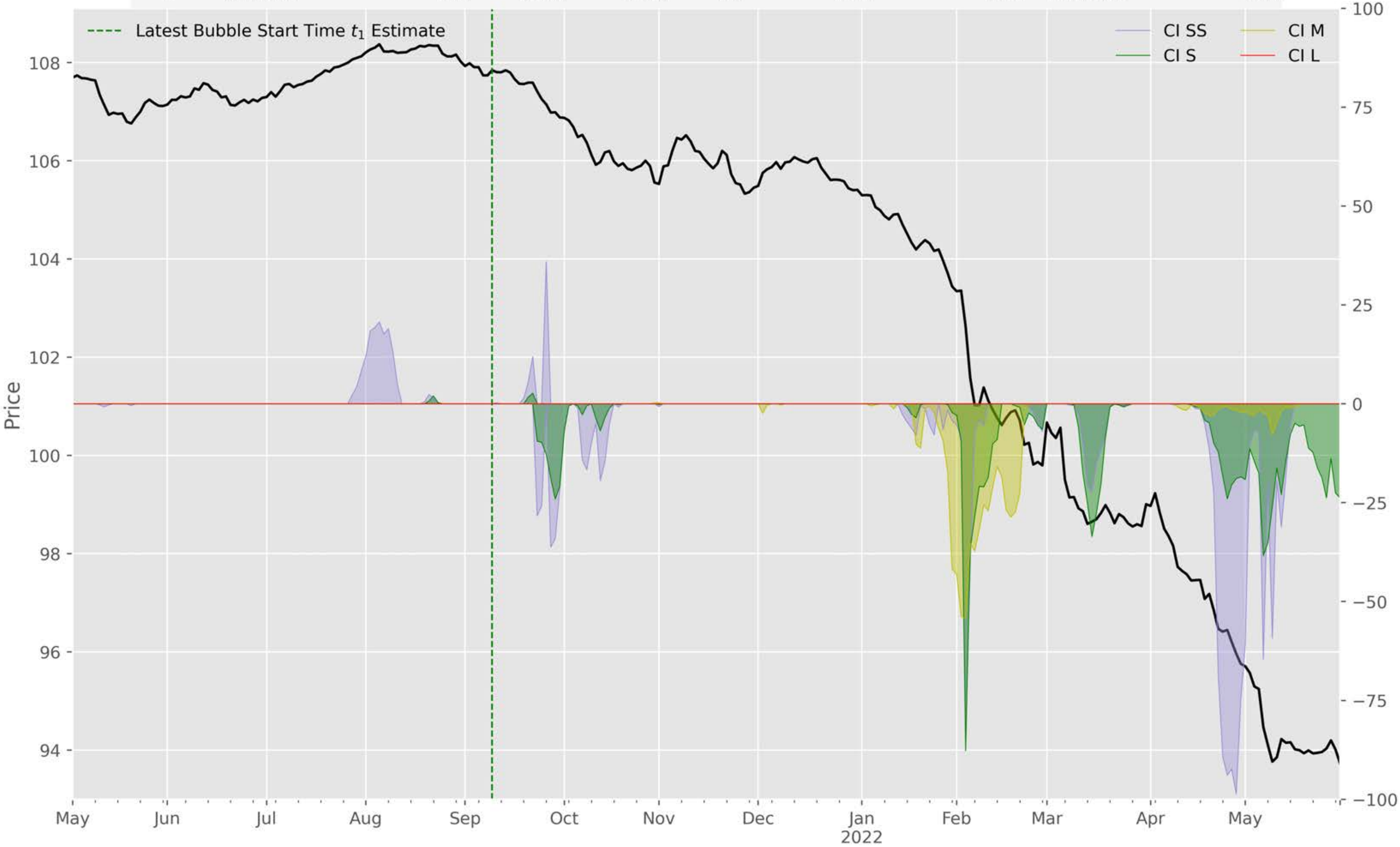
Pax Sinica vs "two empires", with control of digital dollar and yuan, social credit system, health tech, clean energy dominance used to punish economically other nations ('planet rights' replacing efficiently so-called 'human rights'), Electric vehicles and self-driving cars, batteries, robotics, AI dominance and quantum computing come of age.

Fixed Income

No positive bubbles to report

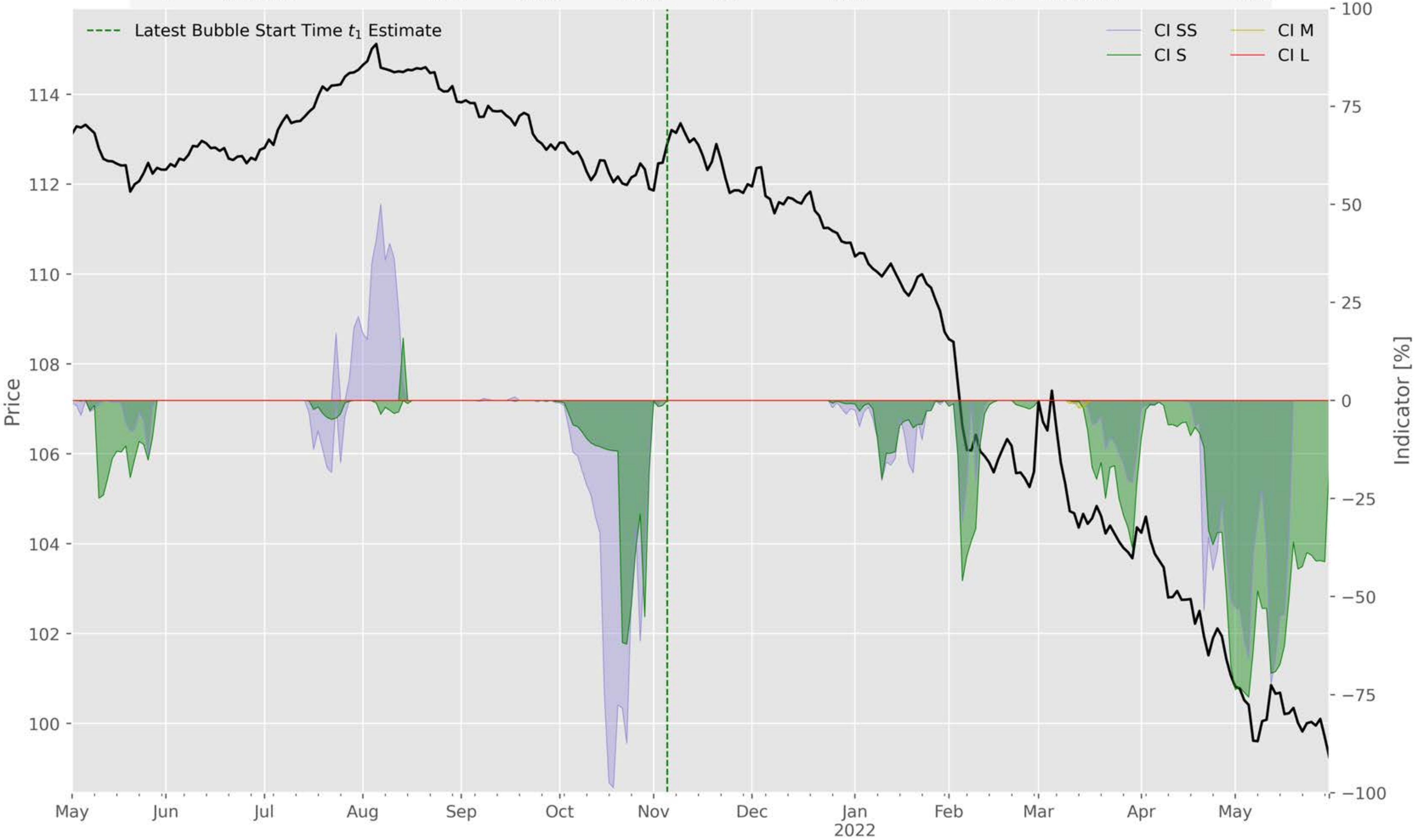
Iboxx Euro Insurance

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
1	Short Term	23.8	-13.09	-17.06	189	89.59	33.13	2022-06-22	92.45



Iboxx Euro Corporates Aaa

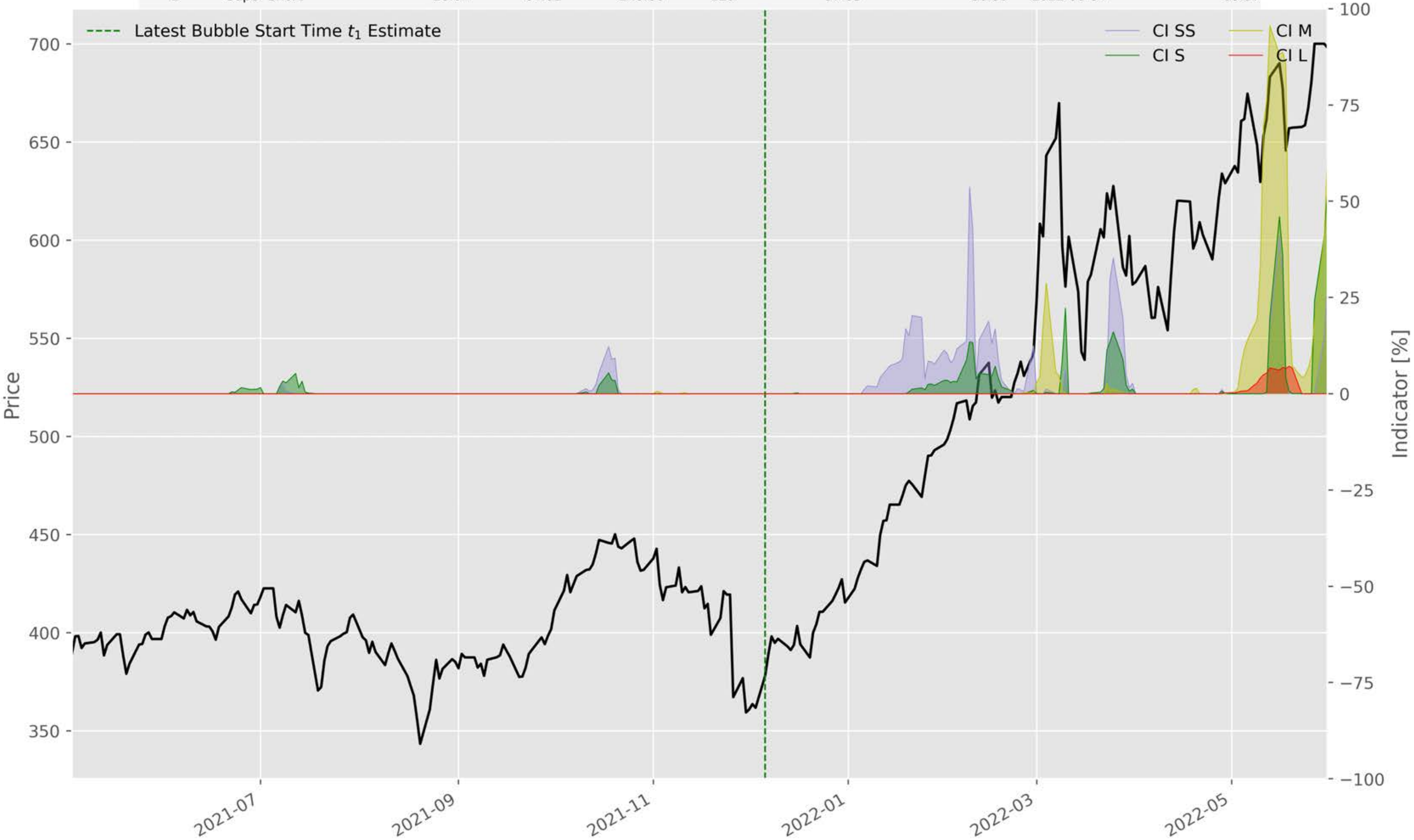
Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
2	Short Term	16.18	-12.08	-19.56	149	68.28	27.86	2022-08-08	99.68



Commodities

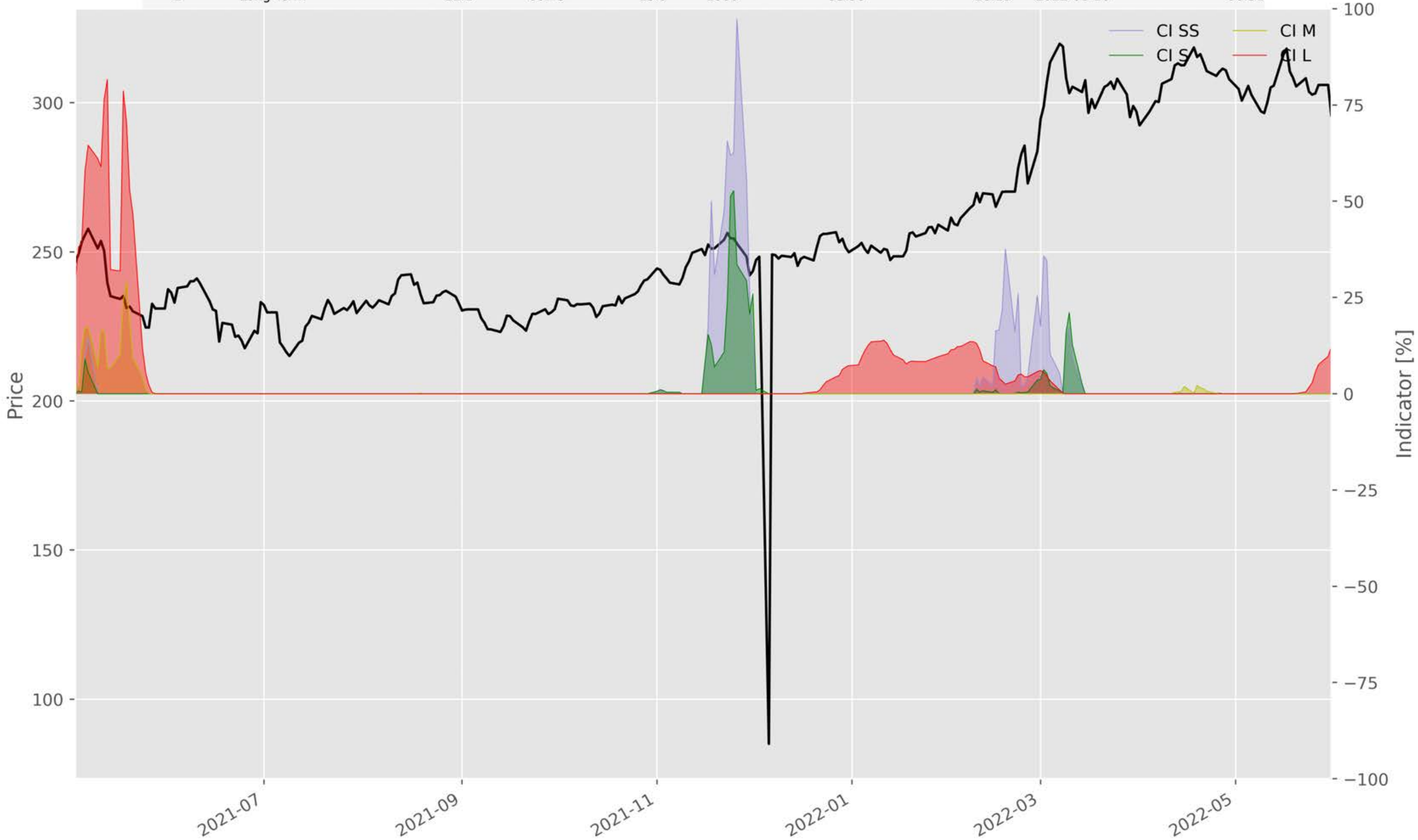
MIcx Gasoline Spot Index

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Super Short	28.04	84.62	240.86	126	97.03	86.86	2022-06-04	99.97



Mlcx Agriculture Spot Index

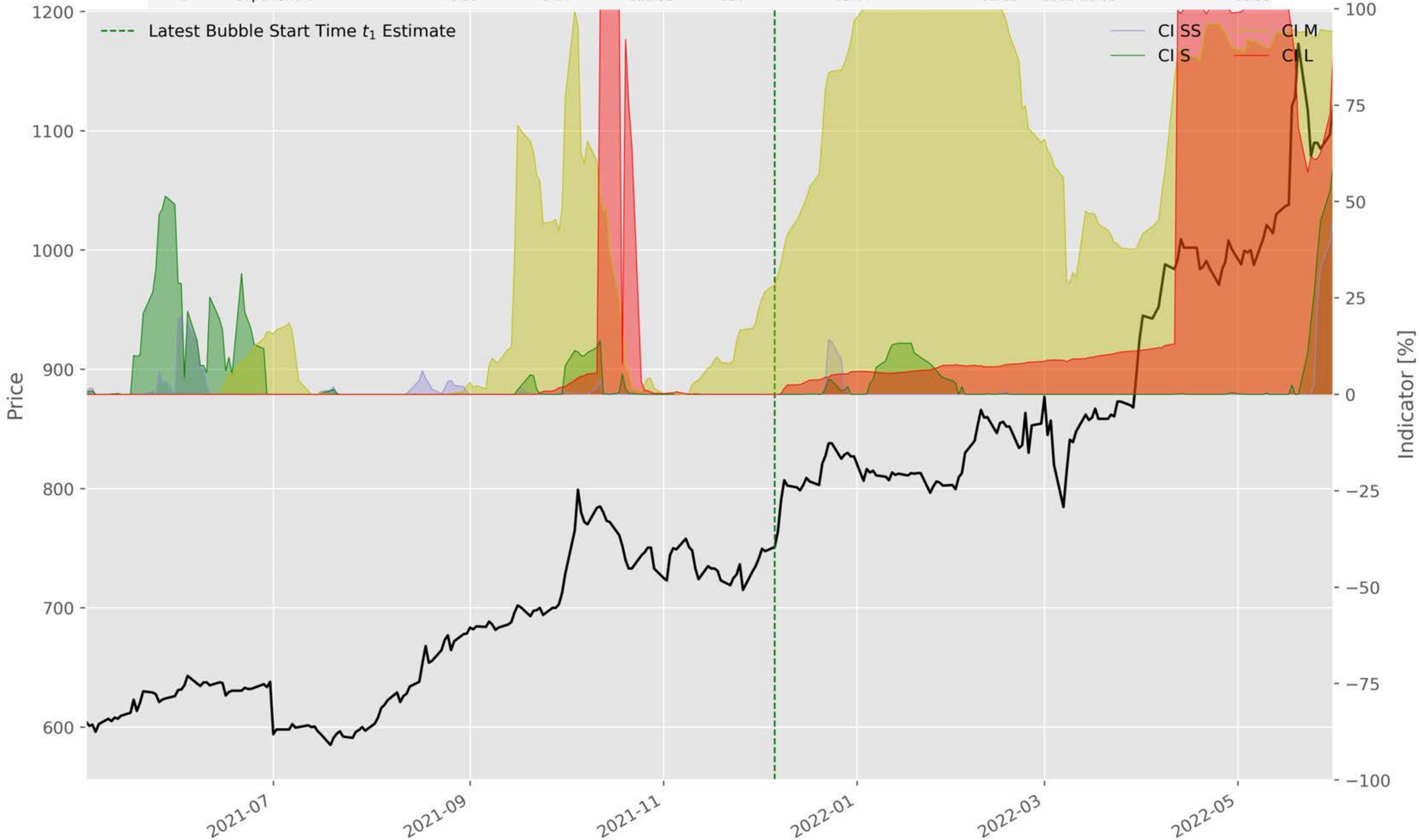
Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Long Term	12.1	80.45	15.9	1008	93.38	26.19	2022-08-10	98.51



Single Stocks

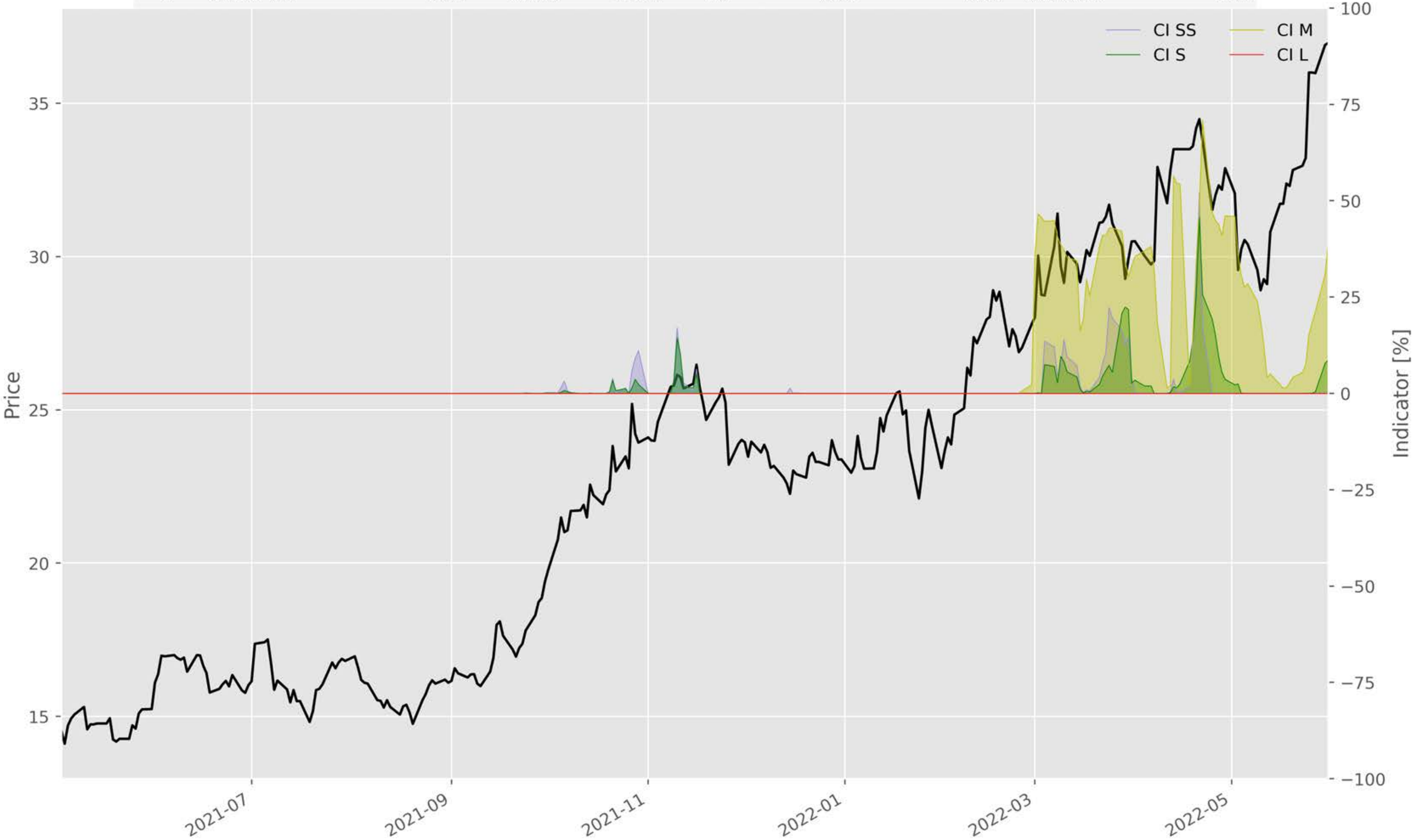
Cez

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Super Short	46.39	49.27	122.81	126	95.64	81.68	2022-06-06	88.59



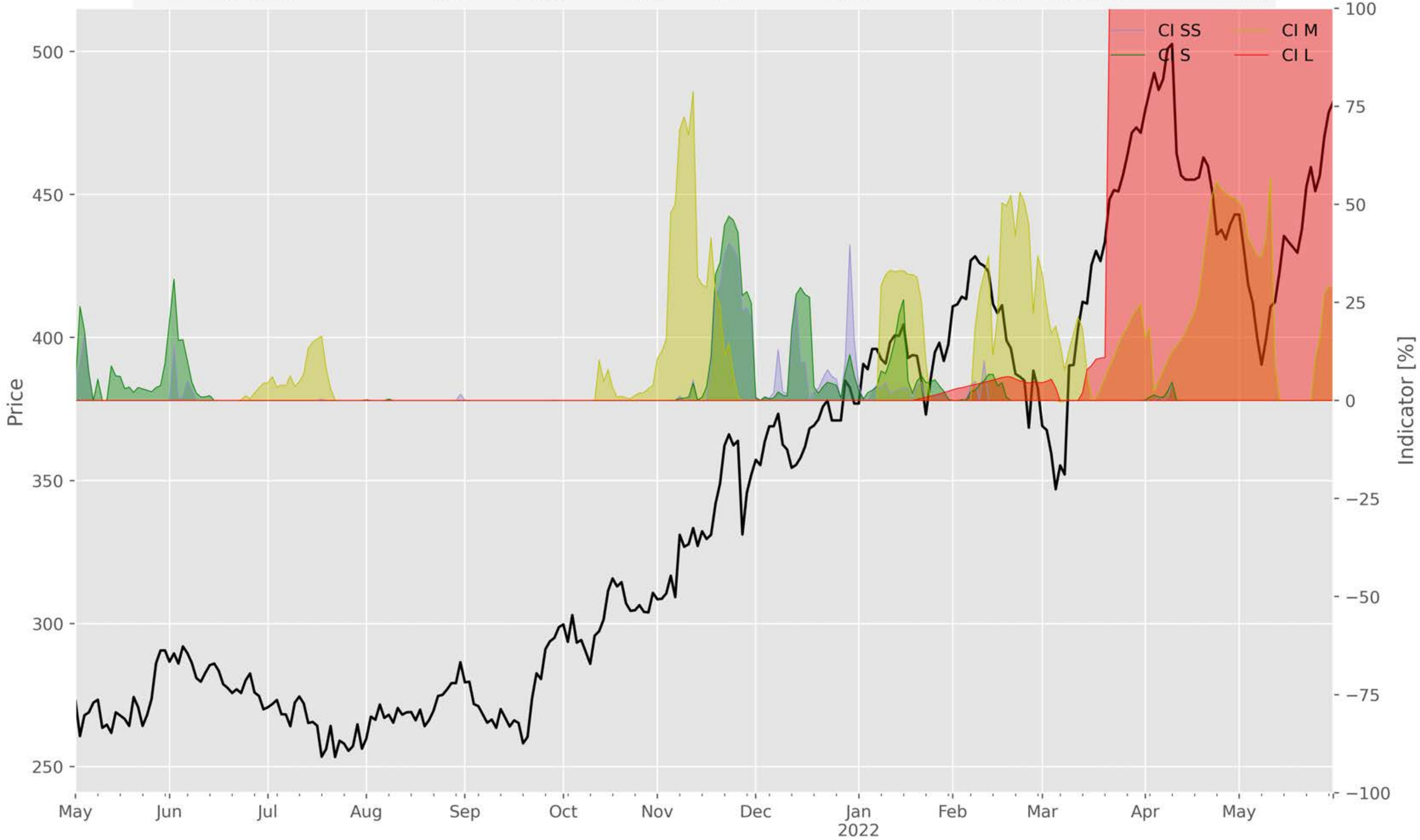
Aker Solutions

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Medium Term	38.67	169.78	111.44	334	75.81	68.88	2022-09-15	84.17



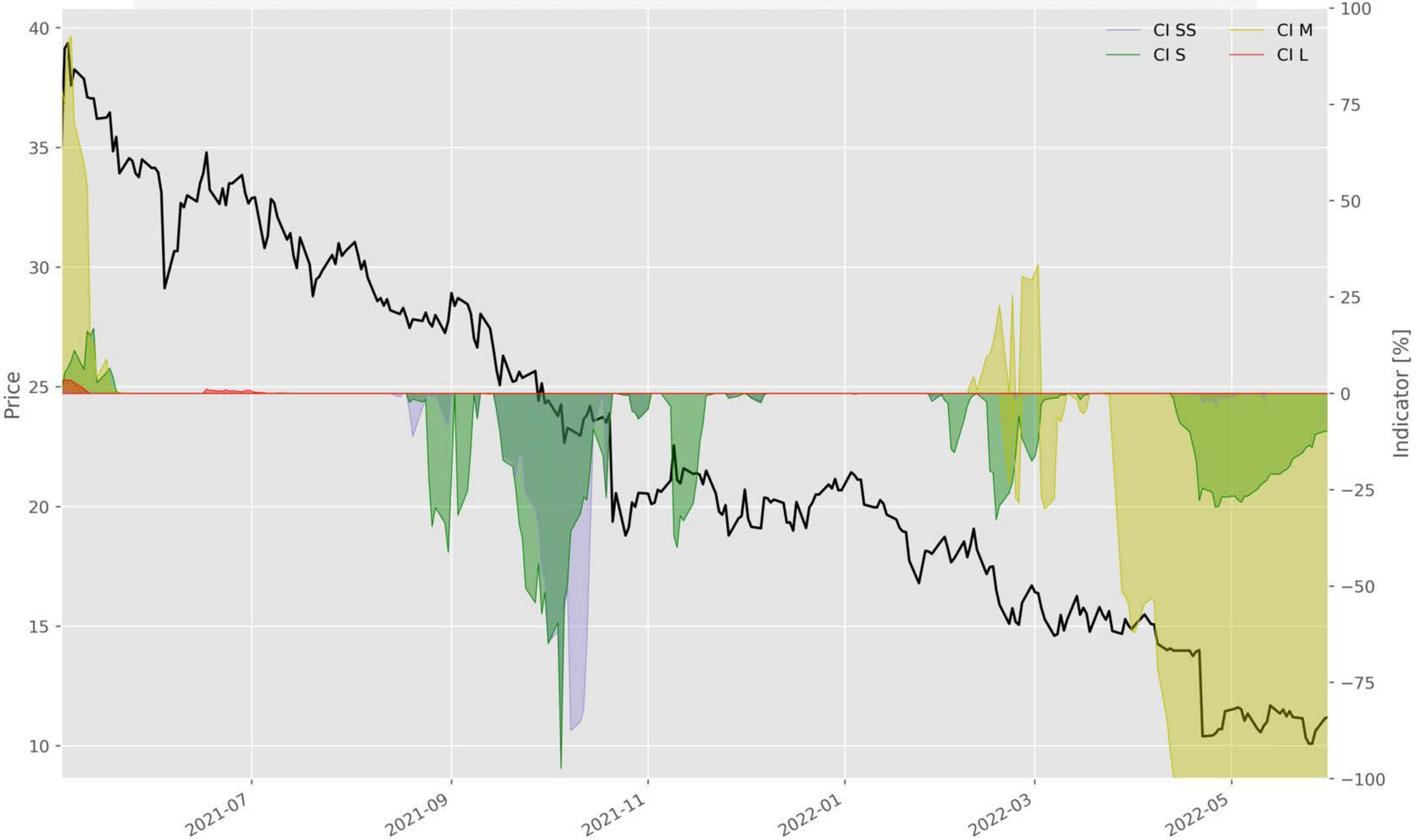
Investec

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
3	Medium Term	29.09	177.86	88.87	405	81.22	59.44	2022-09-02	97.31



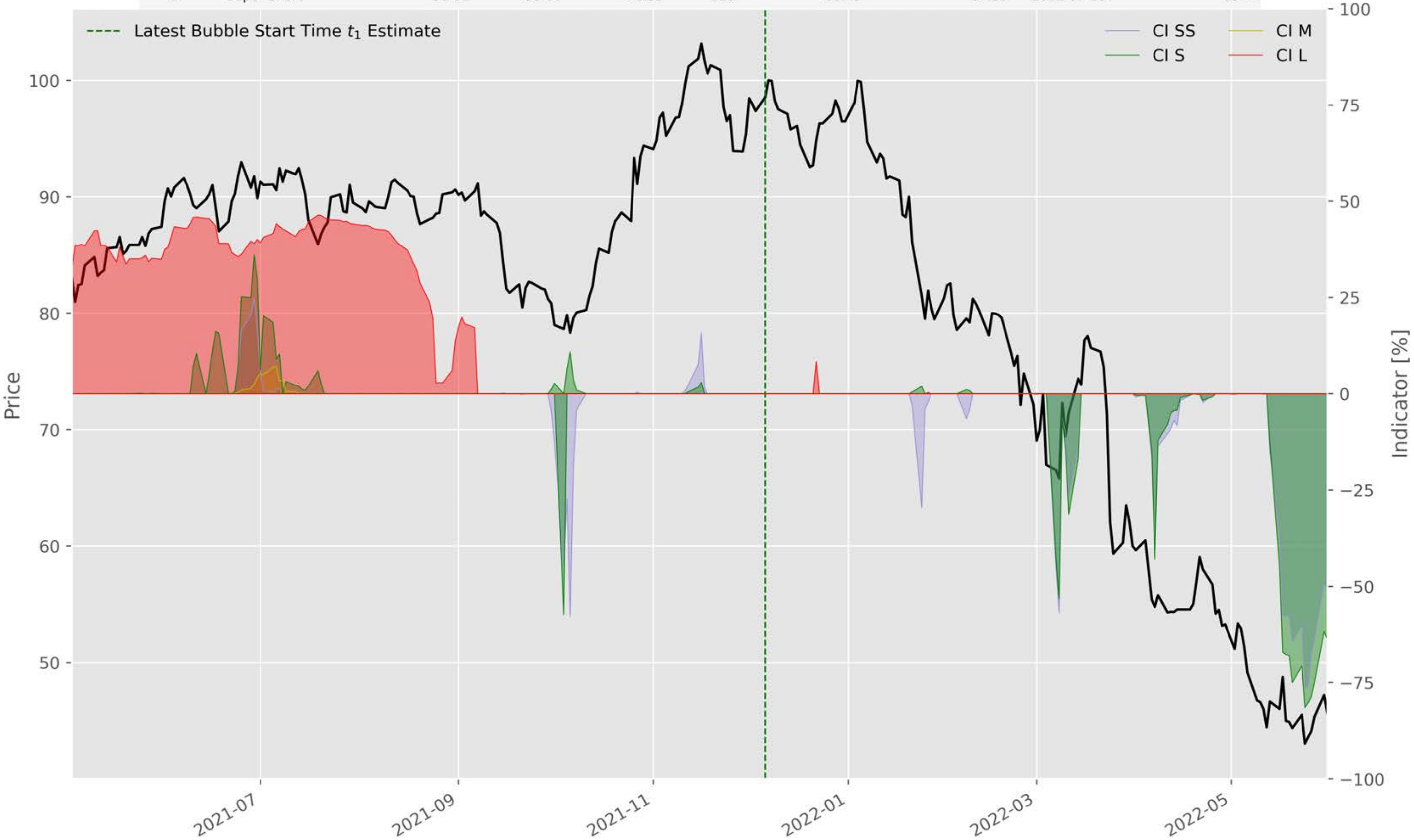
Fingerprint Cards B

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Medium Term	100	-66.16	-61	290	96.71	83.87	2022-06-10	99.98



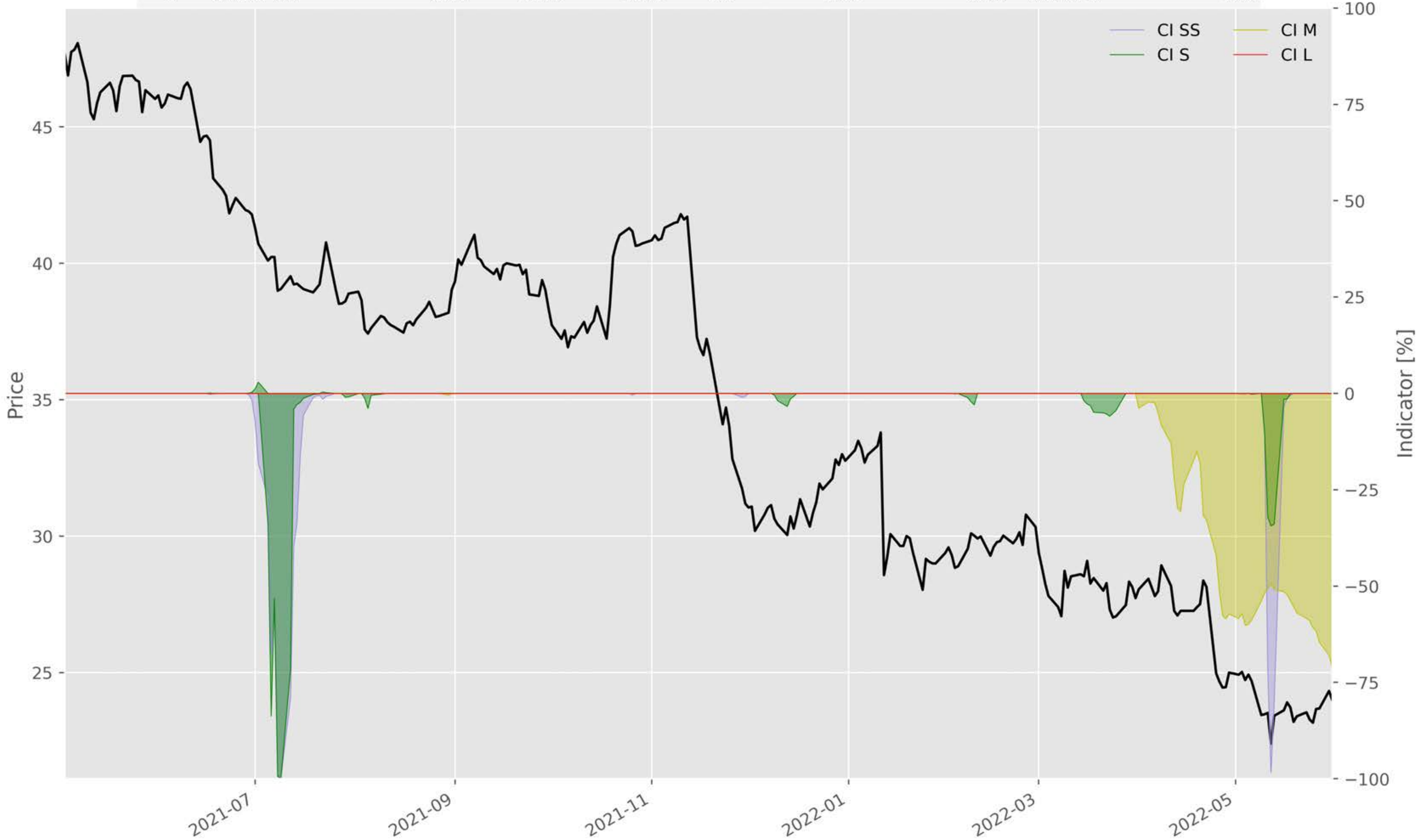
Kion Group (Xet)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Super Short	50.82	-53.66	-78.53	126	68.43	64.88	2022-07-28	99.44



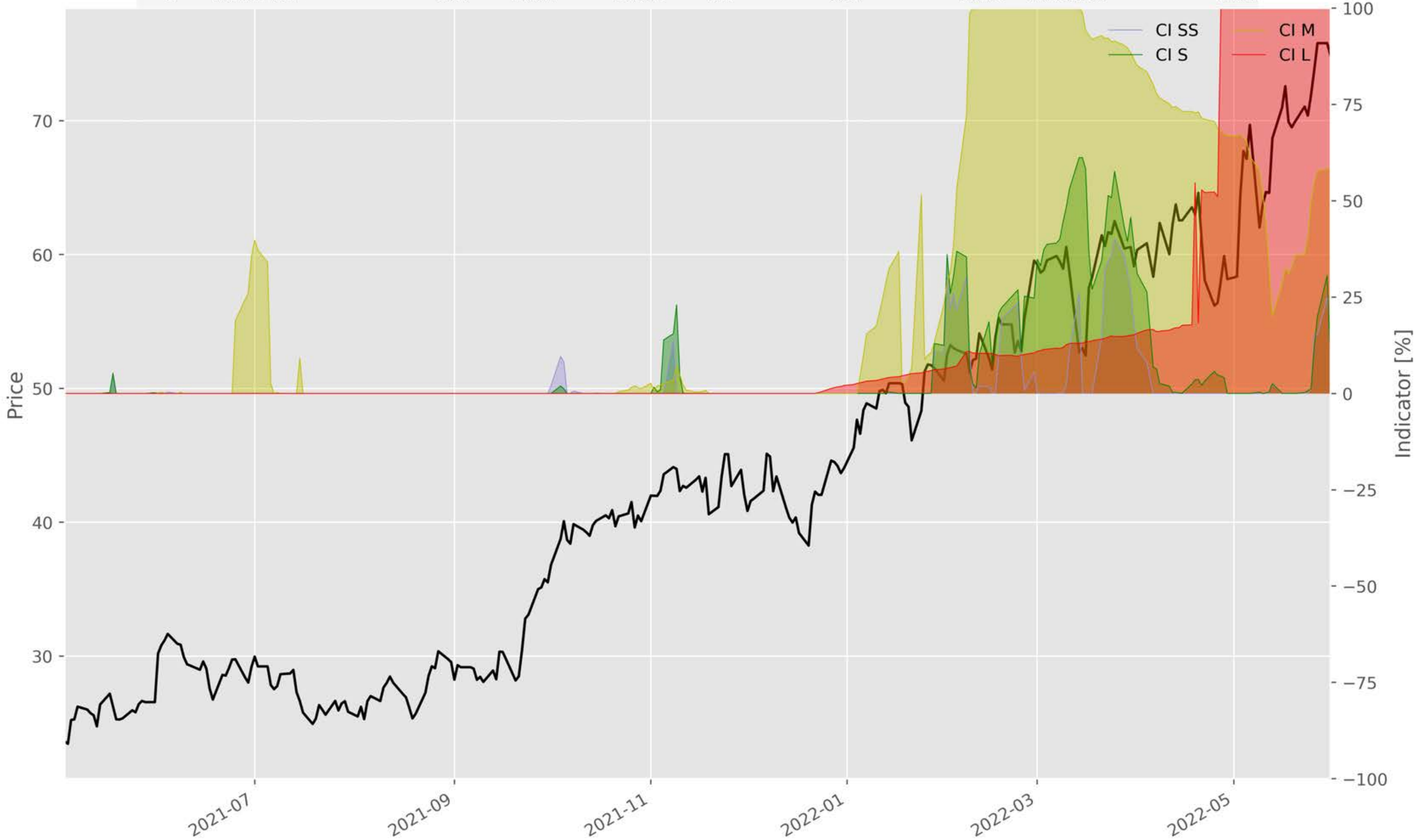
Philips Eltn.Koninklijke

Rank	Strongest Signal	DS-LPPLS Confidence *ci* [%]	Bubble Size *bs* [%]	Bubble CAGR *bg* [%]	Duration [*days*]	Bubble Progress *bp* [%]	Geometric Average (*ci*, *bg*, *bp*)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Medium Term	71.07	-45.48	-33.62	373	88.91	59.67	2022-07-17	96.64



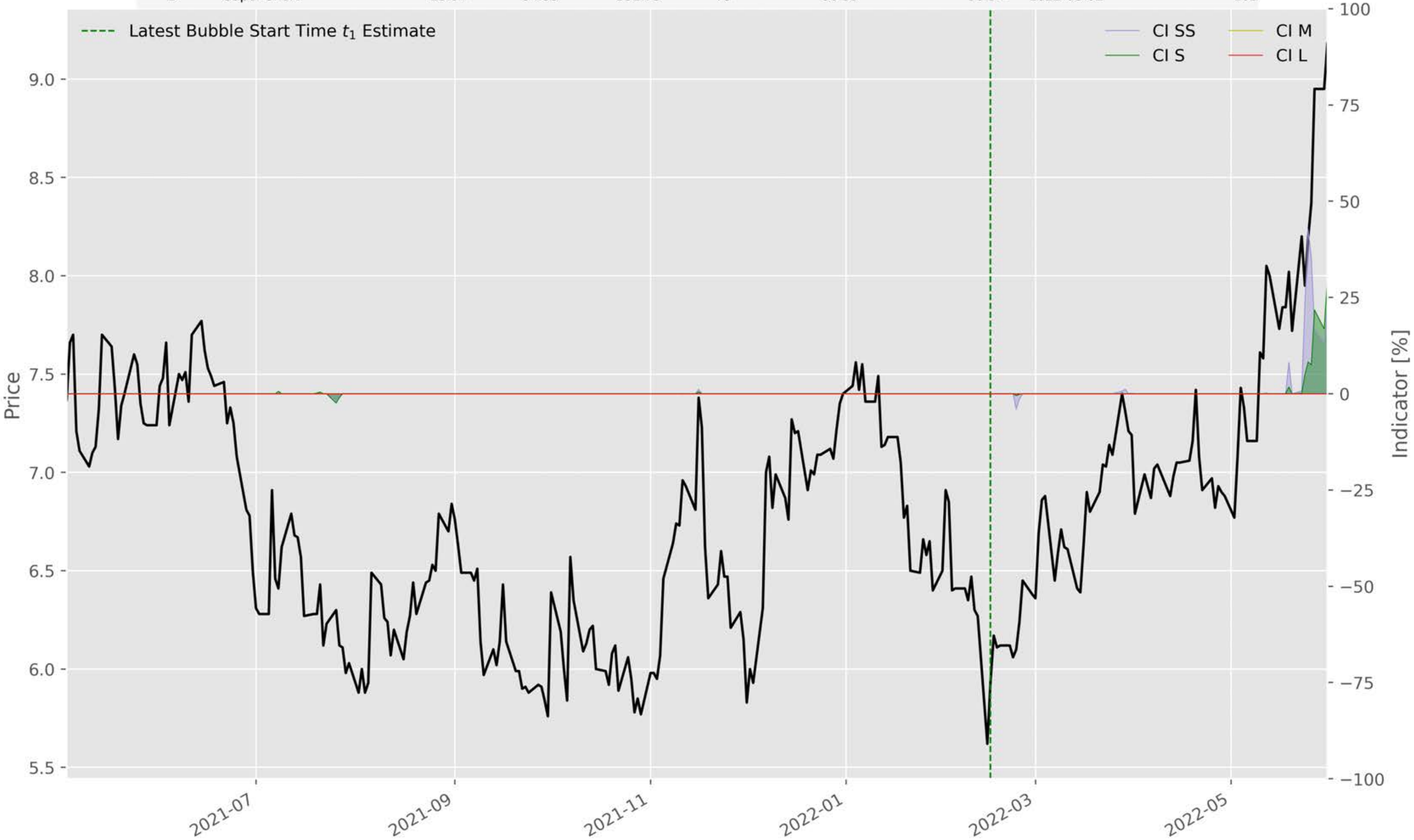
Devon Energy

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Medium Term	58.6	497.77	207.61	401	82.8	100.24	2022-08-22	44.13



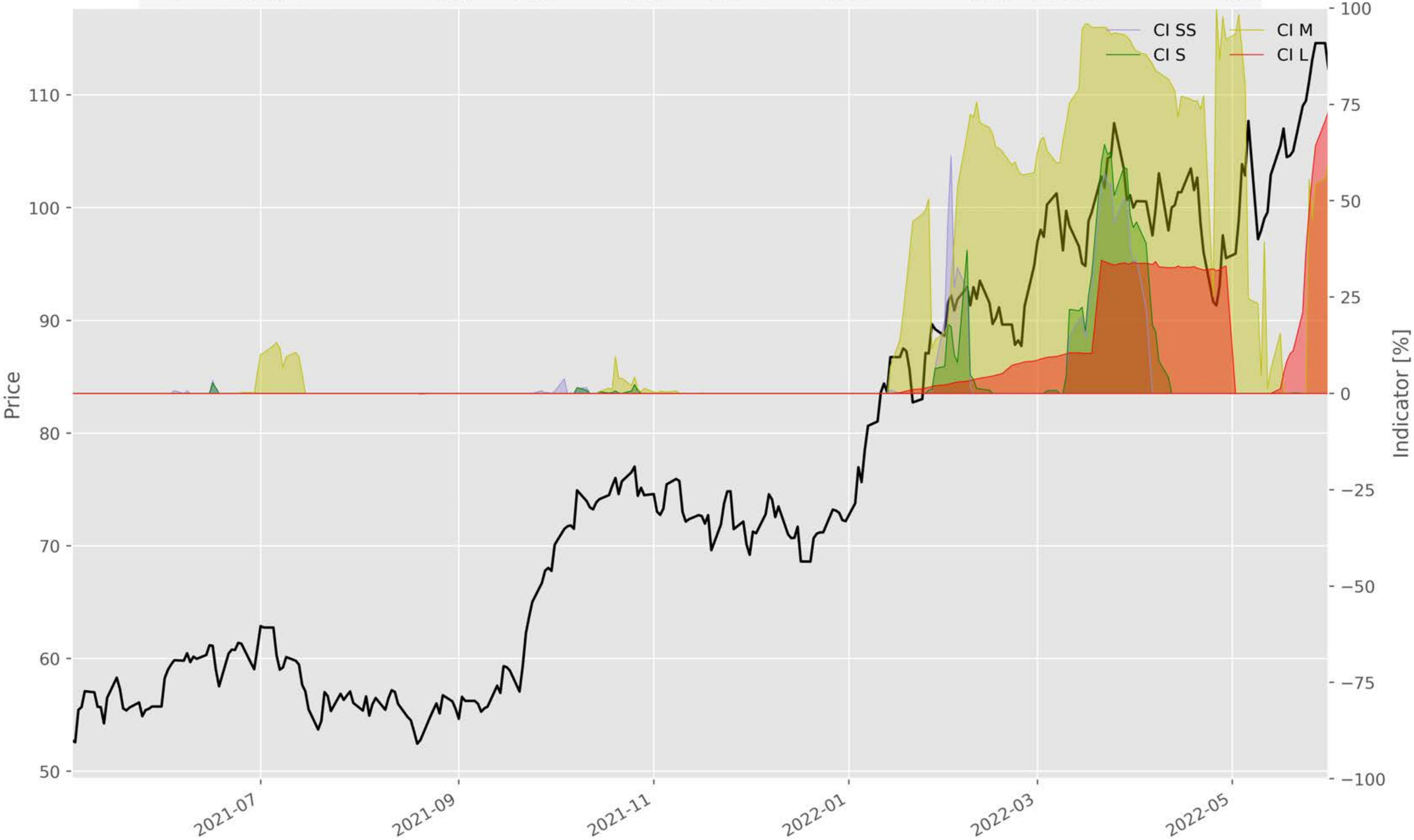
NI Industries

Rank	Strongest Signal	DS-LPPLS Confidence *ci* [%]	Bubble Size *bs* [%]	Bubble CAGR *bg* [%]	Duration [*days*]	Bubble Progress *bp* [%]	Geometric Average (*ci*, *bg*, *bp*)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Super Short	23.04	54.55	331.75	75	98.39	90.94	2022-06-01	100



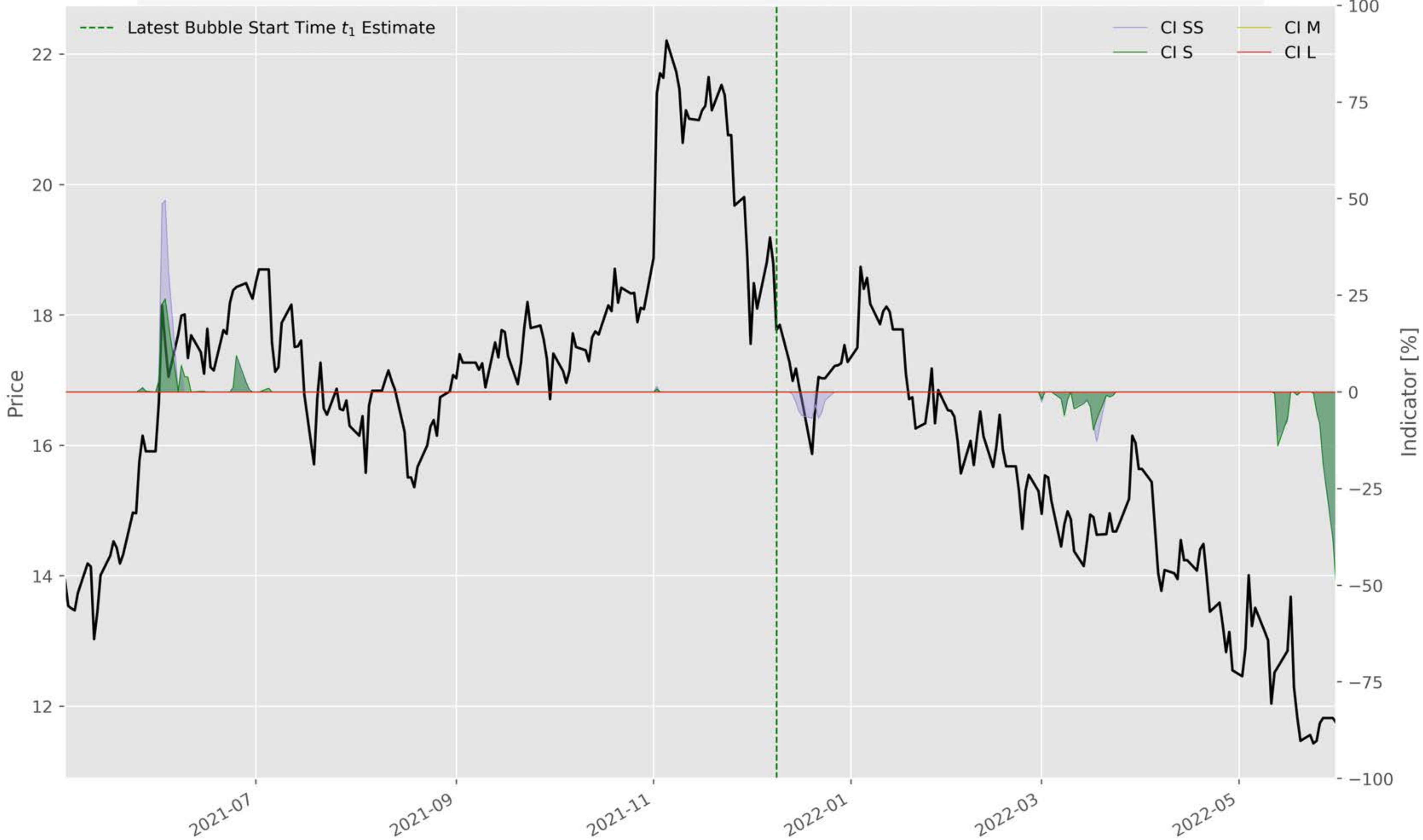
Conocophillips

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Medium Term	60.43	174.12	65.56	504	98.96	73.19	2022-06-05	62.42



Macerich

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Super Short	44.79	-33.88	-57.15	123	91.91	61.73	2022-06-11	83.44

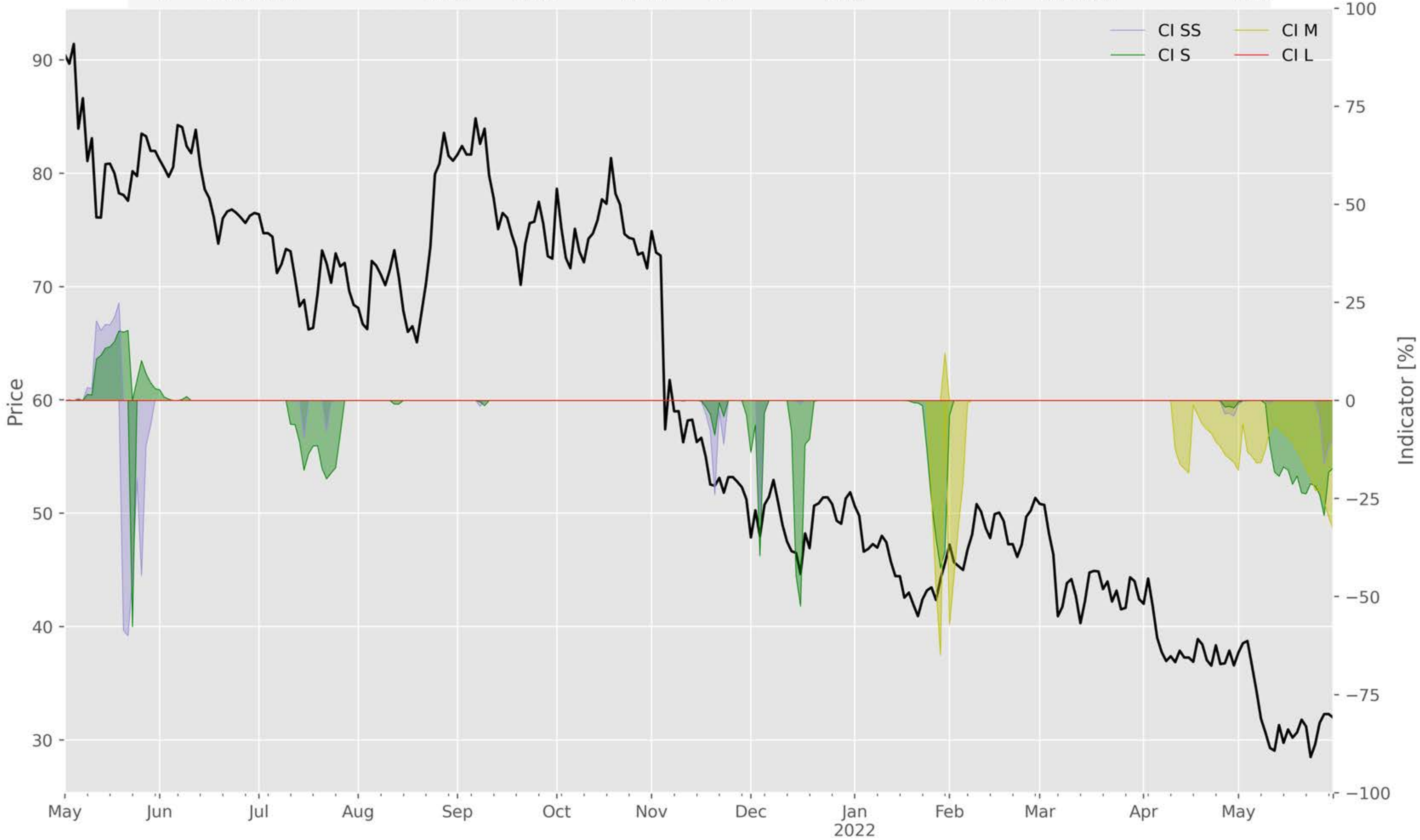


Big Lots

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Short Term	31.92	-64.4	-64.4	252	70.88	52.62	2022-09-12	93.66

Penn Nat.Gaming

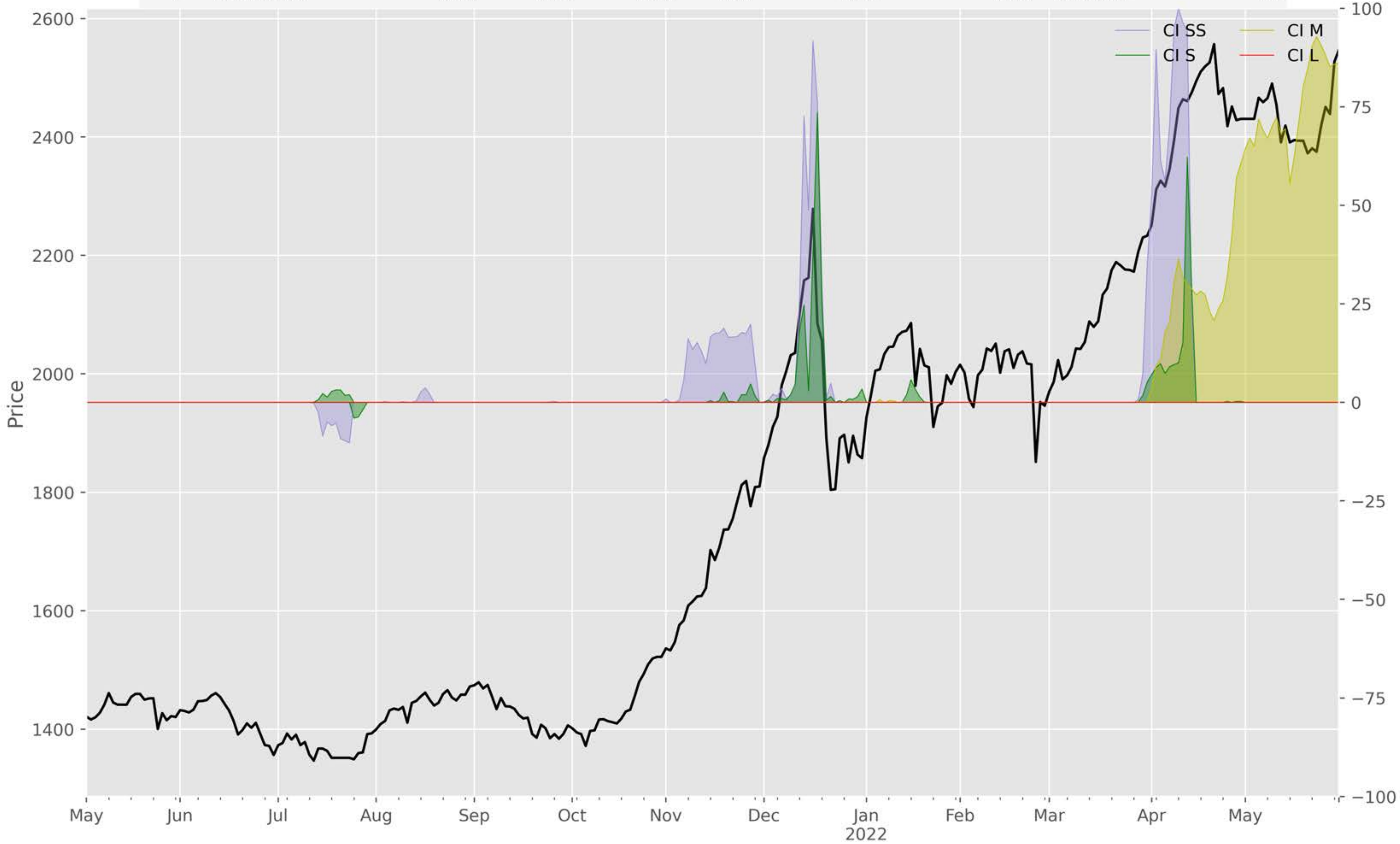
Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Medium Term	32.74	-71.11	-59.51	346	66.69	50.65	2022-11-20	78.56



Equity Indices

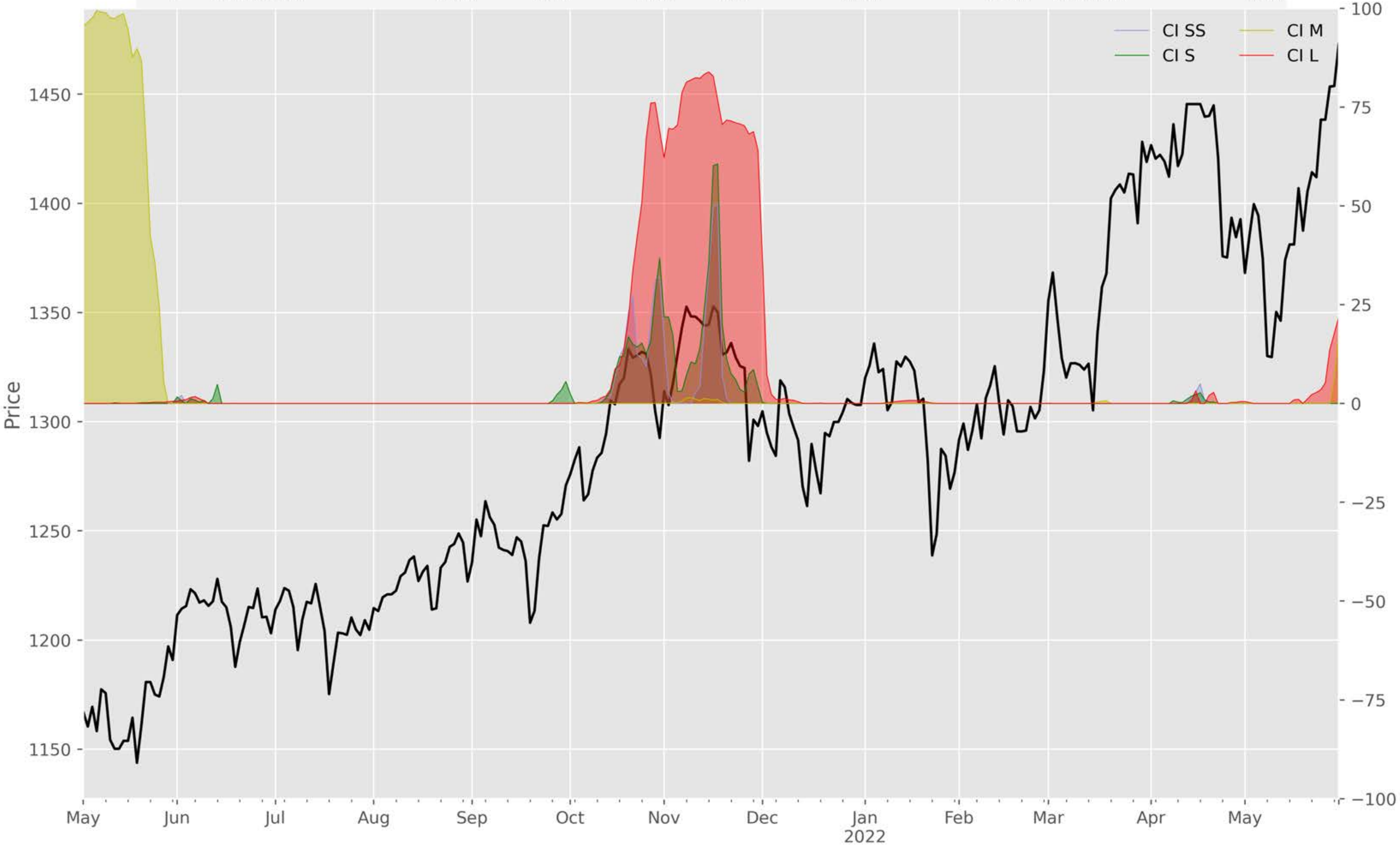
Bist National 100

Rank	Strongest Signal	DS-LPPLS Confidence *ci* [%]	Bubble Size *bs* [%]	Bubble CAGR *bg* [%]	Duration [days]	Bubble Progress *bp* [%]	Geometric Average (*ci*, *bg*, *bp*)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Medium Term	86.17	82.82	66.56	298	86.72	79.23	2022-07-16	68.7



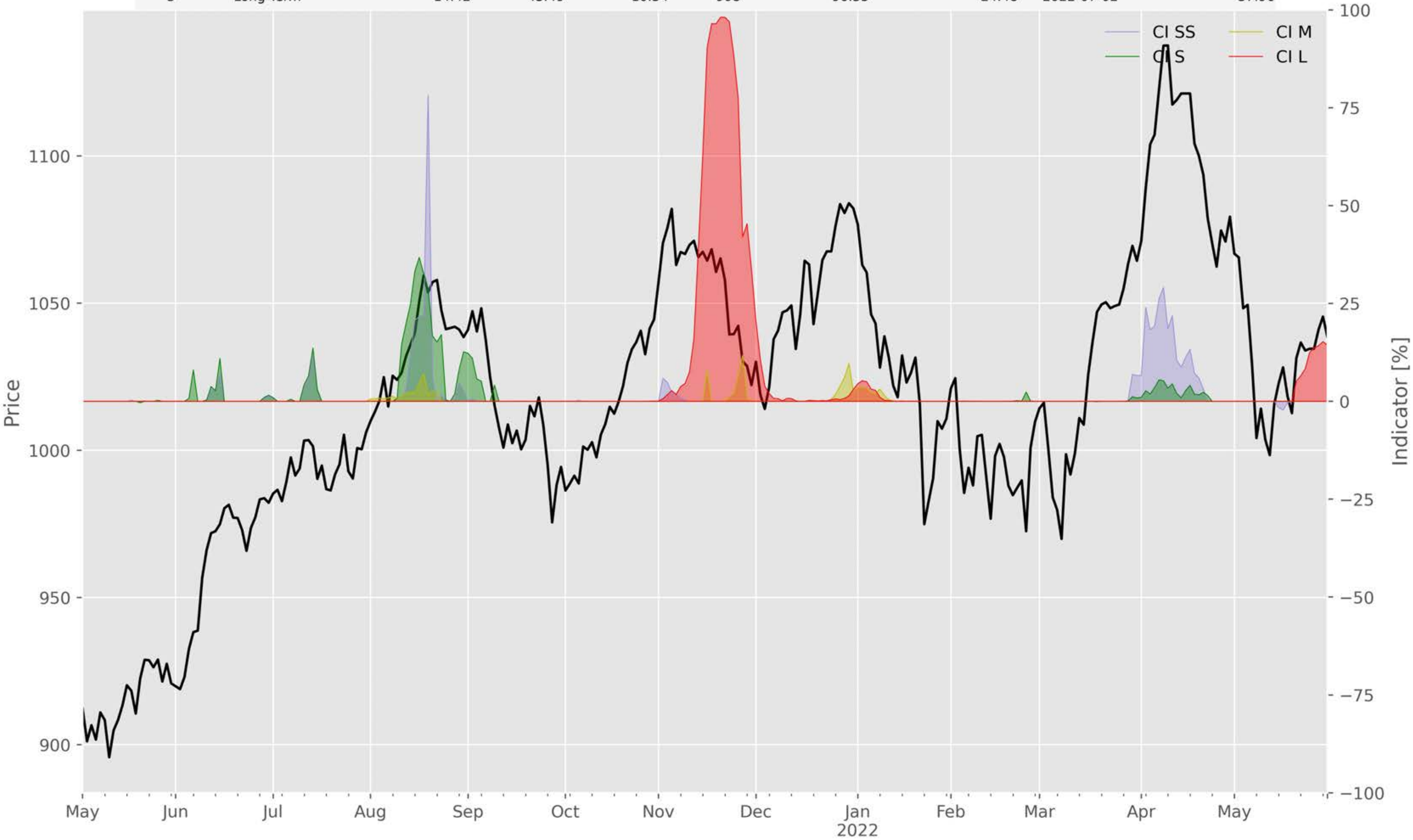
Oslo Exchange All Share

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
2	Medium Term	17.72	39.71	25.97	365	96.78	35.45	2022-06-12	99.99



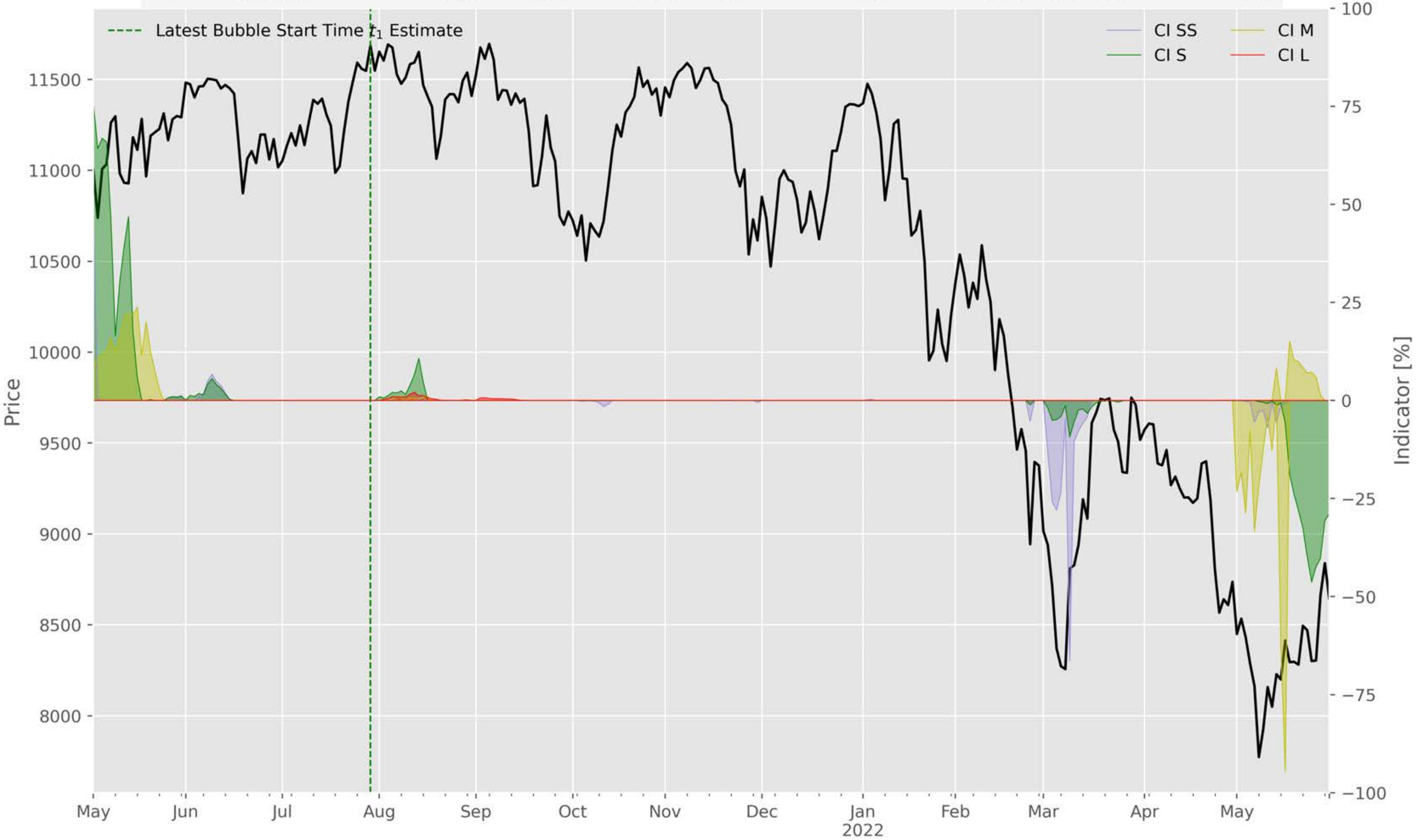
Stoxx Europe 600 Health Care E

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Long Term	14.42	43.49	10.54	908	96.55	24.48	2022-07-02	37.96



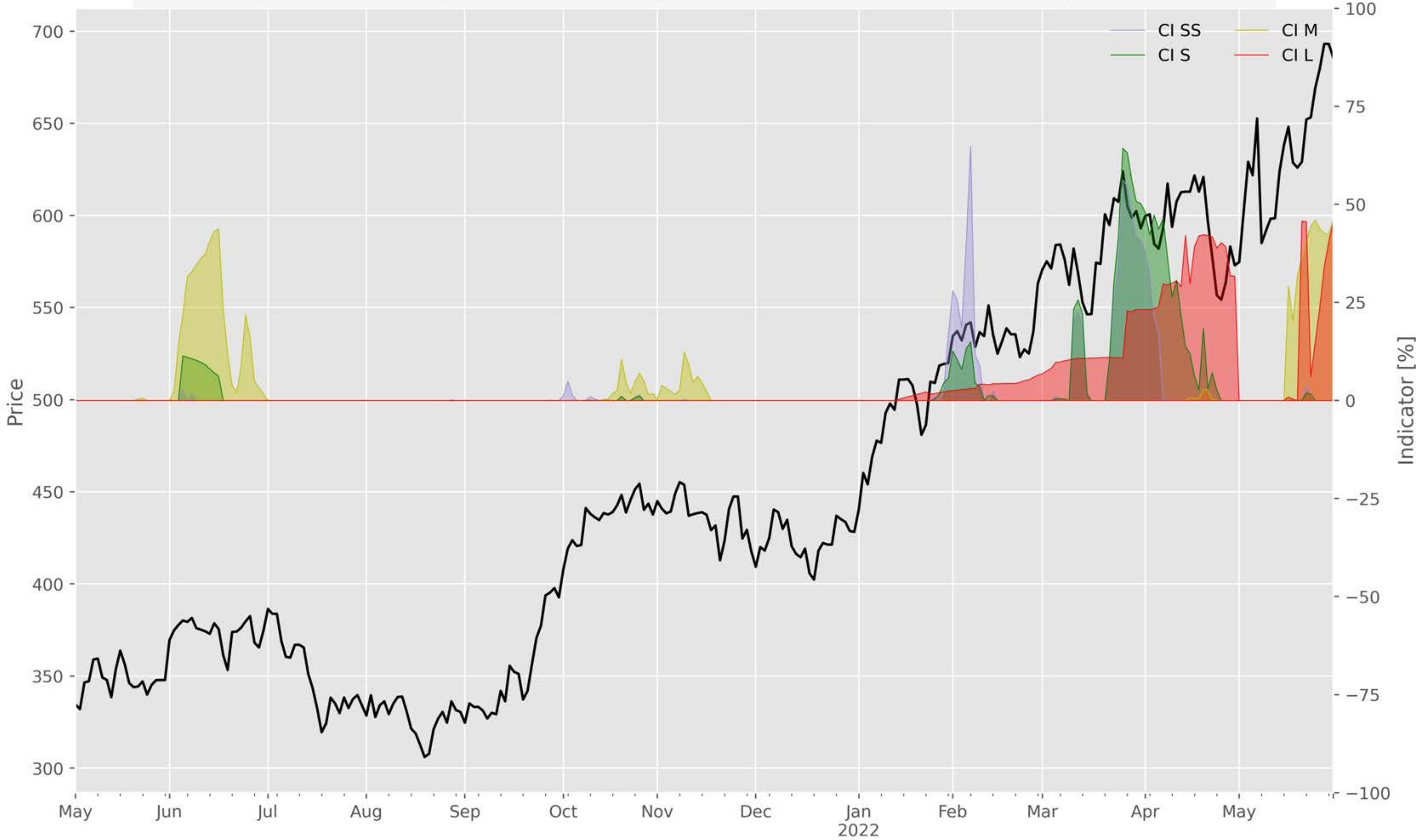
Msci Sweden U$

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Short Term	28.78	-26.02	-29.31	219	89.8	42.31	2022-06-25	96.08



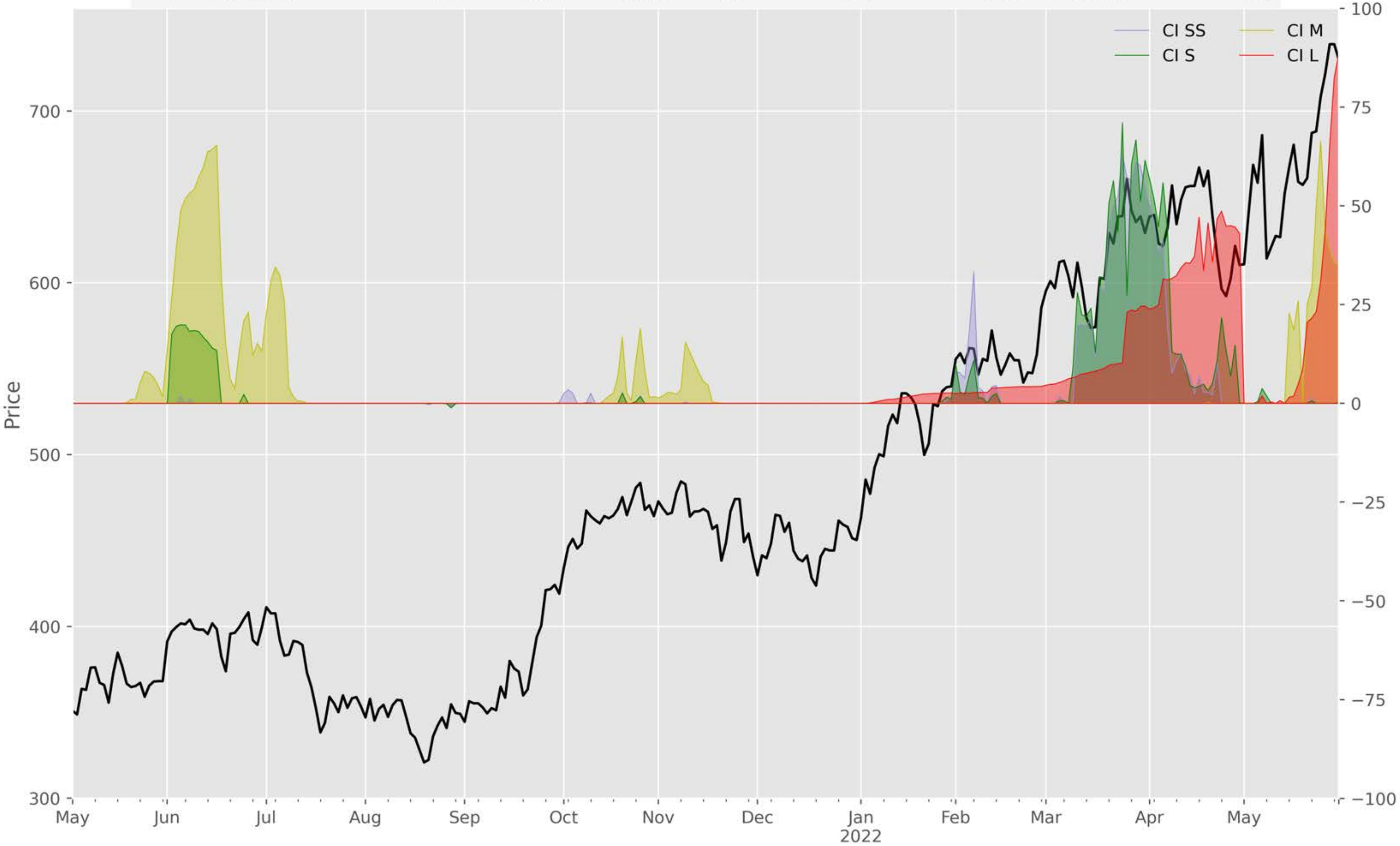
S&P500 Oil & Gas Explor & Prod

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
1	Medium Term	46.16	223.74	108.84	402	97.61	78.86	2022-06-10	99.31



S&P1500 Oil & Gas Explorat & Prod

Rank	Strongest Signal	DS-LPPLS Confidence *ci* [%]	Bubble Size *bs* [%]	Bubble CAGR *bg* [%]	Duration [*days*]	Bubble Progress *bp* [%]	Geometric Average (*ci*, *bg*, *bp*)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Medium Term	35.03	237.47	114.35	402	98.65	73.39	2022-06-05	98.94

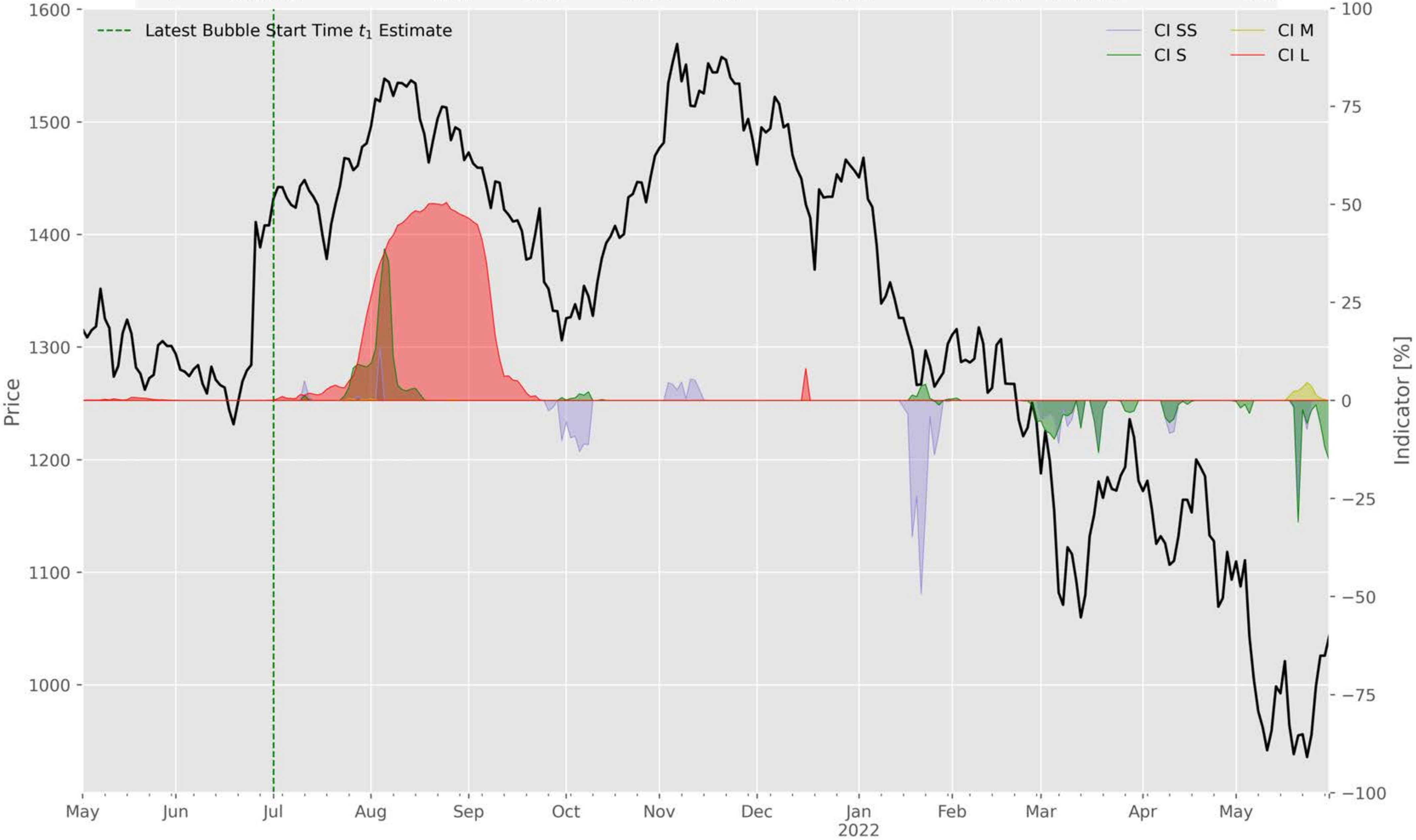


S&P500 Soft Drinks

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
3	Long Term	93.05	33.08	13.67	562	79.91	46.67	2022-10-19	99.51

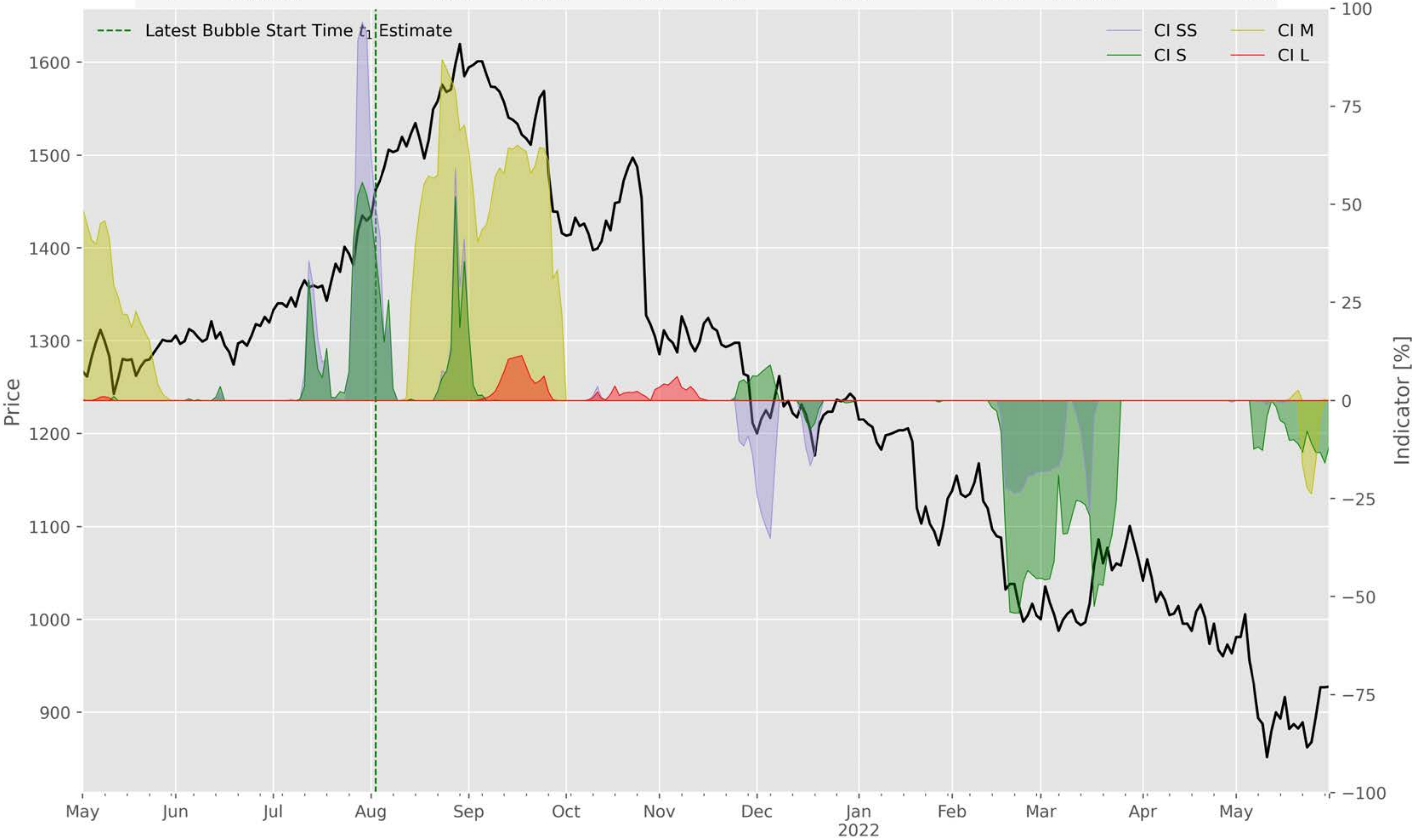
S&P1500 Textiles & Apparel

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Short Term	15.35	-27.12	-28.36	239	93.73	34.43	2022-06-16	73.16



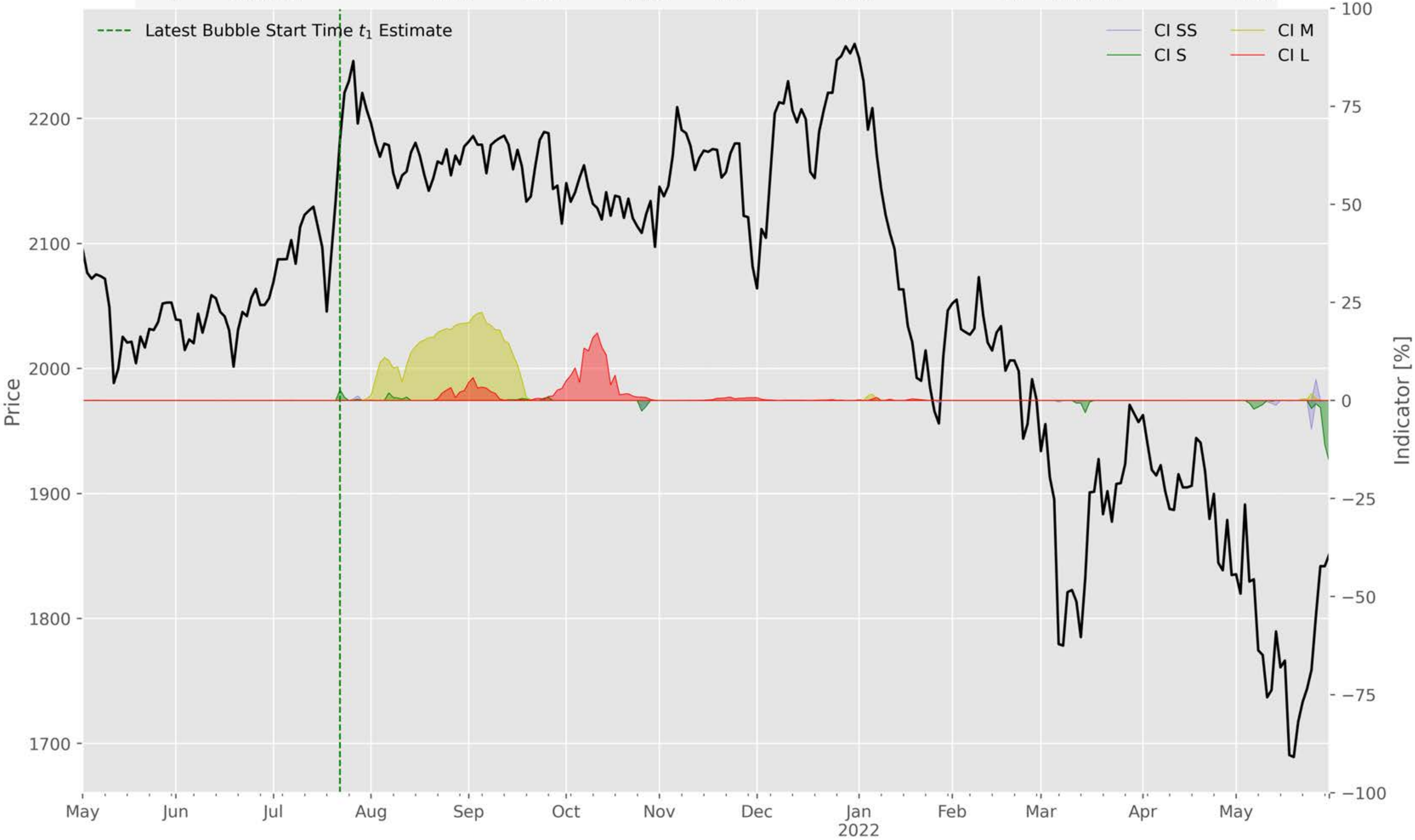
S&P1500 Consumer Electronics

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
2	Short Term	11.47	-36.59	-41.23	216	66.68	31.59	2022-09-16	73.16



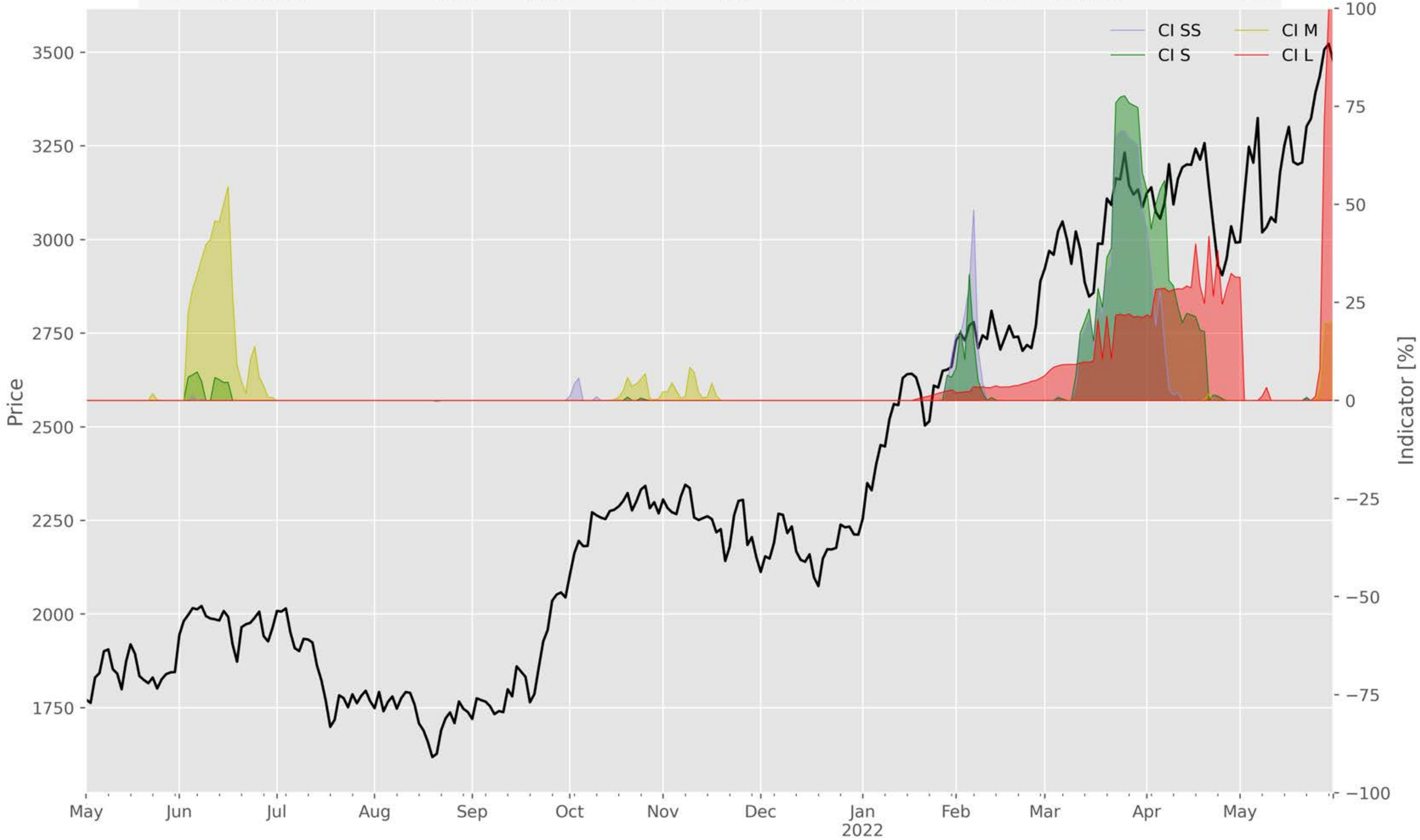
S&P1500 Restaurants

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
3	Short Term	15.68	-15.22	-16.95	224	91.79	29	2022-06-20	64.65



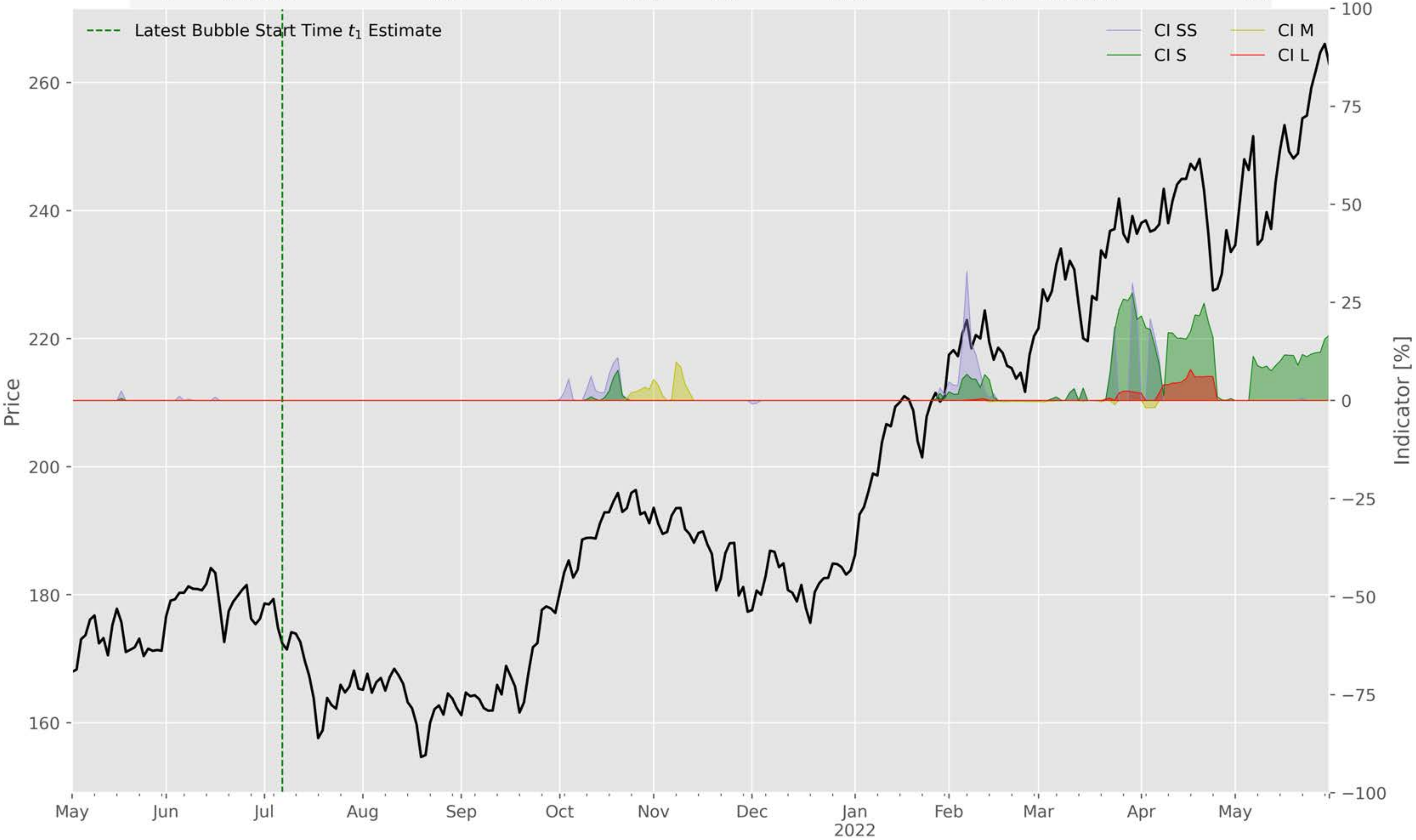
S&P Global 1200 Oil&Gas Exp&Prod

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Medium Term	20.12	183.07	68.25	504	99.81	51.56	2022-06-01	84.57



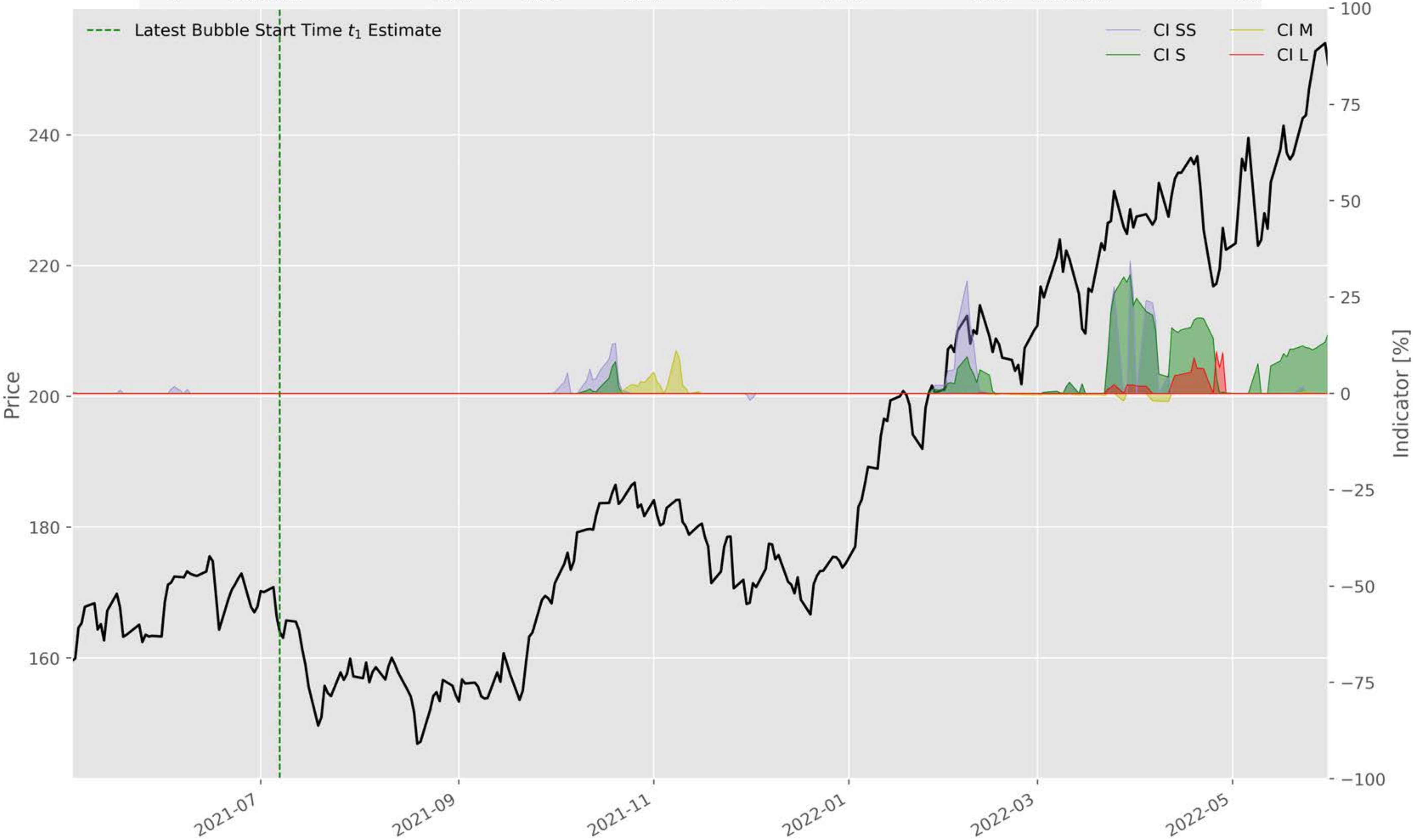
Msci World Oil,Gas&C.Fuel$

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
2	Short Term	16.58	52.51	57.24	235	67.11	39.94	2022-09-23	100



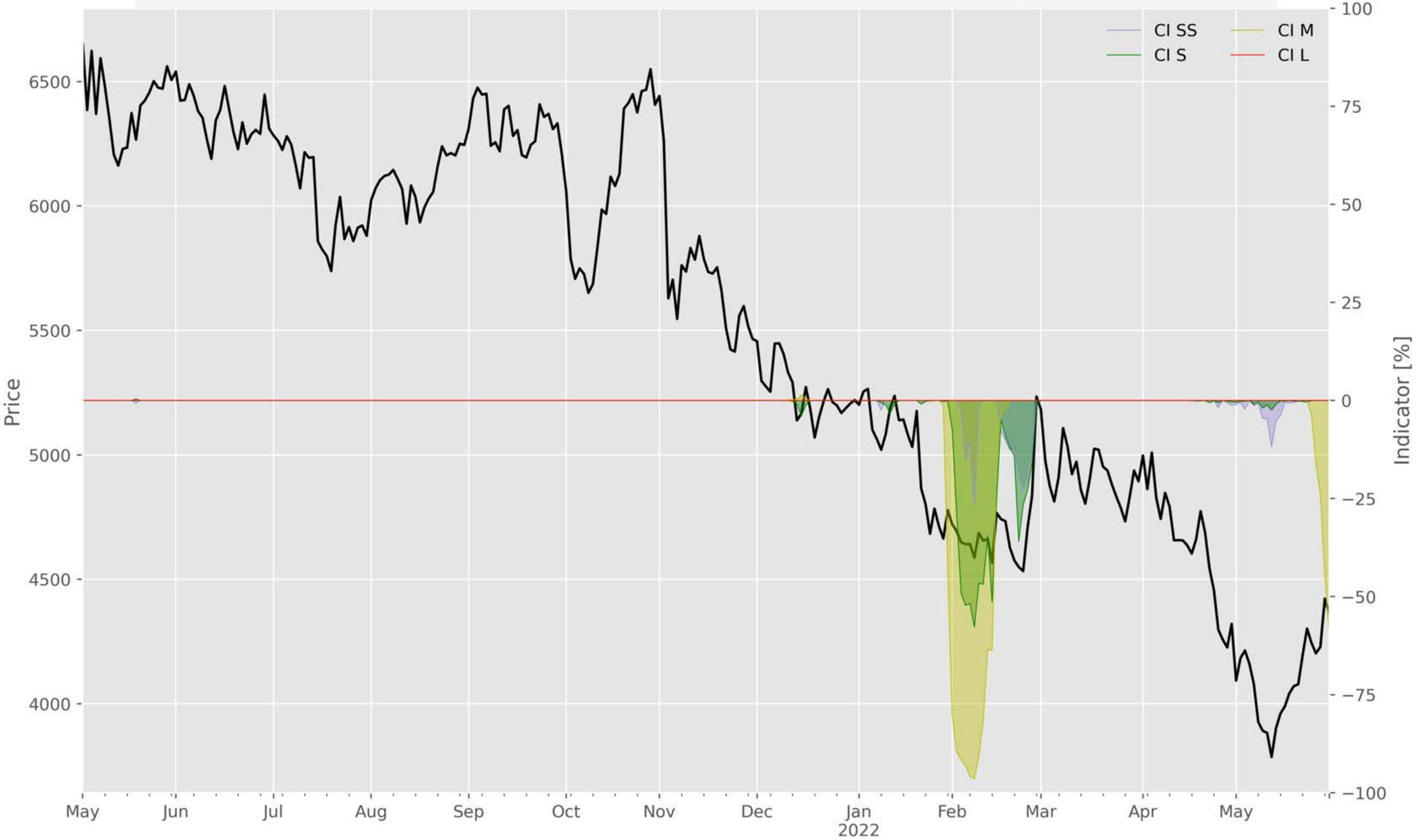
Msci World Energy $

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Short Term	15.69	52.92	57.99	234	66.79	39.32	2022-09-24	100



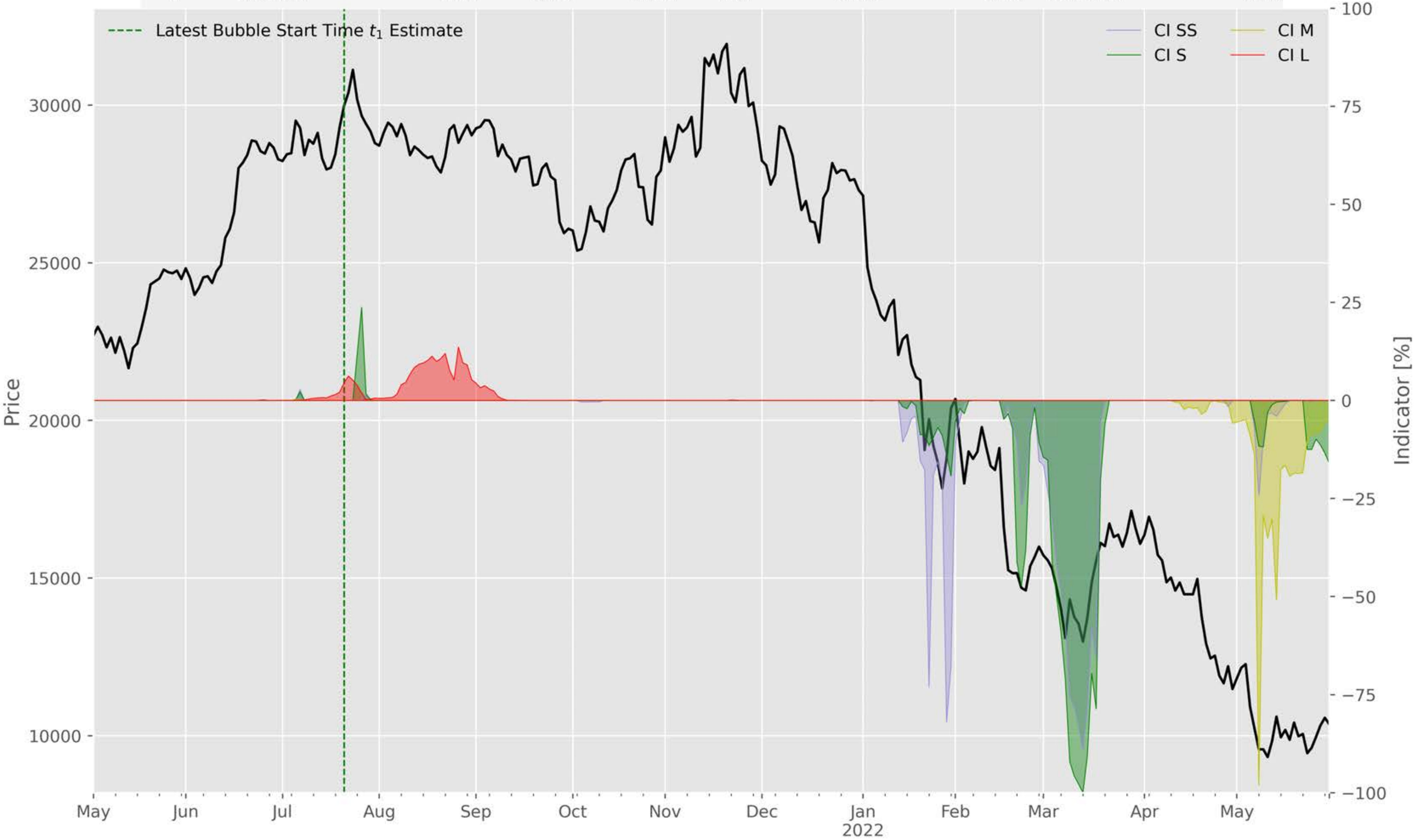
S&P Global 1200 Heavy Elec Eq

Rank	Strongest Signal	DS-LPPLS Confidence *ci* [%]	Bubble Size *bs* [%]	Bubble CAGR *bg* [%]	Duration [*days*]	Bubble Progress *bp* [%]	Geometric Average (*ci*, *bg*, *bp*)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Medium Term	60.24	-36.14	-25.9	377	91.16	52.2	2022-07-07	99.98



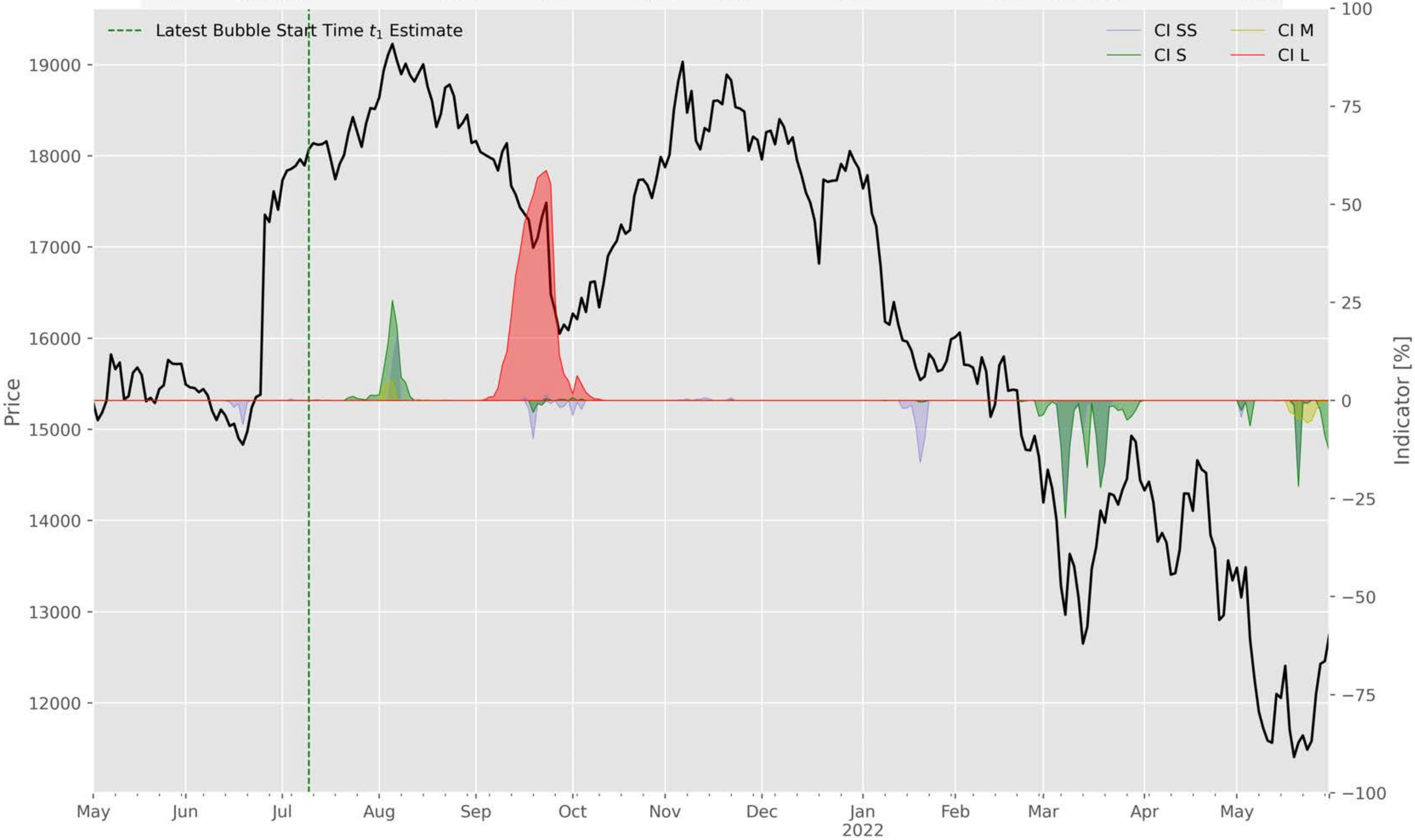
S&P Glb 1200 Inet. Svs & Inf

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
2	Short Term	15.97	-65.43	-69.56	225	90.84	46.55	2022-06-23	99.98



S&P Global 1200 Footwear

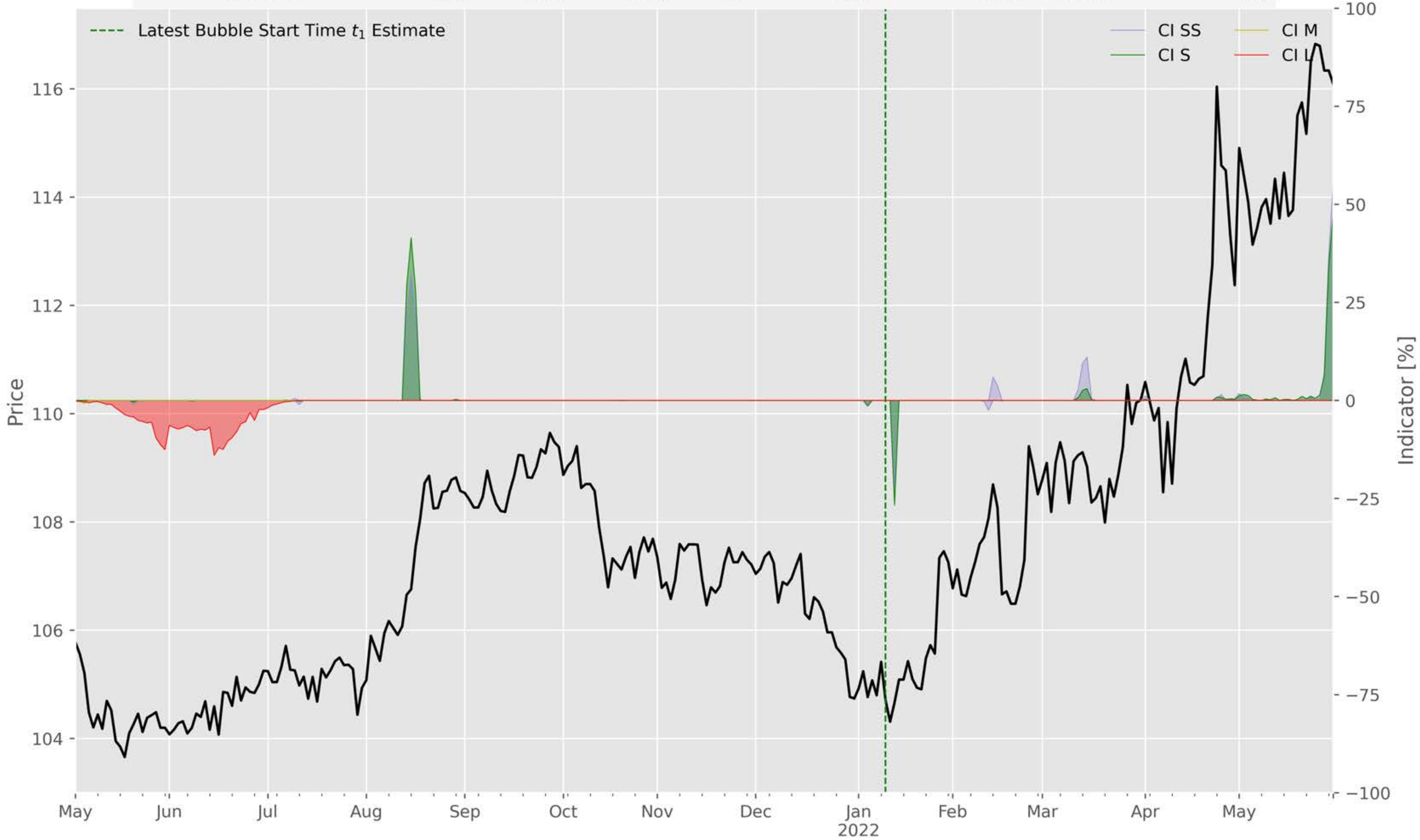
Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Short Term	12.84	-29.47	-31.45	233	91.75	33.33	2022-06-21	74.98



Forex

Jpm Uruguay Reer Cpi (2010=100)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Short Term	47.74	10.88	29.39	101	93.94	50.89	2022-06-07	99.98

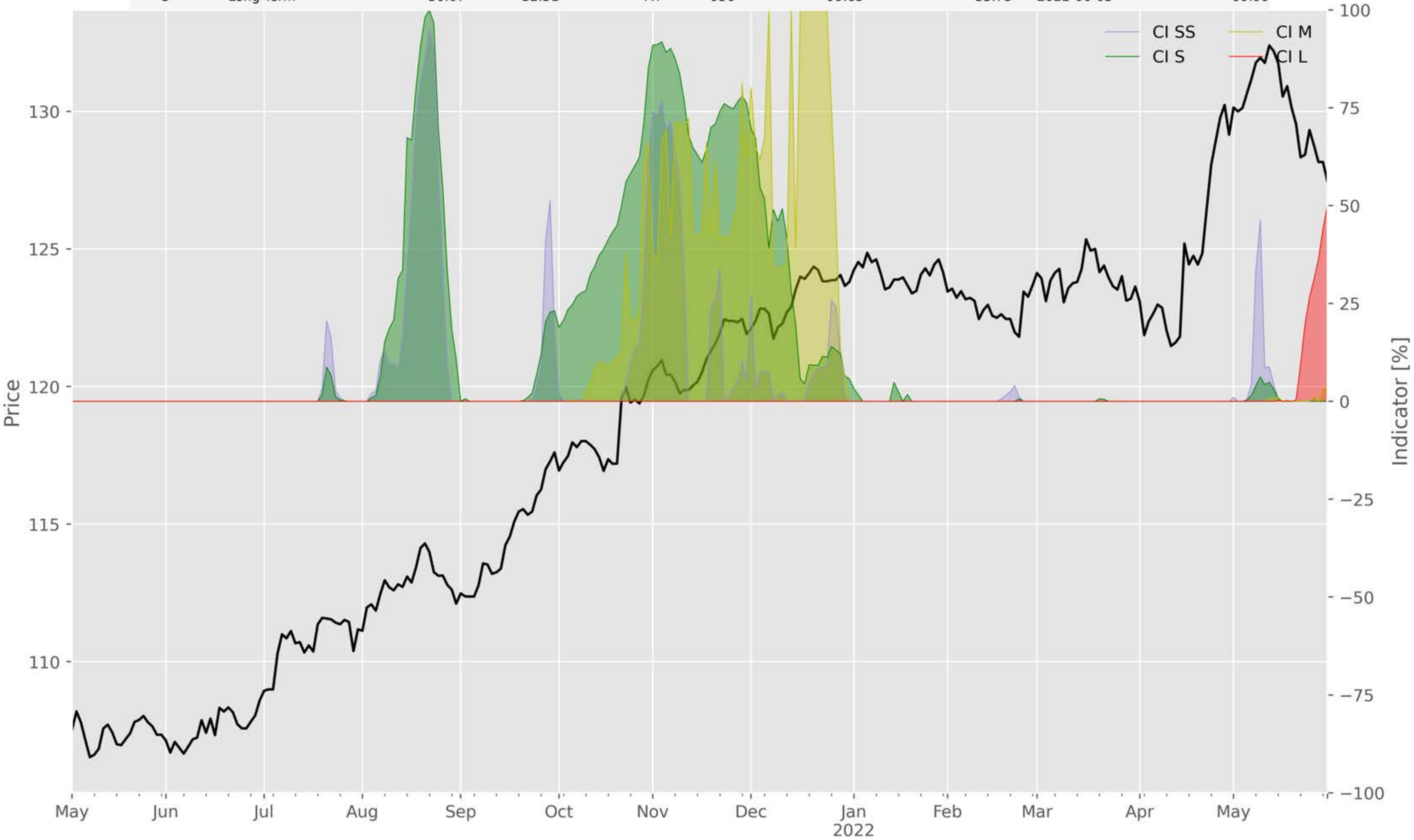


Jpm Mexico Reer Cpi (2010=100)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Super Short	44.43	10.5	22.09	126	96.63	45.61	2022-06-04	99.98

Jpm Argentina Reer Cpi (2010=100)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
3	Long Term	50.07	32.51	7.7	956	99.65	33.75	2022-06-03	99.99

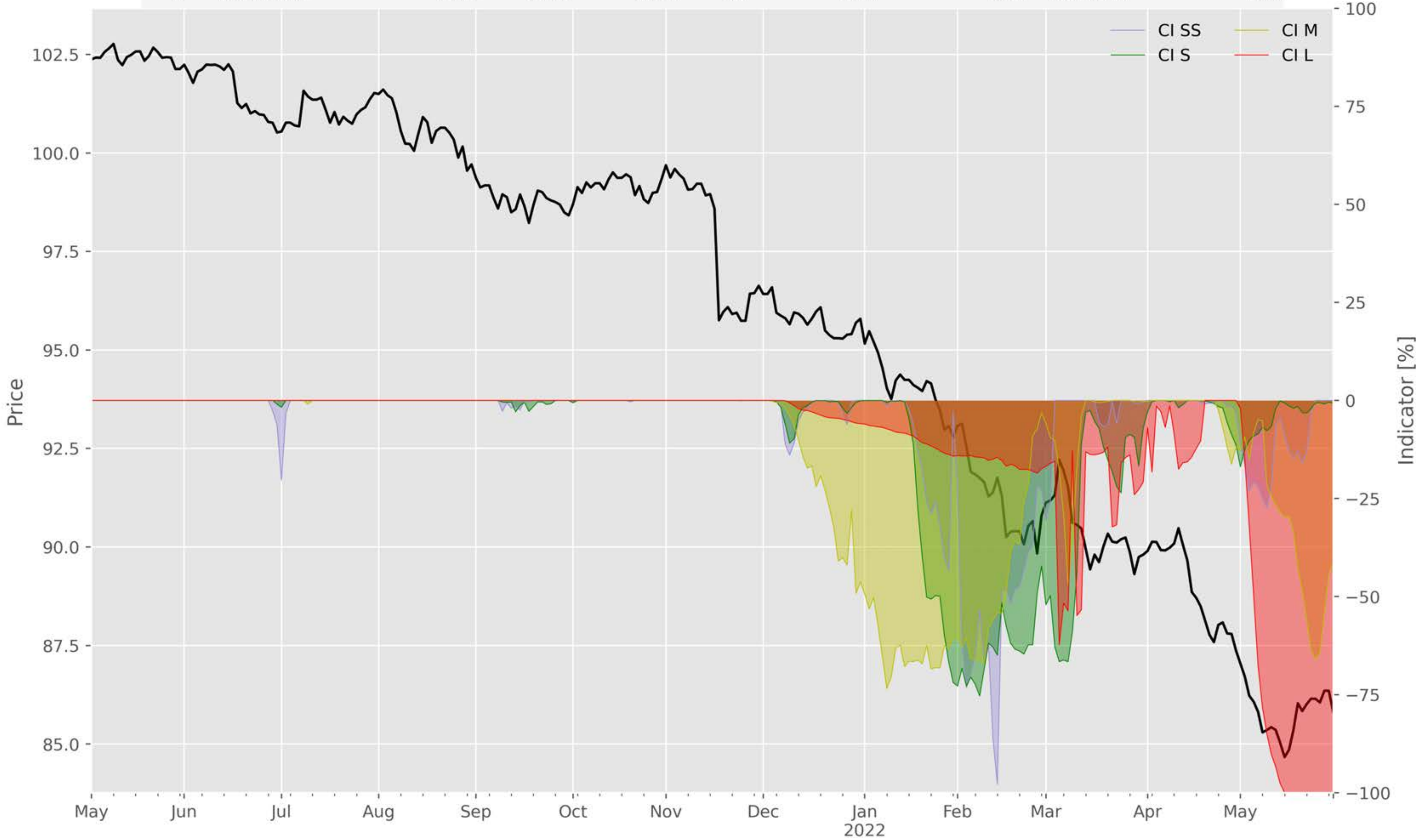


Jpm Hungary Reer Cpi (2010=100)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Short Term	29.88	-15.24	-18.56	203	98.74	37.97	2022-06-03	98.64

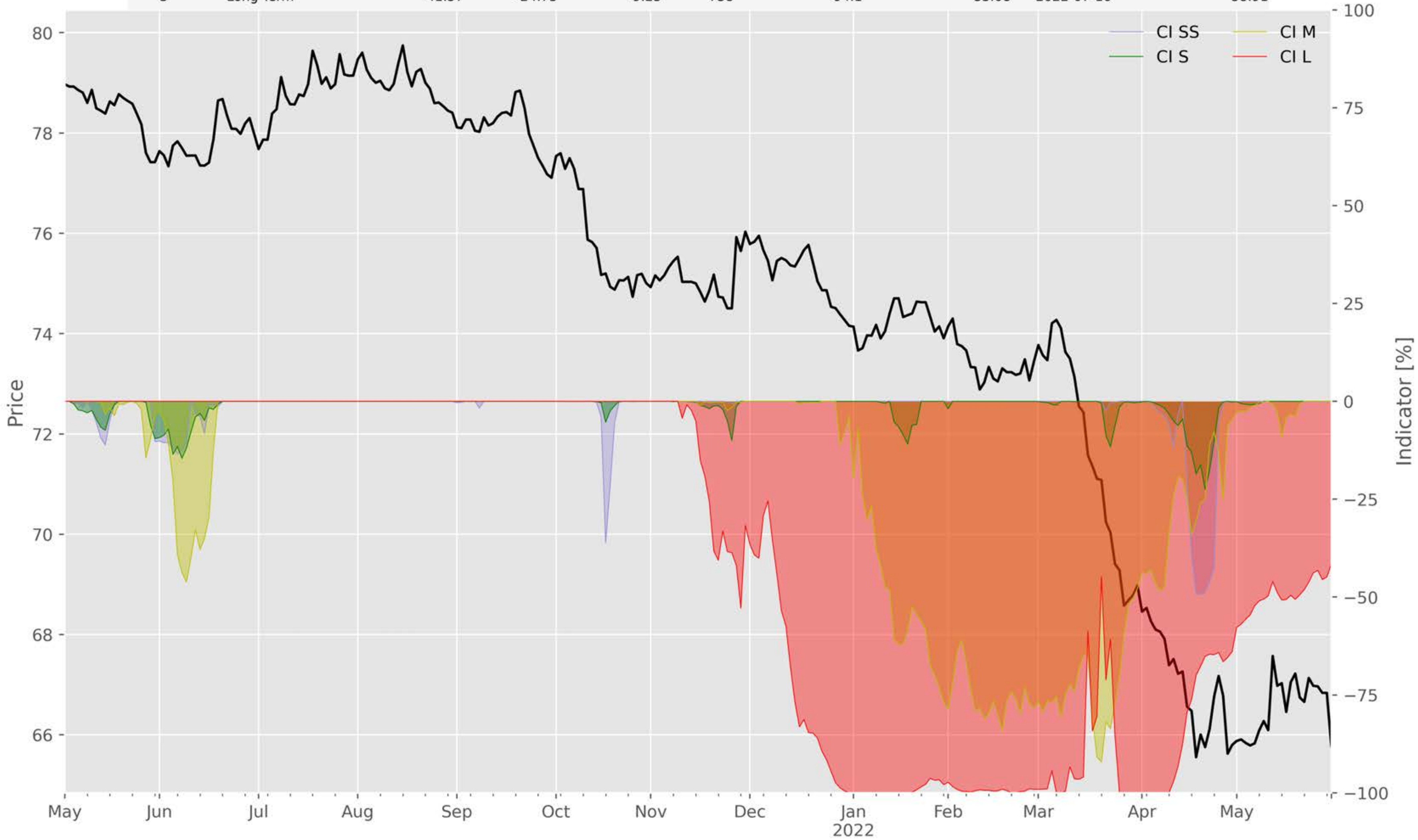
Jpm Switzerland Reer Ppi (2010=100)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Medium Term	41.03	-24.46	-13.11	503	95.37	37.16	2022-06-24	89.04



Jpm Japan Reer Ppi (2010=100)

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Long Term	41.57	-24.75	-9.25	738	94.1	33.08	2022-07-16	58.91

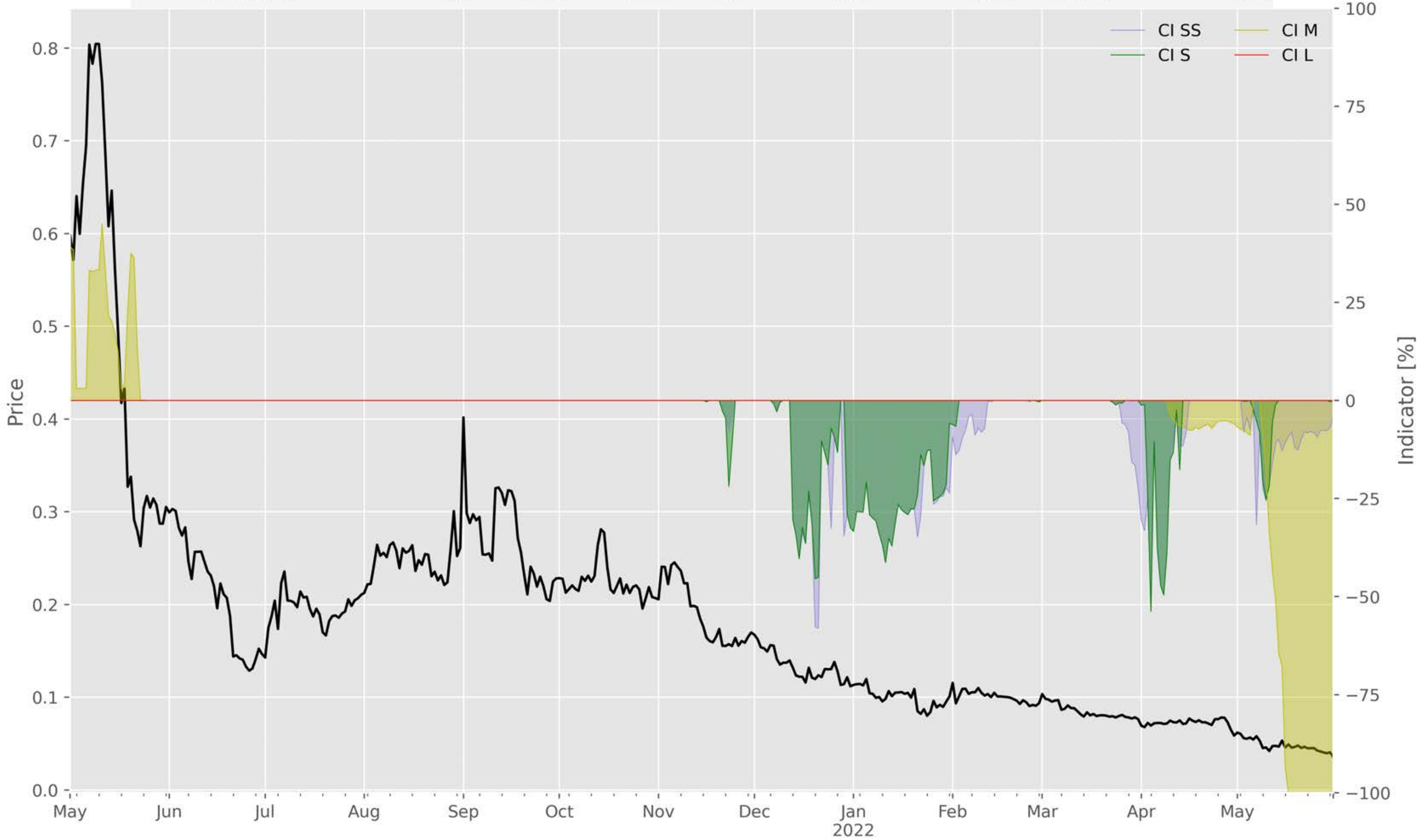


CryptoX

No positive bubbles to report

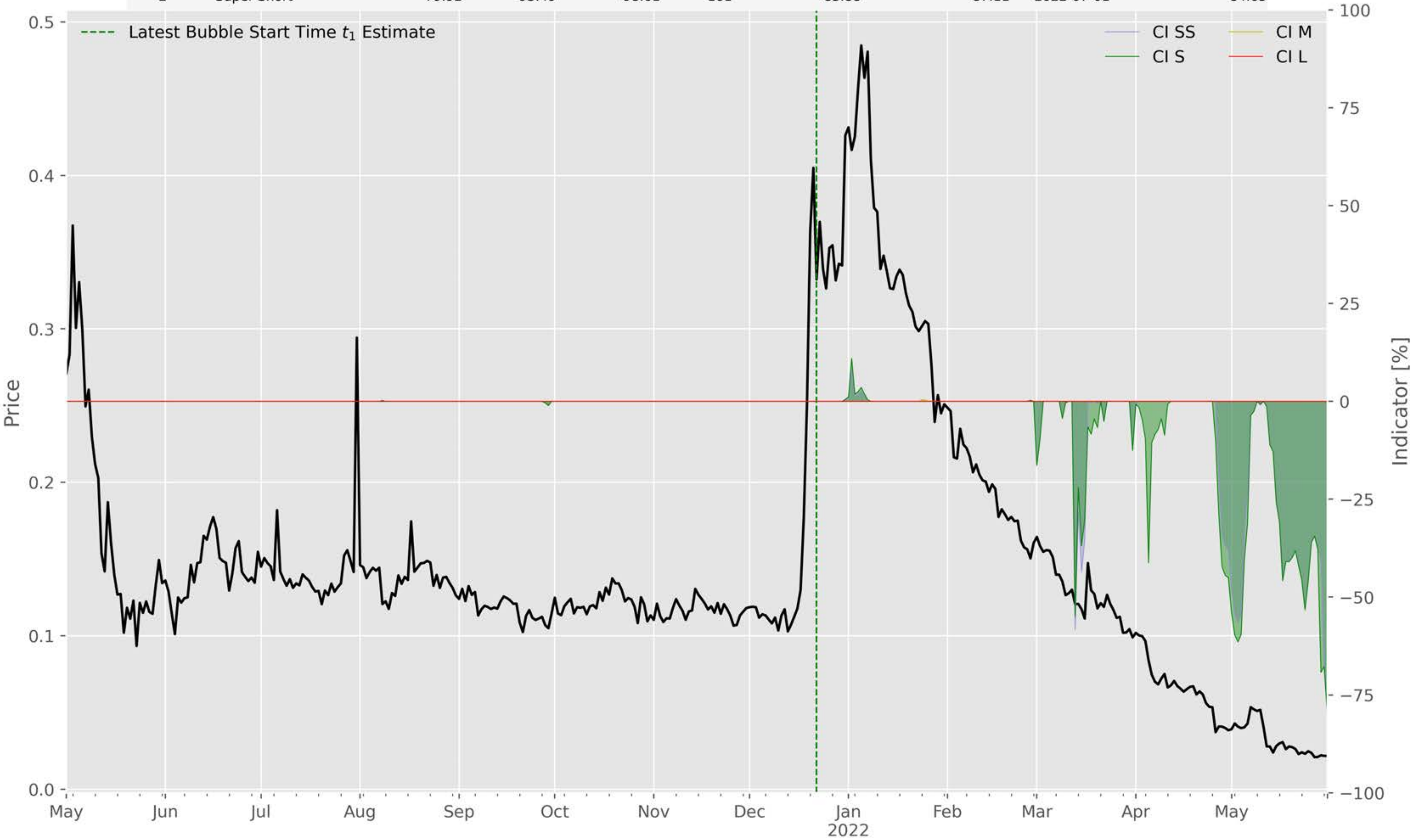
SALT/USD

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Medium Term	100	-94.18	-81.18	429	96.63	92.23	2022-06-15	54.18



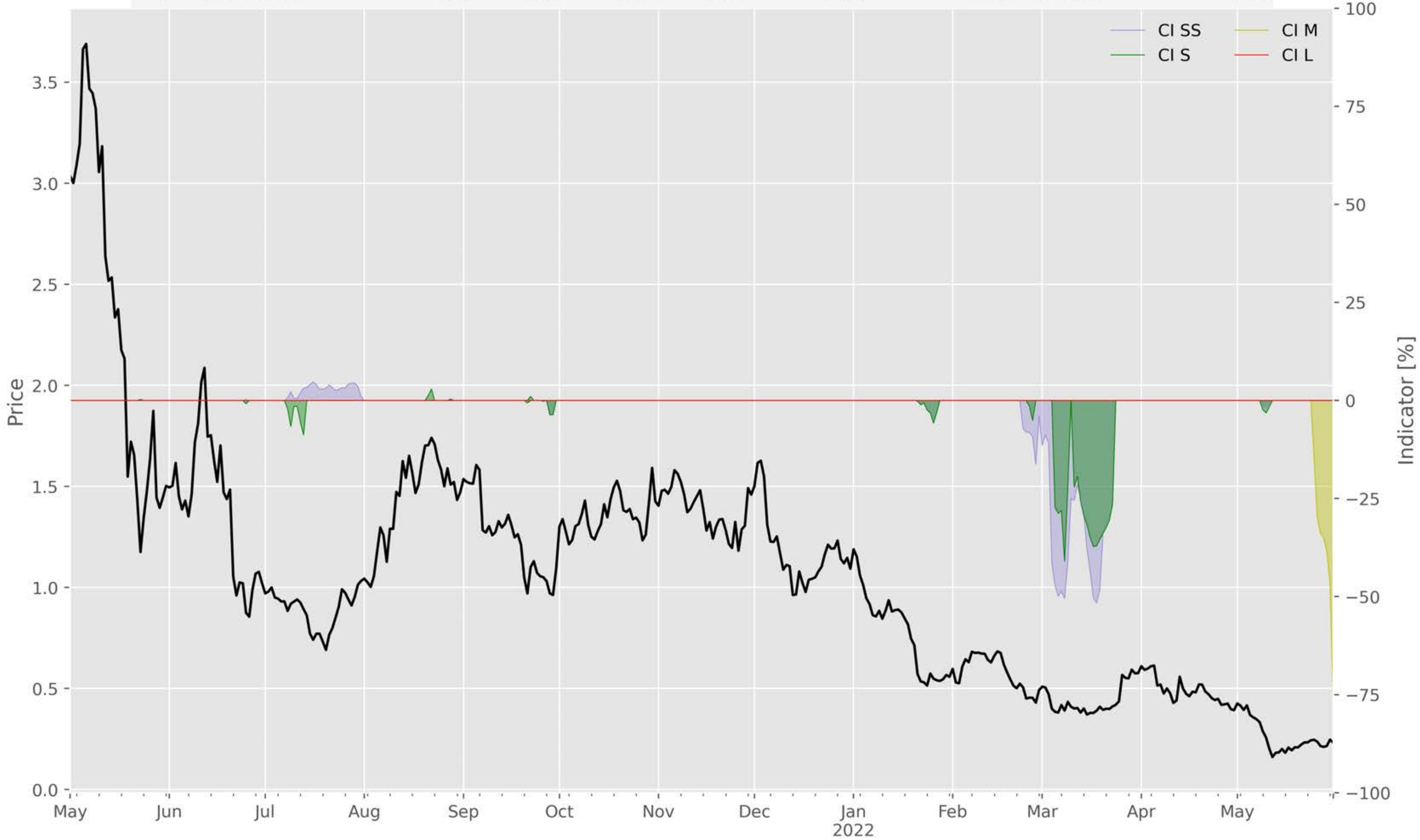
GoldCoin/USD

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Super Short	79.92	-93.49	-98.61	161	83.88	87.11	2022-07-01	84.65



Neblio/USD

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
3	Medium Term	77.34	-90.14	-70.89	473	91.77	79.54	2022-07-12	40.56

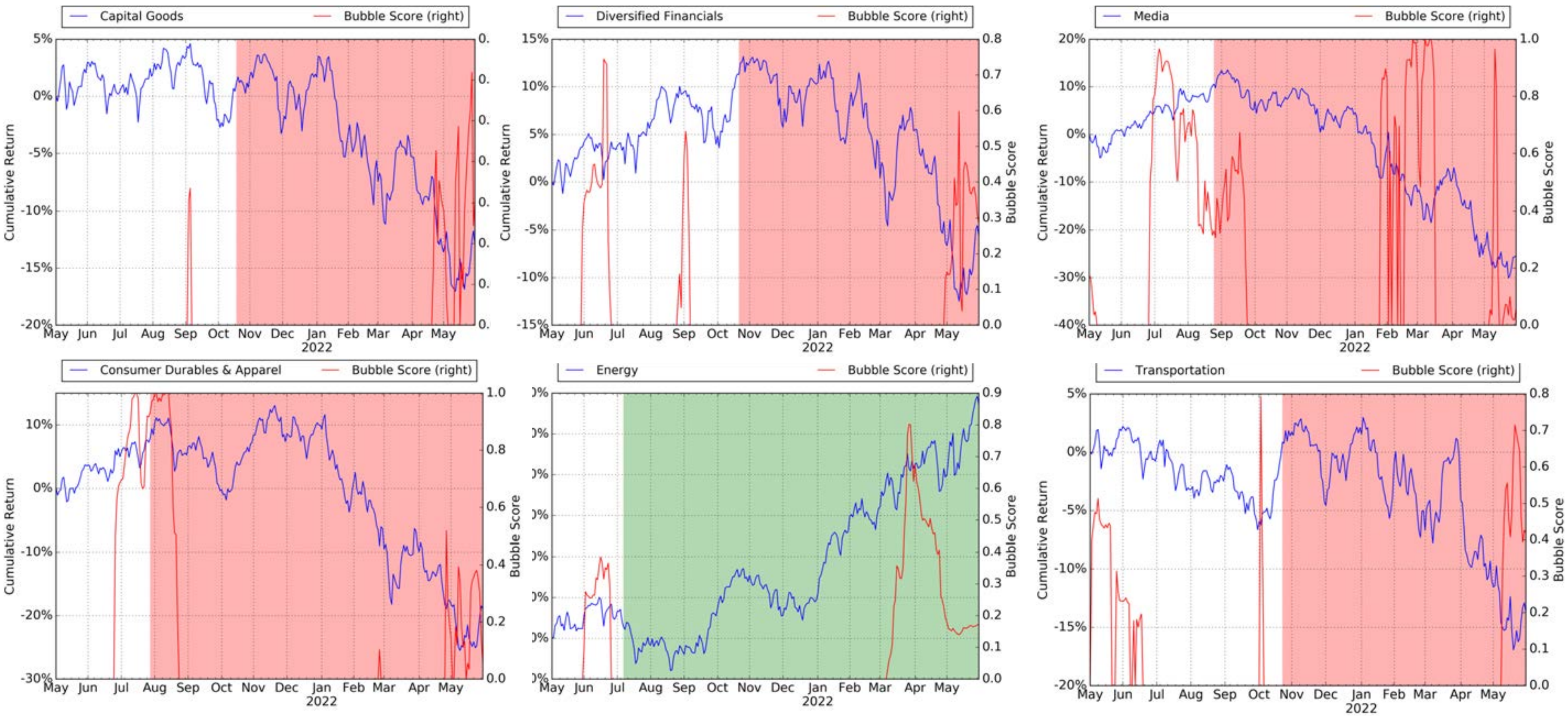


Sectors

GICS Industry Group Name	Yearly Return		Bubble Size		Bubble Score		Value Score		Growth Score	
	Jun 1st	May 1st	Jun 1st	May 1st	Jun 1st	May 1st	Jun 1st	May 1st	Jun 1st	May 1st
Pharmaceuticals, Biotechnology & Life Sciences	2.4%	7.3%	0.0%	0.0%	0.0%	0.0%	60.2%	60.3%	57.5%	57.5%
Consumer Services	-15.9%	-9.2%	0.0%	0.0%	0.0%	0.0%	19.0%	19.1%	61.7%	61.6%
Retailing	-23.7%	-18.4%	0.0%	0.0%	0.0%	0.0%	22.0%	21.8%	42.9%	42.7%
Transportation	-14.5%	-10.5%	-14.4%	0.0%	-41.8%	0.0%	49.2%	49.0%	51.5%	51.8%
Consumer Durables & Apparel	-21.3%	-16.0%	-24.9%	-23.1%	-5.5%	-19.5%	32.6%	34.1%	42.6%	39.8%
Semiconductors & Semiconductor Equipment	-3.3%	4.9%	0.0%	0.0%	0.0%	0.0%	62.7%	65.8%	39.6%	39.0%
Technology Hardware & Equipment	4.8%	14.7%	0.0%	0.0%	0.0%	0.0%	82.2%	82.6%	57.8%	58.8%
Automobiles & Components	-8.3%	8.8%	0.0%	0.0%	0.0%	0.0%	51.5%	48.3%	46.7%	48.8%
Telecommunication Services	-9.0%	-15.4%	0.0%	0.0%	0.0%	0.0%	57.9%	57.6%	41.3%	44.4%
Energy	44.7%	36.7%	52.9%	33.7%	17.2%	19.3%	53.5%	52.9%	52.9%	51.4%
Software & Services	-16.9%	-6.9%	-22.5%	0.0%	-0.5%	0.0%	35.9%	35.5%	52.2%	52.3%
Materials	-4.6%	-5.4%	0.0%	0.0%	0.0%	0.0%	45.9%	45.9%	51.8%	51.7%
Health Care Equipment & Services	-3.4%	-1.2%	0.0%	0.0%	0.0%	0.0%	50.8%	48.5%	57.5%	56.4%
Capital Goods	-14.1%	-12.7%	-14.0%	-13.1%	-15.2%	-14.9%	48.7%	48.7%	49.7%	50.0%
Media & Entertainment	-27.5%	-21.6%	-32.6%	0.0%	-5.8%	0.0%	35.9%	37.5%	34.1%	34.2%
Commercial & Professional Services	-7.7%	-1.8%	0.0%	0.0%	0.0%	0.0%	32.1%	31.1%	55.3%	56.0%
Food & Staples Retailing	0.2%	12.4%	0.0%	0.0%	0.0%	0.0%	41.7%	39.9%	58.8%	59.3%
Household & Personal Products	-10.3%	-5.7%	0.0%	0.0%	0.0%	0.0%	28.4%	27.6%	54.8%	55.0%
Food, Beverage & Tobacco	0.0%	4.5%	0.0%	0.0%	0.0%	0.0%	49.2%	49.1%	60.2%	60.2%
Utilities	3.9%	2.2%	0.0%	0.0%	0.0%	0.0%	45.0%	45.1%	55.3%	53.7%
Insurance	0.4%	-1.1%	0.0%	0.0%	0.0%	0.0%	-	-	-	-
Real Estate	-8.8%	3.5%	0.0%	0.0%	0.0%	0.0%	-	-	-	-
Diversified Financials	-8.8%	-6.3%	-15.2%	-11.8%	-28.5%	-13.0%	-	-	-	-
Banks	-8.6%	-12.1%	0.0%	0.0%	0.0%	0.0%	-	-	-	-

Sectors

- We use the MSCI World Industry Group Indices to calculate bubble size and bubble score of the corresponding sectors. To determine the value scores and growth scores of the sectors, we average over the corresponding values for each stock of a given sector, weighted by market cap.
- This month, the three negative bubbles and one positive bubble we observed last month continued to develop with a moderate negative bubble score. In addition, we observe two industry group index with negative bubble score: *Media & Entertainment,* and *Transportation.* These bubbles have not yet reached the tipping point and one should remain cautious about the future development of the bubble trajectories.



Portfolio Construction & Performance

- Here we illustrate the methodology of the portfolio construction process based on the results of our previous analyses.
- For individual stocks that we identified in the 4 quadrants, we constructed 4 portfolios based on the 4 quadrants defined in the last report. Each portfolio consists of all the stocks listed in the corresponding quadrant.
  1. **Trend-Following Long Stock Portfolio (TFLSP)** is made of the stocks that have a positive bubble signal as well as a strong value score. For instance, TFLSP November consists of all the stocks listed in quadrant 1, identified in slide 37 of November 2017 FCO Report.
  2. **Trend-Following Short Stock Portfolio (TFSSP)** is made of the stocks that have a negative bubble signal as well as a weak value score.
  3. **Contrarian Long Stock Portfolio (CLSP)** is made of the stocks that have a negative bubble signal as well as a strong value score.
  4. **Contrarian Short Stock Portfolio (CSSP)** is made of the stocks that have a positive bubble signal as well as a weak value score.

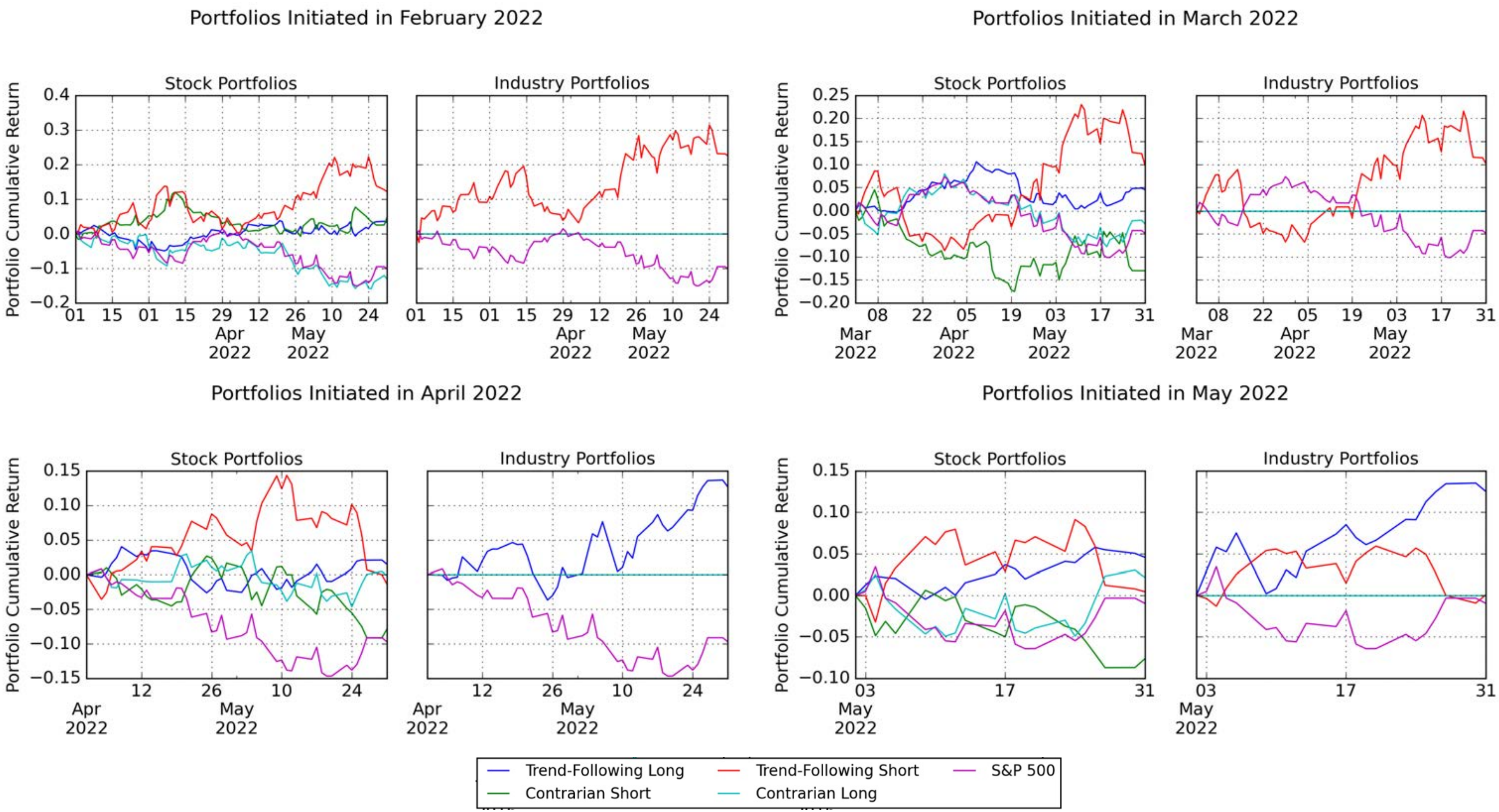
Portfolio Construction & Performance

- At the same time, we also classified 20 industries into 4 quadrants, and constructed 4 type of industry portfolios based on the 4 industry quadrants. Each portfolio consists of all the stocks in the industries listed in the corresponding quadrant. Following the same definitions as above, we have Trend-Following Long Industry Portfolio (TFLIP), Trend-Following Short Industry Portfolio (TFSIP), Contrarian Long Industry Portfolio (CLIP), and Contrarian Short Industry Portfolio (CSIP).

- In each month, we initiated 8 new portfolios based on the updated results. The performance of every 8 portfolios we initiated since November 2017 are presented in the next slide. All of the stocks in our portfolios are weighted by their market capitalizations and we don't consider transaction cost in the portfolio performance.

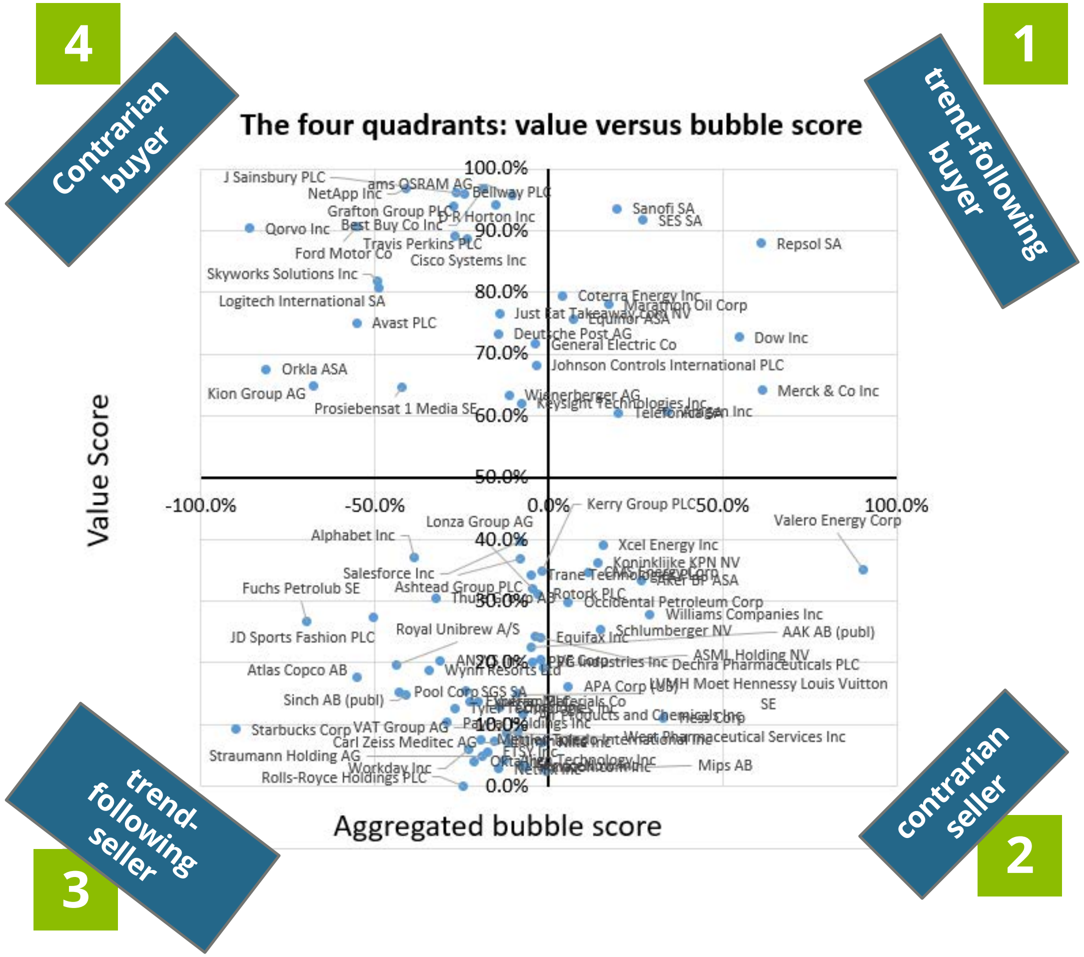
Portfolio Construction & Performance

- The S&P 500 index started to rebound after a further 5% drawdown in the past month, hurting our Short portfolios in the second half of the past month. Some of the Trend-Following Long Industry Portfolios initialed in April and May are outperforming strongly.
- Contrarian Portfolios are more delicate to use due to their sensitivity to timing the expected reversal and exhibit very volatile performances. We expect trend-following positions to perform in the months following the position set-up and then contrarian positions to over-perform over longer time scales over which the predicted corrections play out.



Single Stocks

We can divide the stocks into four quadrants[1)]



*1) A strong positive bubble signal is identified if bubble score is positive, and a strong negative bubble signal is identified if bubble score is negative.
A strong value score is identified if value score is larger than 60%, and a weak value score is identified if value score is smaller than 40%.*

- **Quadrant 1**: Stocks with a strong value score are cheap relative to their earnings potential. The strong positive bubble signal should be interpreted as a momentum indicator possibly the consequence of a repricing based on the fundamentals. As an investor, one could be a **trend-following buyer**. E.g. Dow Inc
- **Quadrant 2**: Stocks with a weak value score are expensive relative to their earnings potential. The strong positive bubble signal is an indication of sentiment and herding increasing the price until it is not linked to fundamentals anymore. As an investor, one could be a **contrarian seller**. E.g. Xcel Energy Inc
- **Quadrant 3**: These stocks are expensive relative to their earnings potential. On top of that, there are clear negative bubble signals. Such stocks should be considered as falling knives. As an investor, one could be a **trend-following seller**. E.g. Alphabet Inc
- **Quadrant 4**: These stocks are cheap relative to their financial performance. The strong negative bubble signal is an indication of sentiment and herding. These stocks can be considered as over-sold. As an investor, one could be a **contrarian buyer**. E.g. Atlas Copco AB

Single Stocks

- For 817 stocks, we calculate the **bubble warning indicators** as well as two financial strength indicators, which indicate the **fundamental value** of the stock and the **growth capability** respectively.
- To analyze the **financial strength of individual stocks**, we have two indicators. Both scores give a value between zero and one, one being the best of the set and zero the worst, so the higher the score, the higher the financial strength.
  - A **value score** that is based on the ROIC (Return on Invested Capital) taking into account the EV (Enterprise Value) to normalize for high/low market valuations and/or high/low debt; Value scores are calculated by comparing ROIC level versus EV/IC in each industry.
  - A **growth score** that has characteristics similar to the PEG ratio, which is the Price to Earnings ratio normalized by the expected growth of the EPS (Earnings per Share).
- The stocks are the constituents of the STOXX Europe 600, the S&P 500 and the Nasdaq 100 indices. From these, all doubles and stocks with incomplete data are removed. Because our financial strength indicators are specifically designed for corporates, all financial institutions are taken out of the set as well.

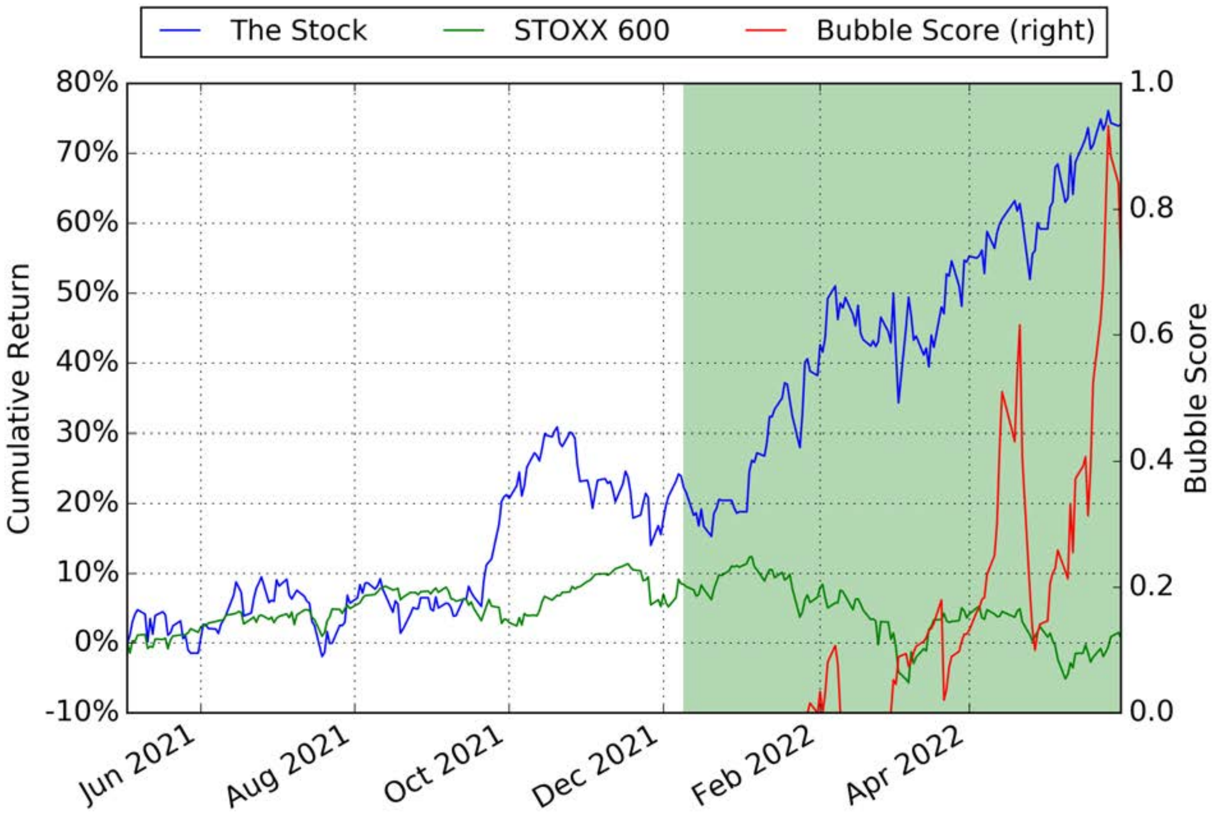
Single Stocks

Quadrant 1 stocks: strong positive bubble signals with strong fundamentals

Company Name	Country of Headquarters	GICS Industry Group Name	Yearly Return	Bubble Size	Bubble Start	Bubble Score	Value Score	Growth Score
Coterra Energy Inc	United States of America	Energy	107.9%	124.7%	Jul-21	3.9%	79.3%	96.4%
Merck & Co Inc	United States of America	Pharmaceuticals, Biotechnology & Life Sciences	22.0%	24.6%	Dec-21	61.3%	64.1%	33.7%
Marathon Oil Corp	United States of America	Energy	134.2%	151.0%	Jul-21	17.1%	78.0%	9.9%
Amgen Inc	United States of America	Pharmaceuticals, Biotechnology & Life Sciences	6.4%	19.1%	Sep-21	33.8%	60.6%	34.5%
Dow Inc	United States of America	Materials	1.5%	23.8%	Nov-21	54.6%	72.7%	26.6%
Telefonica SA	Spain	Telecommunication Services	26.1%	32.0%	Nov-21	20.0%	60.5%	1.2%
Sanofi SA	France	Pharmaceuticals, Biotechnology & Life Sciences	12.4%	16.6%	Oct-21	19.7%	93.5%	29.3%
Repsol SA	Spain	Energy	30.7%	35.5%	Oct-21	61.1%	87.9%	5.2%
Equinor ASA	Norway	Energy	87.7%	61.8%	Nov-21	6.9%	75.6%	83.1%
SES SA	Luxembourg	Media & Entertainment	31.0%	37.8%	Jun-21	27.1%	91.7%	95.0%
Shell PLC	United Kingdom	Energy	63.0%	47.2%	Dec-21	71.3%	79.6%	34.1%
Novartis AG	Switzerland	Pharmaceuticals, Biotechnology & Life Sciences	3.6%	10.4%	Sep-21	38.7%	71.2%	71.0%

Single Stocks

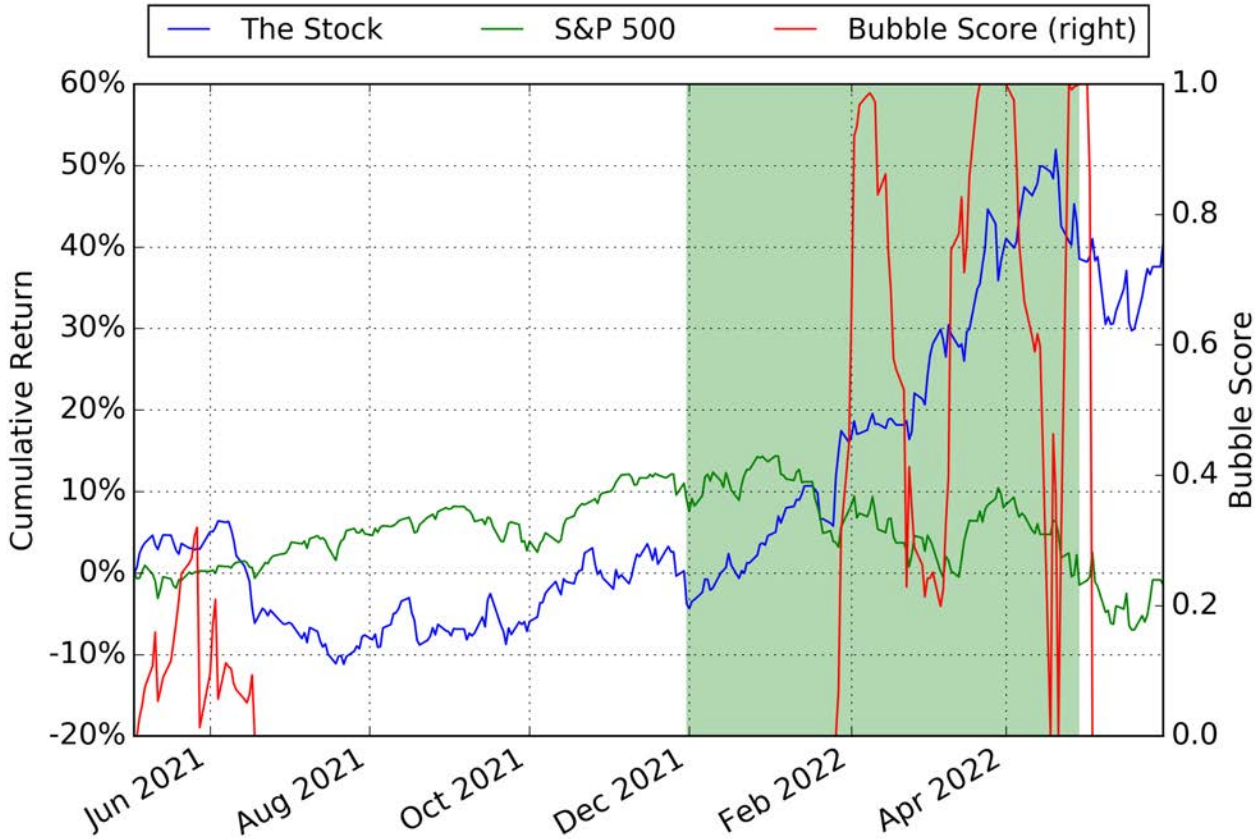
Quadrant 1 Stocks Current Month Example – Shell PLC.



- The above graph shows the one-year cumulative return of the stock in blue (left hand scale), STOXX 600 in green (left hand scale) and the calculated DS LPPLS Bubble Score in red (right hand scale). The green shaded period delineates the time interval within which the positive bubble is identified. The Bubble Score of this six-month bubble has reached 71.3% with a bubble size 47.2%. The high bubble score indicates a high probability of correction in the future due to the unstainable bubble growth.

Single Stocks

Quadrant 1 Stocks Last Month Example – Archer-Daniels-Midland Co.



- The figure above plots the one-year cumulative return of the stock (blue), S&P 500 (green) and LPPLS Bubble Score (red lines on the right y-axis). The green shaded period delineates the time interval within which a strong positive bubble has been identified and reported last month. The stock price has started its correction together with the market drawdown in the past month, in agreement with our DS LPPLS Bubble Signal.

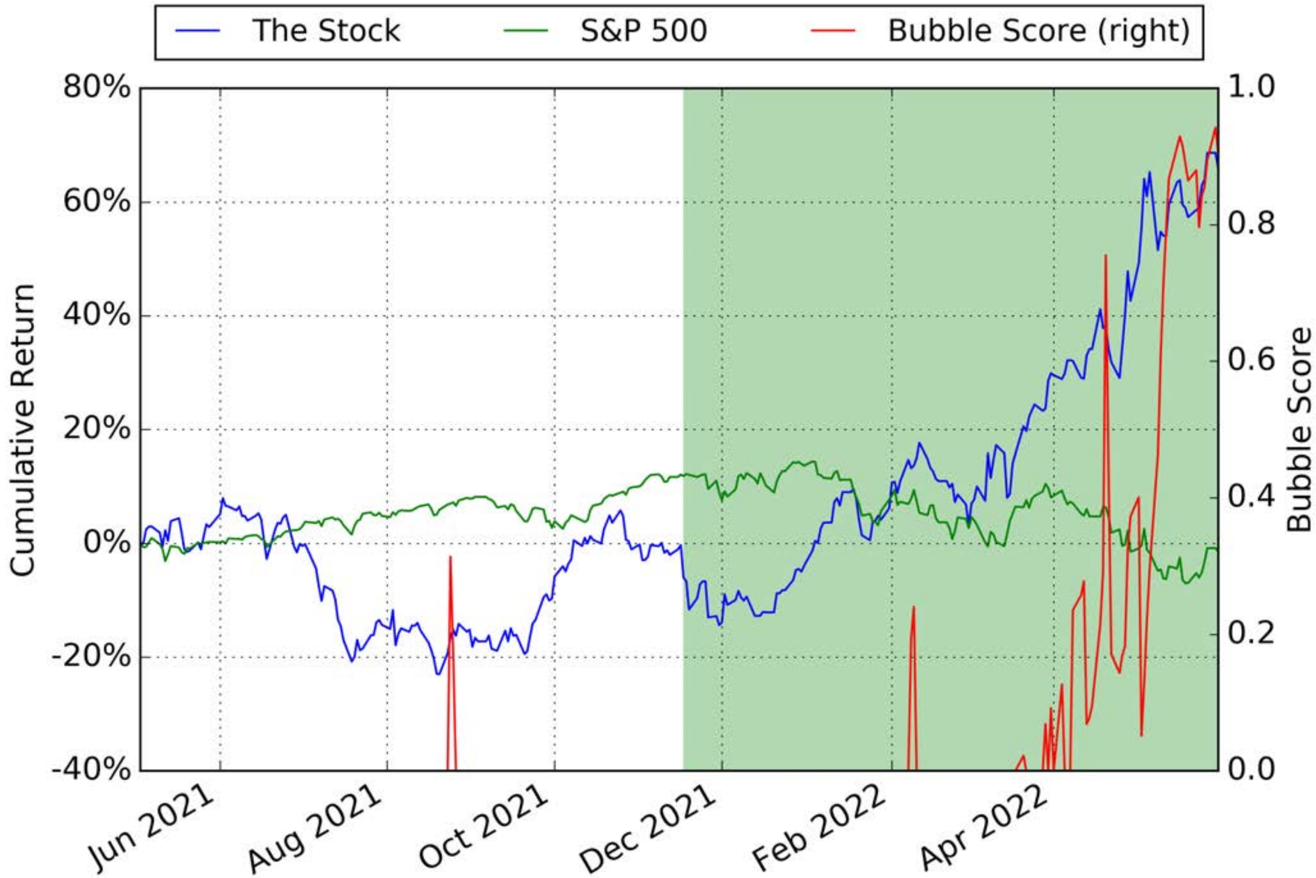
Single Stocks

Quadrant 2 stocks: strong positive bubble signals with weak fundamentals

Company Name	Country of Headquarters	GICS Industry Group Name	Yearly Return	Bubble Size	Bubble Start	Bubble Score	Value Score	Growth Score
Williams Companies Inc	United States of America	Energy	34.7%	54.2%	Aug-21	29.0%	27.8%	91.4%
Occidental Petroleum Corp	United States of America	Energy	145.1%	123.1%	Jan-22	5.6%	29.7%	33.4%
Valero Energy Corp	United States of America	Energy	58.1%	76.4%	Nov-21	90.5%	35.1%	7.3%
Schlumberger NV	United States of America	Energy	35.0%	47.4%	Jul-21	14.9%	25.3%	57.2%
CMS Energy Corp	United States of America	Utilities	17.8%	12.8%	Sep-21	11.4%	34.6%	71.3%
APA Corp (US)	United States of America	Energy	109.7%	162.6%	Aug-21	5.6%	16.1%	84.5%
Hess Corp	United States of America	Energy	38.8%	90.1%	Aug-21	33.0%	11.1%	67.1%
Xcel Energy Inc	United States of America	Utilities	9.1%	15.1%	Sep-21	15.6%	39.1%	71.5%
Aker BP ASA	Norway	Energy	46.5%	32.1%	Nov-21	26.6%	33.3%	3.7%
Koninklijke KPN NV	Netherlands	Telecommunication Services	26.8%	25.0%	Sep-21	14.0%	36.2%	13.5%

Single Stocks

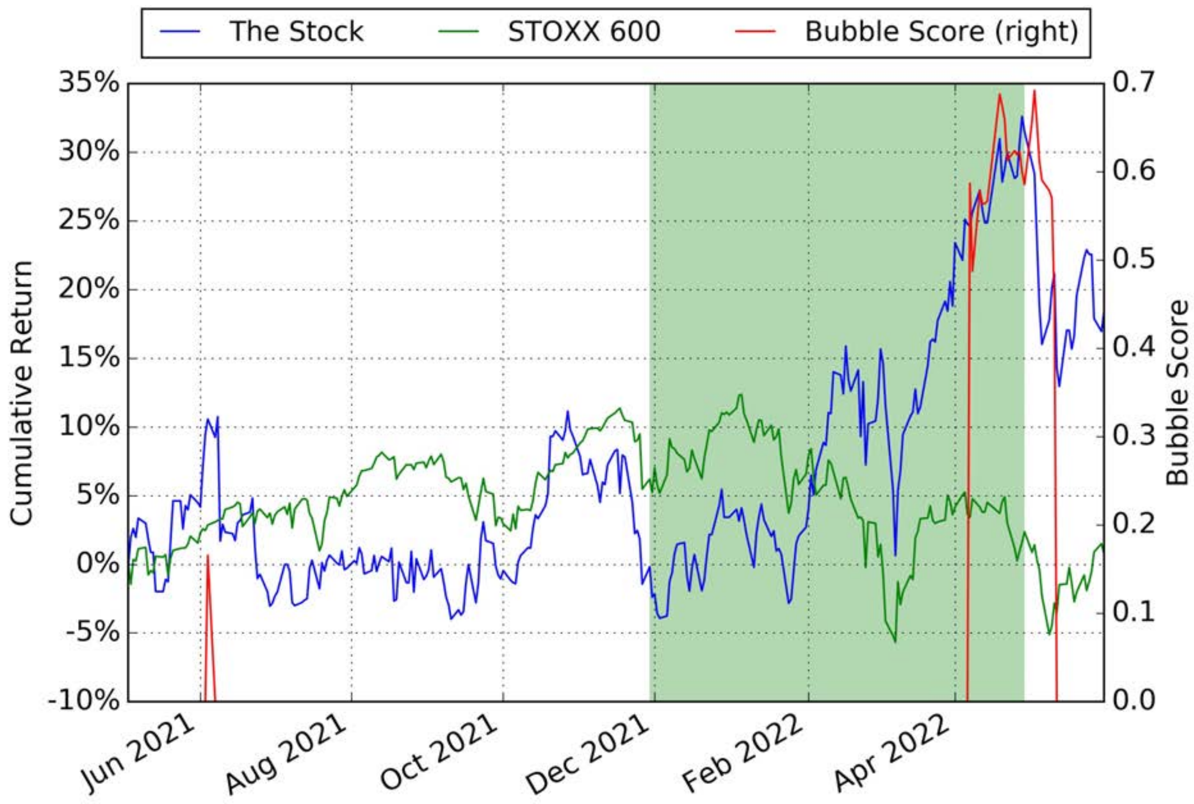
Quadrant 2 Stocks Current Month Example – Valero Energy Corp.



- The above graph shows the one-year cumulative return of the stock in blue (left hand scale), S&P 500 in green (left hand scale) and the calculated DS LPPLS Bubble Score in red (right hand scale). The green shaded period delineates the time interval within which the positive bubble is identified. The Bubble Score of this six-month bubble has reached 90.5% with a bubble size 76.4%. The strong positive bubble signal and weak fundamentals may indicate a high probability of correction in the future.

Single Stocks

Quadrant 2 Stocks Last Month Example – SalMar ASA.



The figure above plots the one-year cumulative return of the stock (blue), STOXX 600 (green) and LPPLS Bubble Score (red lines on the right y-axis). The green shaded period delineates the time interval within which a strong positive bubble has been identified and reported last month. The stock price had a significant drawdown in the past month, agreeing with our DS LPPLS Bubble Signal and weak fundamentals.

Single Stocks

Quadrant 3 stocks: strong negative bubble signals with weak fundamentals

Company Name	Country of Headquarters	GICS Industry Group Name	Yearly Return	Bubble Size	Bubble Start	Bubble Score	Value Score	Growth Score
Pool Corp	United States of America	Retailing	-7.5%	-20.2%	Oct-21	-43.2%	15.2%	29.6%
ServiceNow Inc	United States of America	Software & Services	-8.2%	-18.9%	Jul-21	-7.9%	3.4%	68.9%
Copart Inc	United States of America	Commercial & Professional Services	-8.9%	-20.2%	Jul-21	-14.0%	12.8%	32.6%
Illumina Inc	United States of America	Pharmaceuticals, Biotechnology & Life Sciences	-47.0%	-47.4%	Jun-21	-15.6%	7.3%	43.8%
Align Technology Inc	United States of America	Health Care Equipment & Services	-54.2%	-57.8%	Nov-21	-12.9%	4.2%	79.9%
Equifax Inc	United States of America	Commercial & Professional Services	-11.5%	-26.0%	Sep-21	-2.4%	24.1%	48.7%
Air Products and Chemicals Inc	United States of America	Materials	-17.7%	-17.7%	Jun-21	-7.5%	11.6%	71.7%
ANSYS Inc	United States of America	Software & Services	-23.1%	-28.7%	Jul-21	-31.2%	20.3%	37.2%
Netflix Inc	United States of America	Media & Entertainment	-60.5%	-65.1%	Aug-21	-14.5%	2.8%	28.3%
Nike Inc	United States of America	Consumer Durables & Apparel	-9.5%	-28.8%	Jul-21	-1.9%	7.1%	68.2%
Alphabet Inc	United States of America	Media & Entertainment	-7.1%	-16.2%	Jul-21	-38.9%	37.0%	25.2%
Wynn Resorts Ltd	United States of America	Consumer Services	-47.1%	-29.4%	Nov-21	-34.5%	18.9%	93.0%
Tyler Technologies Inc	United States of America	Software & Services	-18.6%	-27.8%	Jan-22	-27.2%	12.6%	72.6%
Mettler-Toledo International Inc	United States of America	Pharmaceuticals, Biotechnology & Life Sciences	-4.9%	-16.3%	Aug-21	-19.4%	7.6%	55.0%
ETSY Inc	United States of America	Retailing	-52.4%	-68.7%	Dec-21	-17.7%	5.6%	10.0%
PayPal Holdings Inc	United States of America	Software & Services	-69.0%	-70.9%	Jul-21	-29.2%	10.3%	60.9%
PPG Industries Inc	United States of America	Materials	-27.9%	-25.2%	Jul-21	-4.7%	20.2%	64.3%
Vulcan Materials Co	United States of America	Materials	-4.0%	-13.6%	Oct-21	-20.3%	13.7%	59.0%
West Pharmaceutical Services Inc	United States of America	Pharmaceuticals, Biotechnology & Life Sciences	-10.1%	-29.8%	Aug-21	-12.1%	8.6%	55.4%
Trane Technologies PLC	Ireland; Republic of	Capital Goods	-24.4%	-30.8%	Aug-21	-5.0%	34.2%	58.0%
Amazon.com Inc	United States of America	Retailing	-29.0%	-28.9%	Jun-21	-6.7%	3.1%	30.1%
Salesforce Inc	United States of America	Software & Services	-34.9%	-34.4%	Aug-21	-8.3%	39.8%	40.9%
Starbucks Corp	United States of America	Consumer Services	-30.2%	-32.9%	Nov-21	-89.9%	9.2%	38.3%
VF Corp	United States of America	Consumer Durables & Apparel	-38.0%	-29.8%	Sep-21	-2.2%	20.4%	84.9%
Workday Inc	United States of America	Software & Services	-32.7%	-42.0%	Aug-21	-23.1%	5.9%	42.3%

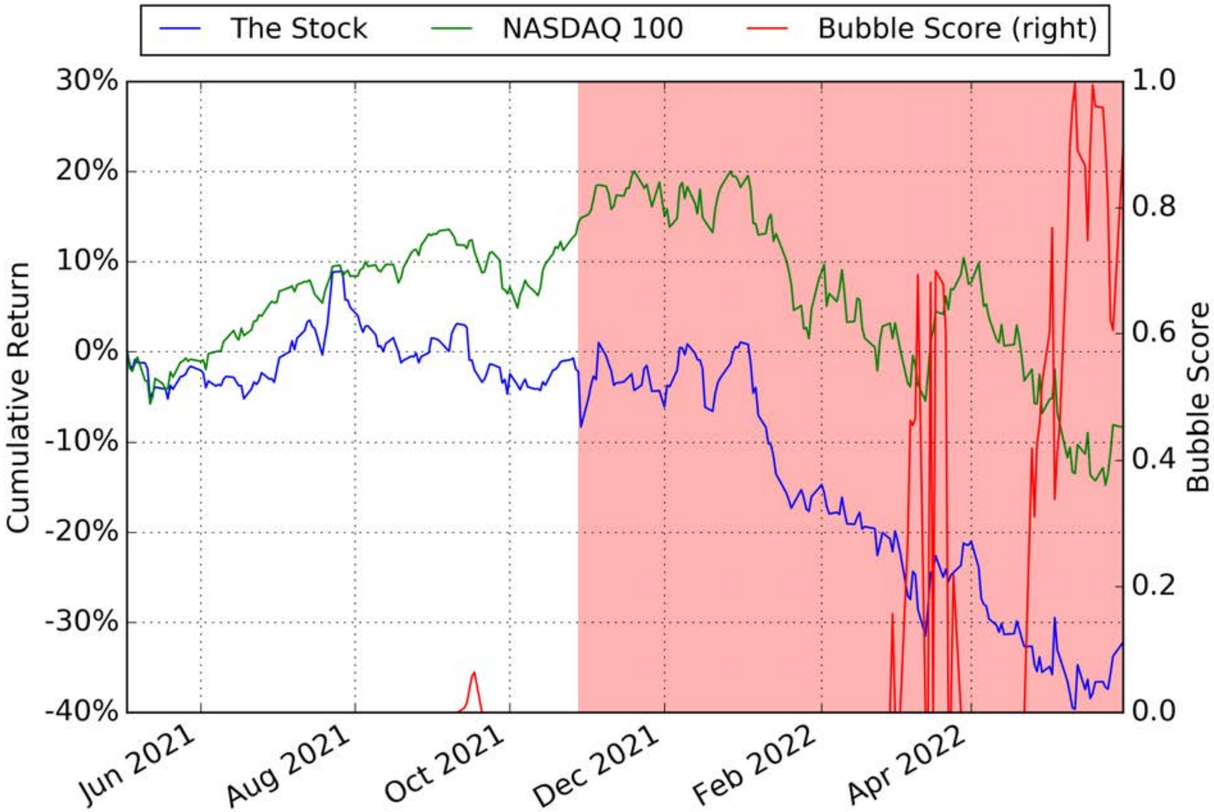
Single Stocks

Quadrant 3 stocks: strong negative bubble signals with weak fundamentals

Company Name	Country of Headquarters	GICS Industry Group Name	Yearly Return	Bubble Size	Bubble Start	Bubble Score	Value Score	Growth Score
Okta Inc	United States of America	Software & Services	-63.5%	-68.3%	Aug-21	-21.4%	4.0%	46.4%
ASML Holding NV	Netherlands	Semiconductors & Semiconductor Equipment	-18.8%	-31.4%	Aug-21	-1.5%	19.1%	35.2%
Sinch AB (publ)	Sweden	Software & Services	-68.5%	-72.7%	Jul-21	-41.0%	14.8%	13.8%
Ashtead Group PLC	United Kingdom	Capital Goods	-18.3%	-27.0%	Aug-21	-8.1%	36.9%	79.1%
AAK AB (publ)	Sweden	Food, Beverage & Tobacco	-15.4%	-18.4%	Jul-21	-5.1%	22.6%	85.5%
Fuchs Petrolub SE	Germany	Materials	-33.3%	-32.0%	Nov-21	-69.6%	26.7%	52.4%
Carl Zeiss Meditec AG	Germany	Health Care Equipment & Services	-19.4%	-34.6%	Sep-21	-11.8%	7.0%	83.7%
SGS SA	Switzerland	Commercial & Professional Services	-15.5%	-17.8%	Sep-21	-23.9%	15.4%	67.8%
Rotork PLC	United Kingdom	Capital Goods	-17.1%	-20.2%	Sep-21	-3.2%	31.1%	55.3%
Mips AB	Sweden	Consumer Durables & Apparel	-4.1%	-31.3%	Aug-21	-0.7%	2.4%	77.5%
Kerry Group PLC	Ireland; Republic of	Food, Beverage & Tobacco	-9.4%	-22.5%	Jul-21	-1.8%	34.8%	72.2%
Lonza Group AG	Switzerland	Pharmaceuticals, Biotechnology & Life Sciences	-13.2%	-24.3%	Aug-21	-4.8%	31.9%	87.2%
Thule Group AB	Sweden	Consumer Durables & Apparel	-15.3%	-30.9%	Aug-21	-32.6%	30.5%	64.0%
Atlas Copco AB	Sweden	Capital Goods	-13.3%	-19.2%	Oct-21	-55.1%	17.7%	76.5%
JD Sports Fashion PLC	United Kingdom	Retailing	-32.8%	-40.8%	Dec-21	-50.5%	27.3%	43.9%
LVMH Moet Hennessy Louis Vuitton SE	France	Consumer Durables & Apparel	-11.2%	-15.8%	Nov-21	-9.0%	15.0%	18.1%
Dechra Pharmaceuticals PLC	United Kingdom	Pharmaceuticals, Biotechnology & Life Sciences	-15.8%	-30.6%	Aug-21	-4.1%	24.3%	75.3%
Straumann Holding AG	Switzerland	Health Care Equipment & Services	-13.1%	-32.0%	Sep-21	-19.1%	5.0%	44.7%
Experian PLC	Ireland; Republic of	Commercial & Professional Services	-1.8%	-24.8%	Dec-21	-22.6%	13.8%	71.8%
Rolls-Royce Holdings PLC	United Kingdom	Capital Goods	-19.3%	-39.4%	Nov-21	-24.5%	0.0%	3.9%
VAT Group AG	Switzerland	Capital Goods	-1.3%	-21.6%	Aug-21	-9.0%	8.9%	72.8%
Royal Unibrew A/S	Denmark	Food, Beverage & Tobacco	-23.8%	-26.1%	Jul-21	-43.8%	19.6%	74.2%
Nordic Semiconductor ASA	Norway	Semiconductors & Semiconductor Equipment	-21.9%	-39.7%	Sep-21	-35.1%	6.9%	50.9%
Future PLC	United Kingdom	Media & Entertainment	-32.7%	-46.5%	Sep-21	-5.0%	27.9%	25.4%

Single Stocks

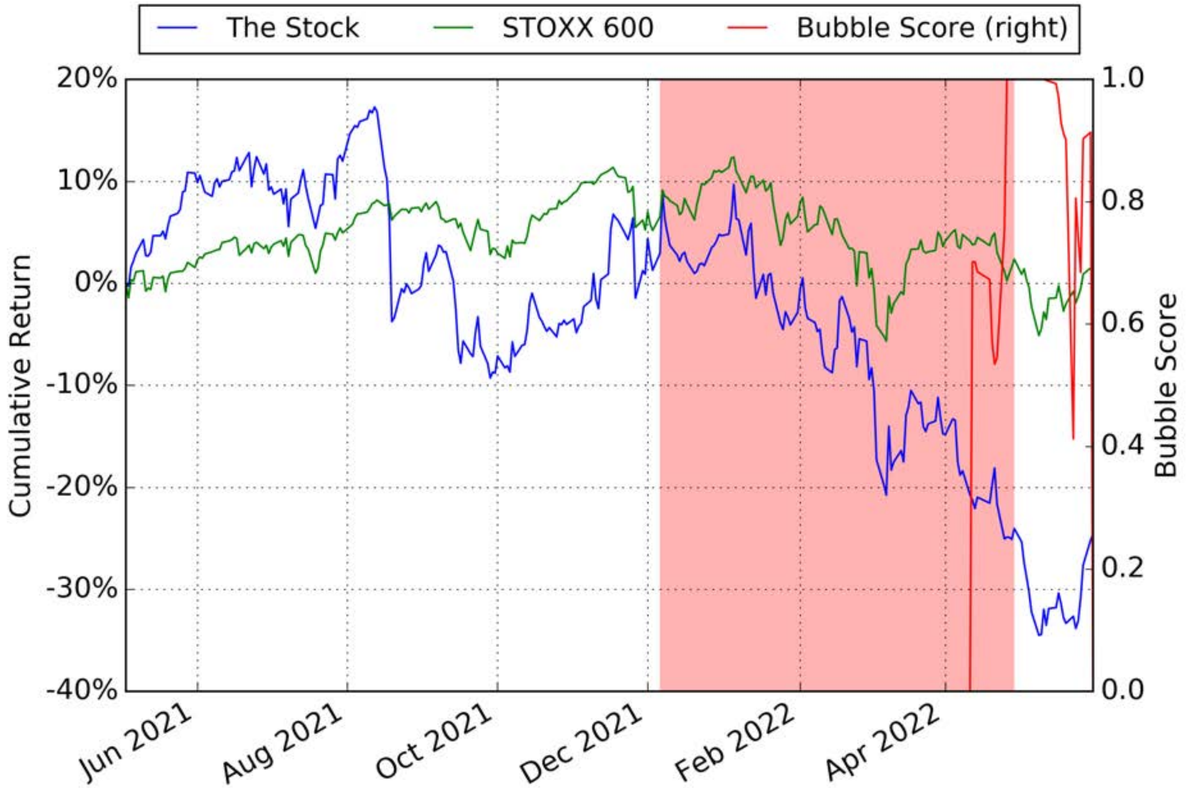
Quadrant 3 Stocks Current Month Example – Starbucks Corp.



- The above graph shows the one-year cumulative return of the stock in blue (left hand scale), NASDAQ 100 in green (left hand scale) and the calculated DS LPPLS Bubble Score in red (right hand scale). The red shaded period delineates the time interval within which the negative bubble is identified. The Bubble Score of this seven-month bubble has reached 89.9% with a bubble size -32.9%.

Single Stocks

Quadrant 3 Stocks Last Month Example – Kering SA.



- The figure above plots the one-year cumulative return of the stock (blue), STOXX 600 (green) and LPPLS Bubble Score (red line on the right y-axis). The red shaded period delineates the time interval within which the strong negative bubble was identified and reported last month. The stock price started a rebound in recent weeks after a further drawdown in the past month. The negative bubble score remains significant this month, indicating the regime shift has not yet finished. One should remain cautious however, given the weak fundamentals of this stock, which may indicate a continuation of the negative bubble in the future.

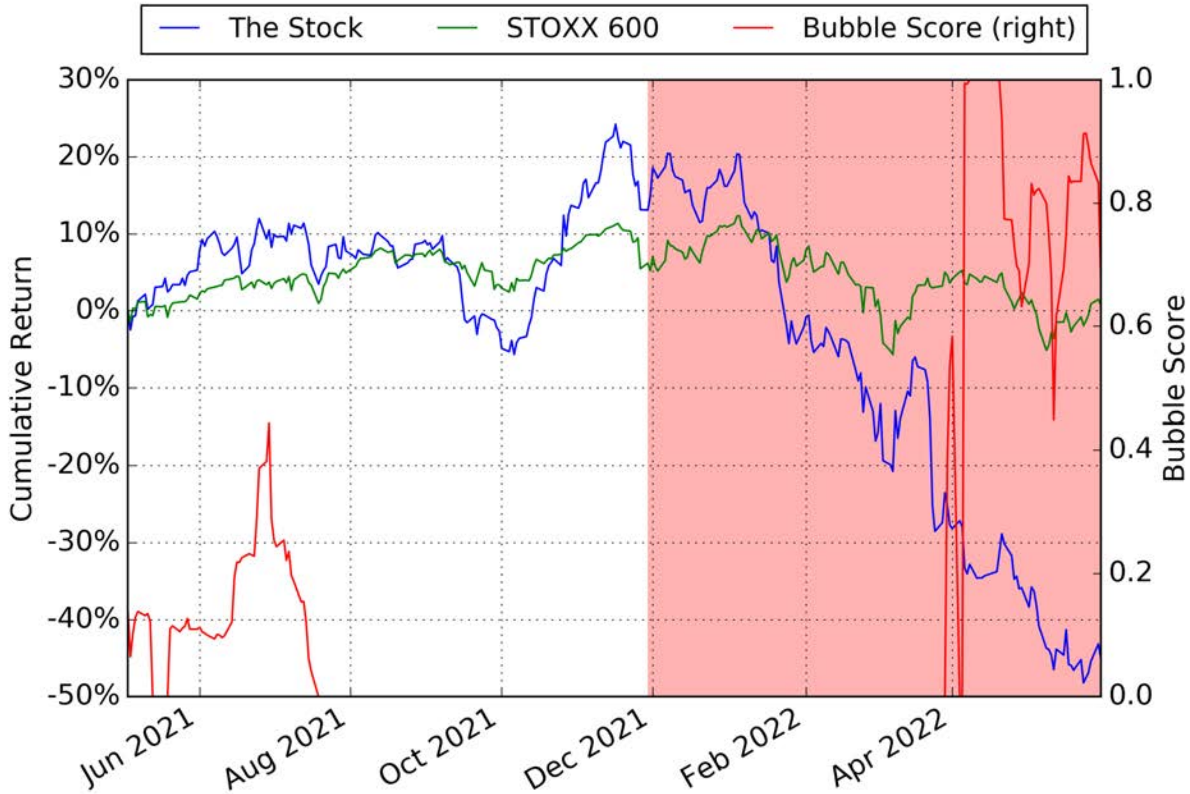
Single Stocks

Quadrant 4 stocks: strong negative bubble signals with strong fundamentals

Company Name	Country of Headquarters	GICS Industry Group Name	Yearly Return	Bubble Size	Bubble Start	Bubble Score	Value Score	Growth Score
Qorvo Inc	United States of America	Semiconductors & Semiconductor Equipment	-39.6%	-26.7%	Dec-21	-86.0%	90.3%	22.2%
D R Horton Inc	United States of America	Consumer Durables & Apparel	-14.6%	-22.9%	Nov-21	-15.4%	94.1%	10.2%
Keysight Technologies Inc	United States of America	Technology Hardware & Equipment	-2.0%	-16.8%	Aug-21	-8.0%	62.0%	35.6%
NetApp Inc	United States of America	Technology Hardware & Equipment	-13.3%	-20.0%	Sep-21	-40.9%	96.9%	30.2%
Ford Motor Co	United States of America	Automobiles & Components	-8.0%	-32.1%	Nov-21	-55.0%	90.6%	90.2%
General Electric Co	United States of America	Capital Goods	-27.3%	-26.2%	Aug-21	-3.8%	71.6%	77.1%
Cisco Systems Inc	United States of America	Technology Hardware & Equipment	-16.8%	-26.2%	Dec-21	-27.0%	89.0%	47.2%
Best Buy Co Inc	United States of America	Retailing	-28.0%	-28.0%	Jun-21	-18.9%	96.7%	45.7%
Johnson Controls International PLC	Ireland; Republic of	Capital Goods	-17.6%	-26.9%	Aug-21	-3.6%	68.2%	65.1%
Skyworks Solutions Inc	United States of America	Semiconductors & Semiconductor Equipment	-37.4%	-35.9%	Oct-21	-49.5%	81.7%	31.3%
Bellway PLC	United Kingdom	Consumer Durables & Apparel	-32.9%	-28.1%	Nov-21	-26.5%	96.3%	11.2%
ams OSRAM AG	Austria	Semiconductors & Semiconductor Equipment	-38.1%	-36.9%	Aug-21	-10.6%	95.6%	81.4%
Travis Perkins PLC	United Kingdom	Capital Goods	-24.4%	-31.2%	Aug-21	-23.4%	88.8%	39.5%
Orkla ASA	Norway	Food, Beverage & Tobacco	-16.2%	-12.2%	Nov-21	-81.2%	67.6%	25.5%
Logitech International SA	Switzerland	Technology Hardware & Equipment	-49.5%	-23.6%	Oct-21	-48.8%	80.7%	5.4%
Wienerberger AG	Austria	Materials	-24.7%	-24.0%	Nov-21	-11.5%	63.4%	1.8%
J Sainsbury PLC	United Kingdom	Food & Staples Retailing	-10.0%	-22.2%	Nov-21	-24.3%	95.9%	84.2%
Just Eat Takeaway.com NV	Netherlands	Retailing	-71.9%	-54.9%	Jan-22	-14.3%	76.5%	99.3%
Prosiebensat 1 Media SE	Germany	Media & Entertainment	-42.7%	-32.4%	Nov-21	-42.2%	64.6%	8.7%
Avast PLC	United Kingdom	Software & Services	-0.3%	-20.9%	Dec-21	-55.3%	75.1%	32.2%
Deutsche Post AG	Germany	Transportation	-32.3%	-33.0%	Nov-21	-14.6%	73.3%	88.5%
Grafton Group PLC	Ireland; Republic of	Capital Goods	-17.4%	-33.6%	Aug-21	-27.4%	94.0%	4.6%
Kion Group AG	Germany	Capital Goods	-49.1%	-53.6%	Dec-21	-67.6%	64.8%	1.9%
Bucher Industries AG	Switzerland	Capital Goods	-24.7%	-23.1%	Oct-21	-21.8%	71.7%	55.1%

Single Stocks

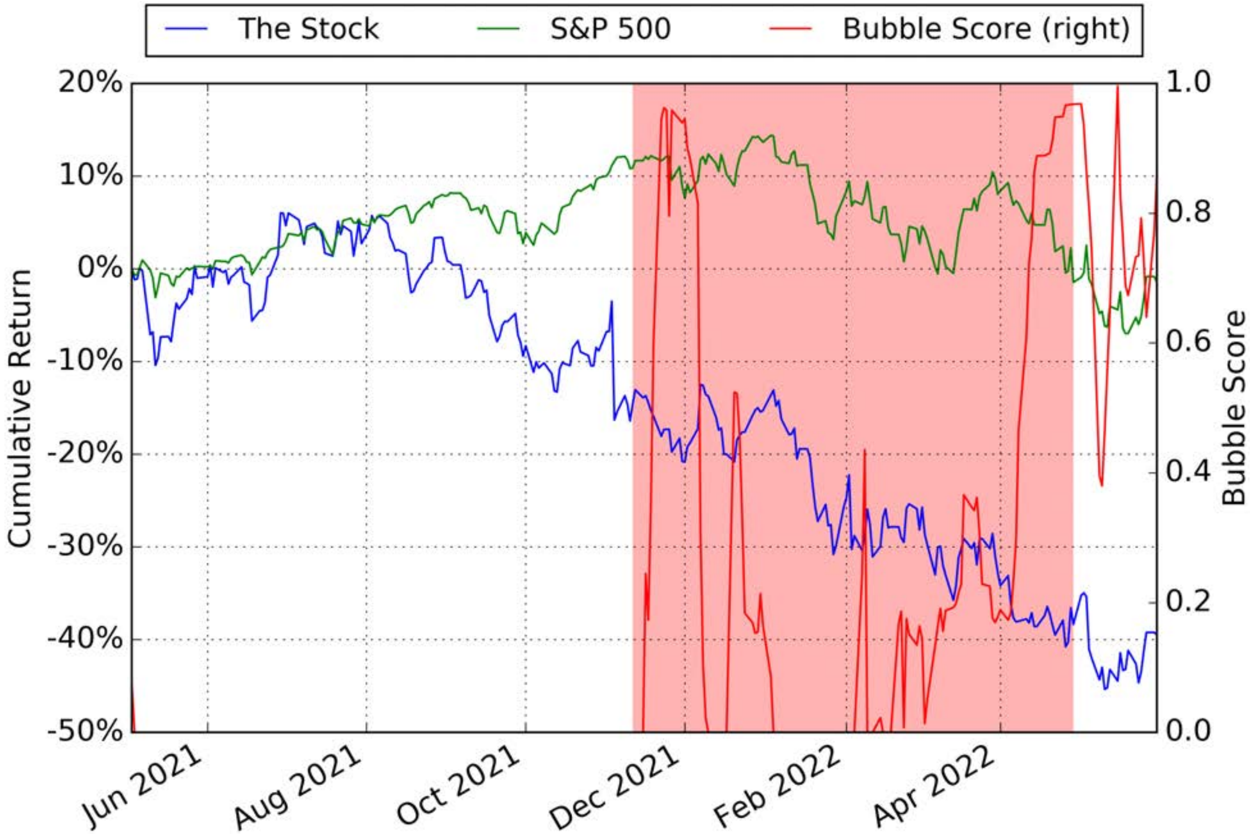
Quadrant 4 Stocks Current Month Example – Kion Group AG.



- The above graph shows the one-year cumulative return of the stock in blue (left hand scale), STOXX 600 in green (left hand scale) and the calculated DS LPPLS Bubble Score in red (right hand scale). The red shaded period delineates the time interval within which the strong negative bubble is identified. The Bubble Score of this six-month bubble has reached 67.6% with a negative bubble size -53.6%. We expect a rebound in the future, given our diagnostic of a negative bubble signal with strong fundamentals, calling for a contrarian buyer position.

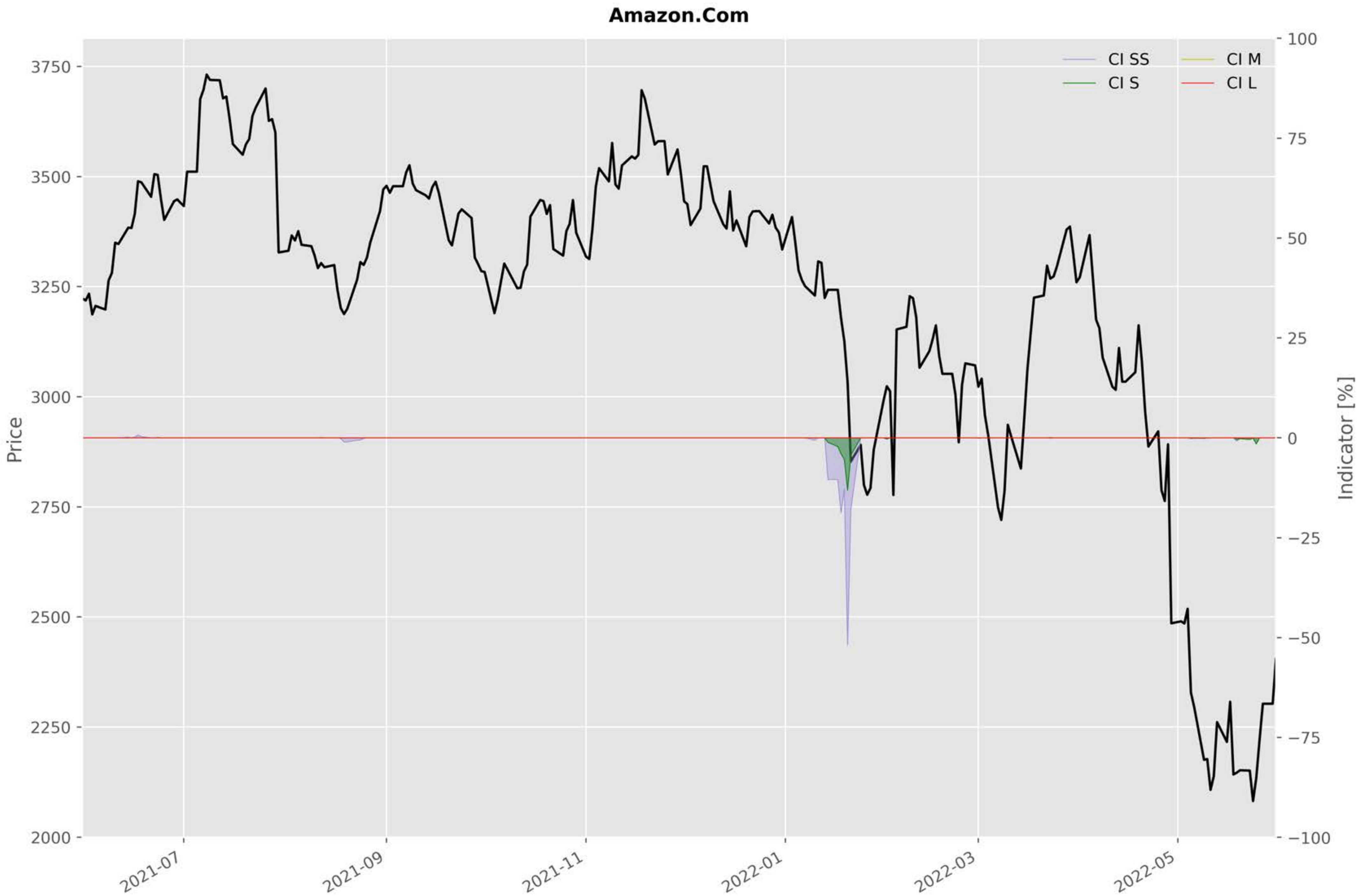
Single Stocks

Quadrant 4 Stocks Last Month Example – Qorvo Inc.



The figure above plots the one-year cumulative return of the stock (blue), S&P 500 (green) and LPPLS Bubble Score (red line on the right y-axis). The red shaded period delineates the time interval within which the strong negative bubble was identified and reported last month. The stock price continued its downward trend in the past month with the bubble signals remaining significant this month. This means that the signal last month was false positive, and the negative bubble may continue to evolve into a larger bubble. Another scenario is that the recent upward correction may be the start of the expected rebound signaling the end of this negative bubble.

Appendix


Amazon.Com
CI SS
CI M
CI S
CI L
Price
Indicator [%]
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3500
3250
3000
2750
2500
2250
2000
100
75
50
25
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−50
−75
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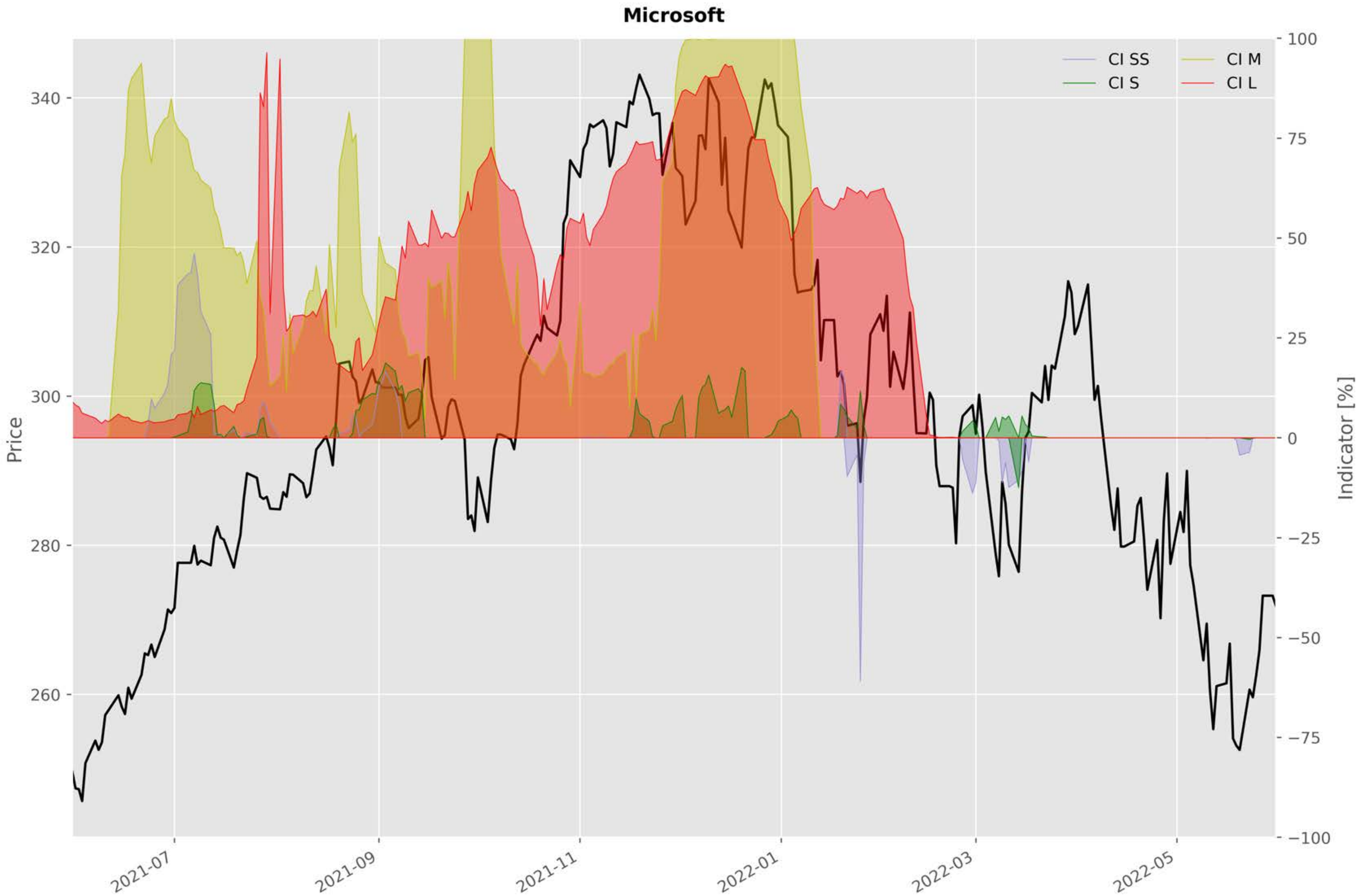
2021-09
2021-11
2022-01
2022-03
2022-05



Microsoft
CI SS
CI M
CI S
CI L
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320
300
280
260
Price
100
75
50
25
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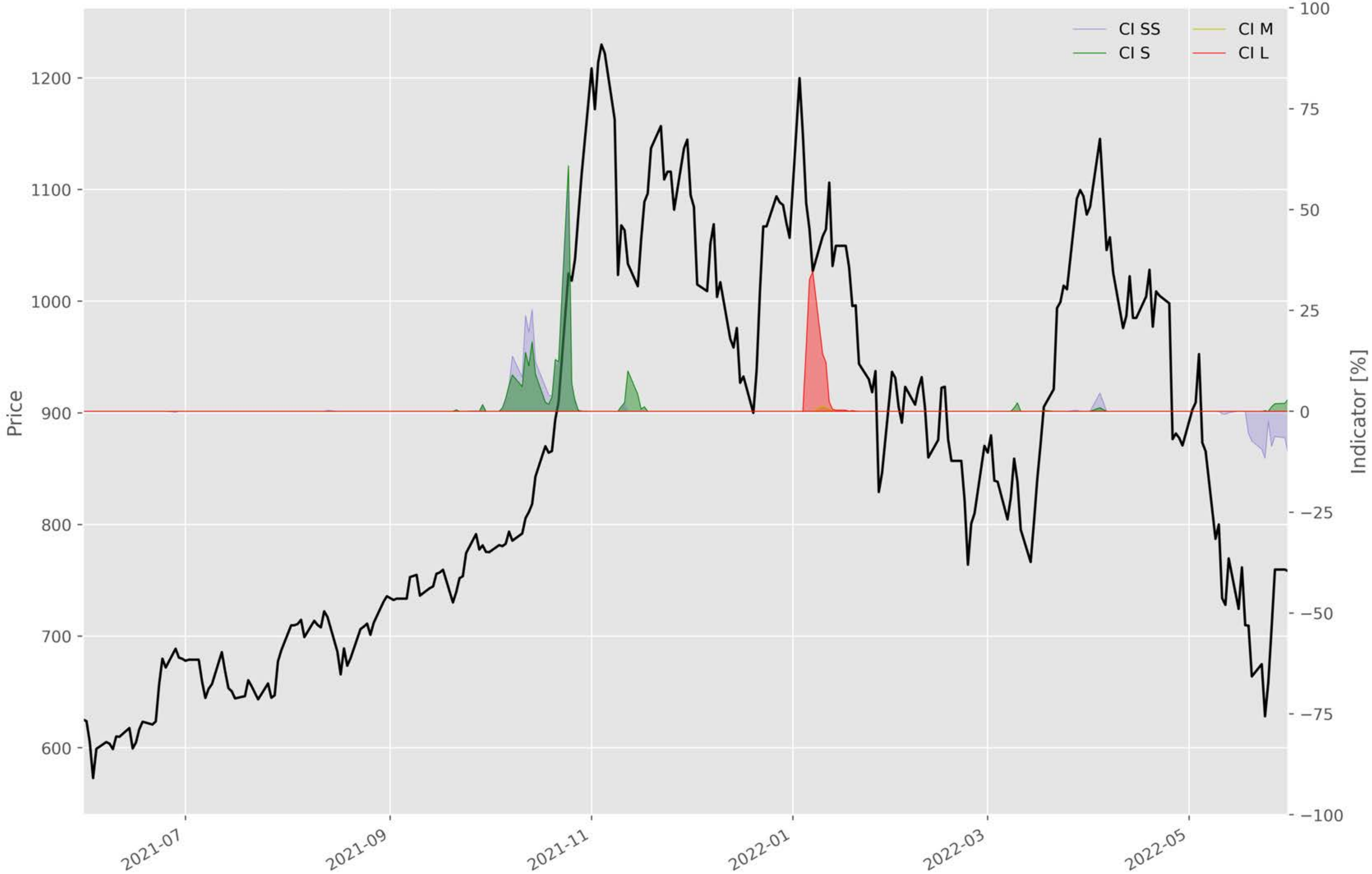
2022-03
2022-05

Tesla



CI SS
CI S
CI M
CI L
Price
Indicator [%]
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1000
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800
700
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100
75
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−75
−100
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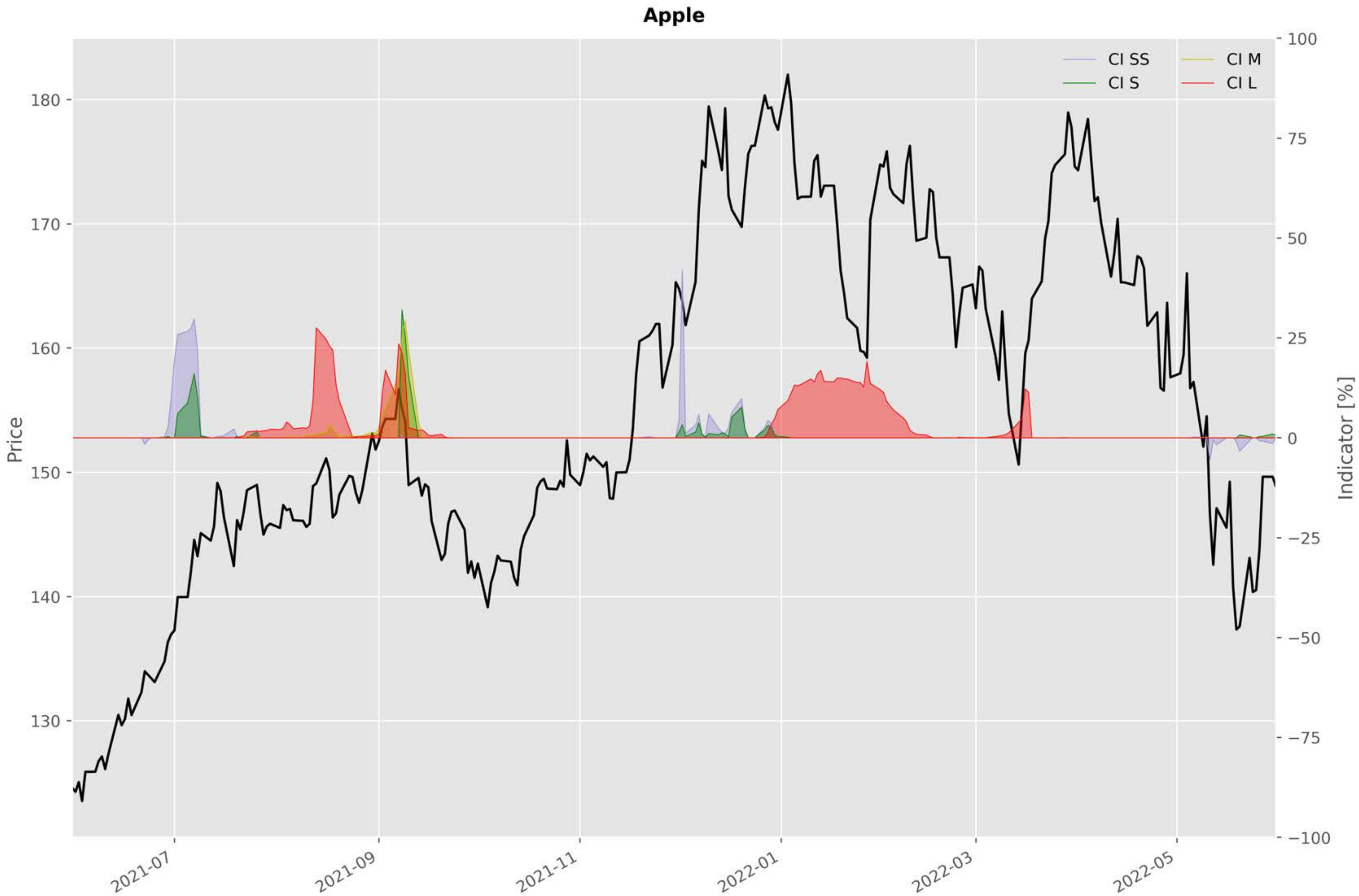
2021-09
2021-11
2022-01
2022-03
2022-05


Apple
CI SS
CI M
CI S
CI L
Price
Indicator [%]
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170
160
150
140
130
100
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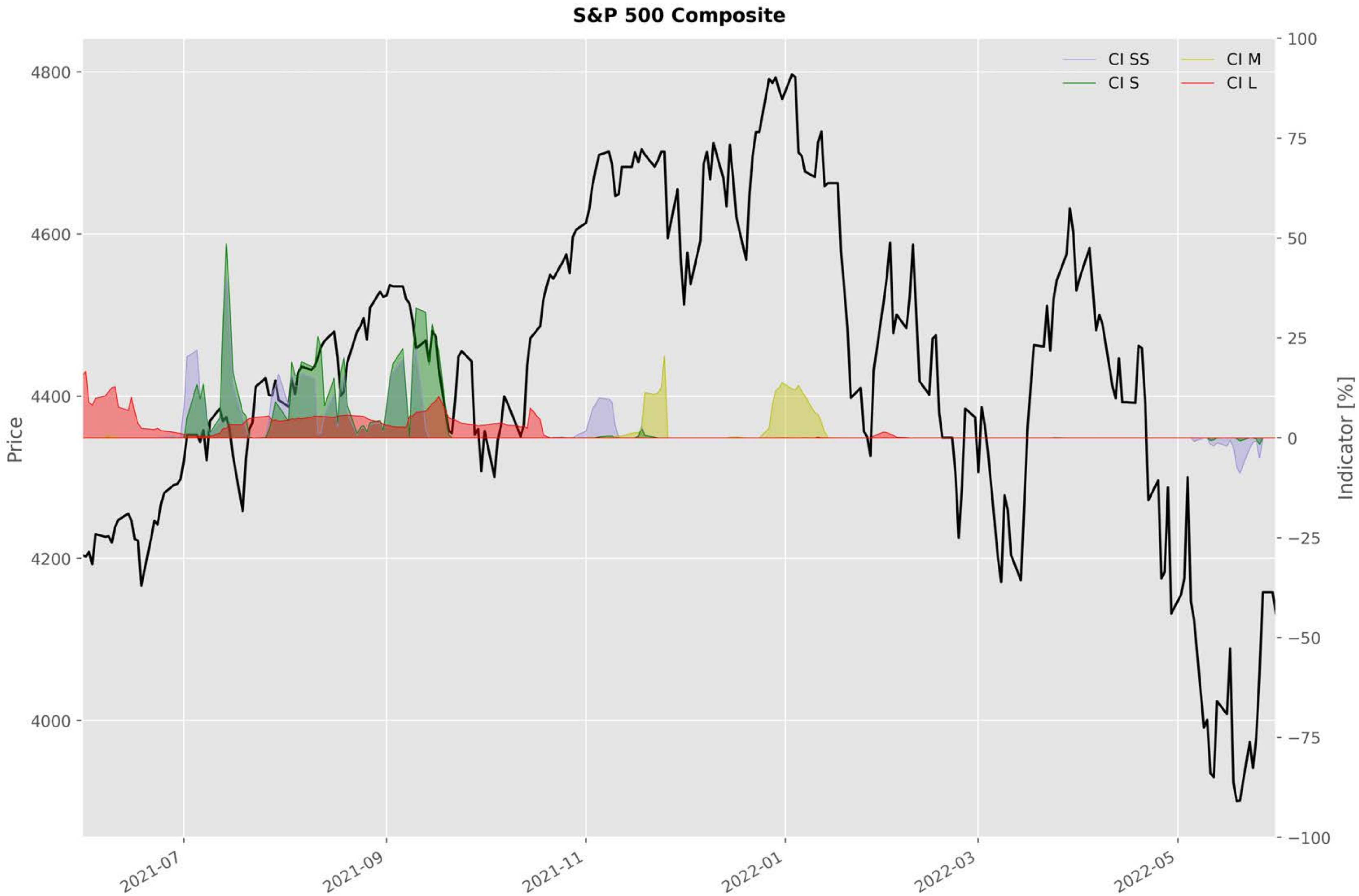
2022-03
2022-05


S&P 500 Composite
CI SS
CI S
CI M
CI L
Price
Indicator [%]
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75
50
25
0
−25
−50
−75
−100
2021-07
2021-09
2021-11
2022-01
2022-03
2022-05

Bitcoin/USD
CI SS
CI M
CI S
CI L
Price
Indicator [%]
65000
60000
55000
50000
45000
40000
35000
30000
100
75
50
25
0
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Jun
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May

More Information

- Visit the **Financial Crisis Observatory** for more information

  http://www.er.ethz.ch/financial-crisis-observatory.html

- **Contacts ETH Zurich**

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